

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
8	0003549590	ENGINEERING RELEASED	2014-12-19

X304 MLB SCHEMATIC - DVT

Fri Dec 19 12:14:48 2014

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Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-1573	1	SCHEM, MLB, X304	SCH	CRITICAL	
820-4924	1	PCBF, MLB, X304	PCB	CRITICAL	

DRAWING TITLE		SCHEM, MLB, X304	
 Apple Inc.	DRAWING NUMBER		SIZE
	051-1573		D
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BOM Groups

BOM GROUP	BOM OPTIONS
X304_COMMON	ALTERNATE,COMMON,X304_COMMON1,X304_COMMON2,X304_COMMON3,X304_COMMON4,X304_PROGPARTS
X304_COMMON1	TBTHV:P15V,SKIP_5V3V3:AUDIBLE,PANEL:NEW,SSD_CLKREQ:BT
X304_COMMON2	EDP,EDP_LS_CAP,CAMERA_3V3:S0,CAM_WAKE:NO,CAM_XTAL:NO,VCORE_FETS
X304_COMMON3	XDP,SAMCONN,BKLT:PR0D,CPUHRM:ALRT,LOADRC:NO,OTHERRC:NO,DDRRC:NO,BTTRC:NO,BMONRC:NO,TPADRC:NO
X304_PROGPARTS	SMC_PROG:PROTO00,BOOTROM_PROG,TBTROM_PROG
X304_DEVEL:ENG	ALTERNATE,ENGISNS,XDP_CONN,DBGLED
X304_DEVEL: DVT	ALTERNATE,ENGISNS,XDP_CONN,S0PG00D_ISL
X304_DEVEL:PVT	ALTERNATE
ENGISNS	LOADISNS,OTHERISNS,DDRISNS,TBTISNS,BMONISNS,TPADISNS

Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
337S00107	1	CPU, BW, SR26K, PRQ, F0-B2, 2.7, 2BW, 1.05, 1168	U0500	CRITICAL	CPU_BDW23:2.7G
337S00108	1	CPU, BW, SR26H, PRQ, F0-B2, 2.9, 2BW, 1.1, 1168	U0500	CRITICAL	CPU_BDW23:2.9G
337S00109	1	CPU, BW, SR26E, PRQ, F0-B2, 3.1, 2BW, 1.1, 1168	U0500	CRITICAL	CPU_BDW23:3.1G

DVT

998-7866	1	INTERPOSER, BGA1168P, SINGLE SIDE	U0500	CRITICAL	CPU_SOCKET
338S1247	1	IC, TBT, FR-4C, AB, PQ, CIO, SR13C, FCBGA288	U2800	CRITICAL	
338S1264	1	IC, BCM1570A2KFEB4G, S2 CMRA, 8XB, 208FCBGA	U3900	CRITICAL	
376S1194	2	MOSFET, N-CH, 30V, 15.3A, 12M, 8P 3.3X3.3 DFN	Q7310, Q7320	CRITICAL	VCORE_FET:VSHY
376S1193	2	MOSFET, N-CH, 30V, 22A, 6.0M, 8P 3.3X3.3 DFN	Q7311, Q7321	CRITICAL	VCORE_FET:VSHY
376S00036	2	MOSFET, N-CH, 30V, 52A, 5.9MO, 3.3X3.3 DFN8	Q7310, Q7320	CRITICAL	VCORE_FET:ONSMI
376S00037	2	MOSFET, N-CH, 30V, 64A, 3.5MO, 3.3X3.3 DFN8	Q7311, Q7321	CRITICAL	VCORE_FET:ONSMI

Programmables (All Builds)

TBT

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
341S00192	1	T29,EPROM,FALCON RIDGE (V27.1) EVT2,X304	U2890	CRITICAL	TBTROM_PROG

SMC

341S3982	1	IC,SMC-B1,EXT(V2.21A5) PROTO 0,X304	U5000	CRITICAL	SMC_PROG:PROT00
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EFI ROM

341S00235	1	EFI ROM,MLB (V0145) DVT,X304	U6100	CRITICAL	BOOTROM_PROG
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Variable BOM Groups

BOM GROUP	BOM OPTIONS
X304_COMMON4	SMCBOARDID:16

Development/Base BOMs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-1314	1	X304 MLB COMMON BOM	BASE	CRITICAL	BASE_BOM
985-1319	1	X304 MLB DEVEL BOM	DEVEL	CRITICAL	DEVEL_BOM

Sub - BOMs

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
685-1318	1	VCORE FET, VSHY, X304	VCOREFETS	CRITICAL	VCORE_FETS

Main DRAM SPD Straps

BOM GROUP	BOM OPTIONS
RAM_16G_HYNIX_1600	16G_HYNIX_1600, RAMCFG3:L, RAMCFG2:L, RAMCFG1:H, RAMCFG0:L
RAM_16G_HYNIX_1866	16G_HYNIX_1866, RAMCFG3:L, RAMCFG2:H, RAMCFG1:H, RAMCFG0:L
RAM_8G_HYNIX_1600	8G_HYNIX_1600, RAMCFG3:L, RAMCFG2:L, RAMCFG1:L, RAMCFG0:H
RAM_8G_HYNIX_1866	8G_HYNIX_1866, RAMCFG3:L, RAMCFG2:H, RAMCFG1:L, RAMCFG0:H
RAM_4G_HYNIX_1600	4G_HYNIX_1600, RAMCFG3:L, RAMCFG2:L, RAMCFG1:L, RAMCFG0:L
RAM_4G_HYNIX_1866	4G_HYNIX_1866, RAMCFG3:L, RAMCFG2:H, RAMCFG1:L, RAMCFG0:L
RAM_16G_ELPIDA_1600	16G_ELPIDA_1600, RAMCFG3:H, RAMCFG2:L, RAMCFG1:H, RAMCFG0:L
RAM_16G_ELPIDA_1866	16G_ELPIDA_1866, RAMCFG3:H, RAMCFG2:H, RAMCFG1:H, RAMCFG0:L
RAM_8G_ELPIDA_1600	8G_ELPIDA_1600, RAMCFG3:H, RAMCFG2:L, RAMCFG1:L, RAMCFG0:H
RAM_8G_ELPIDA_1866	8G_ELPIDA_1866, RAMCFG3:H, RAMCFG2:H, RAMCFG1:L, RAMCFG0:H
RAM_4G_ELPIDA_1600	4G_ELPIDA_1600, RAMCFG3:H, RAMCFG2:L, RAMCFG1:L, RAMCFG0:L
RAM_4G_ELPIDA_1866	4G_ELPIDA_1866, RAMCFG3:H, RAMCFG2:H, RAMCFG1:L, RAMCFG0:L
RAM_8G_SAMSUNG_1600	8G_SAMSUNG_1600, RAMCFG3:H, RAMCFG2:L, RAMCFG1:H, RAMCFG0:H
RAM_8G_SAMSUNG_1866	8G_SAMSUNG_1866, RAMCFG3:H, RAMCFG2:H, RAMCFG1:H, RAMCFG0:H
RAM_4G_SAMSUNG_1600	4G_SAMSUNG_1600, RAMCFG3:L, RAMCFG2:L, RAMCFG1:H, RAMCFG0:H
RAM_4G_SAMSUNG_1866	4G_SAMSUNG_1866, RAMCFG3:L, RAMCFG2:H, RAMCFG1:H, RAMCFG0:H

Strategic Silicon

PART#	STRATEGIC VALUE	COMMENT
337S00068	08	CPU
337S00069	08	CPU
337S00070	08	CPU
337S00071	08	CPU
353S00260	07	TPAD ELEC FUSE
333S0786	07	SYS MEMORY HYNIX
333S0784	07	SYS MEMORY HYNIX
333S0792	07	SYS MEMORY MICRON
333S0790	07	SYS MEMORY MICRON
333S00004	07	SYS MEMORY SAMSUNG
311S0597	02	KEYBOARD I2C EXPANDER
359S0197	01	GREEN CLOCK
338S1247	01	FALCON RIDGE
353S3931	01	TBT PWR MUX
353S3812	01	TBT MUX
353S3814	01	TBT MUX
353S00095	01	DDC CROSSBAR
353S3328	01	DDC CROSSBAR
343S0511	01	PCI-E DELAY IC
338S1264	01	S2
333S0760	01	S2 MEMORY
333S0704	01	S2 MEMORY
353S3054	01	USB POWER/SAFETY
343S0649	01	SMC RESET CHIP
353S4080	01	AUDIO
353S2888	01	AUDIO AMPS
353S2958	01	AUDIO AMPS
353S2929	01	BAT CHARGER
353S00036	01	VR12.6 CONTROLLER
353S4160	01	BEN
343S0666	01	SAK, HDMI SELECT
341S3982	01	SMC
341S00192	01	T29 ROM
341S00235	01	EFI ROM

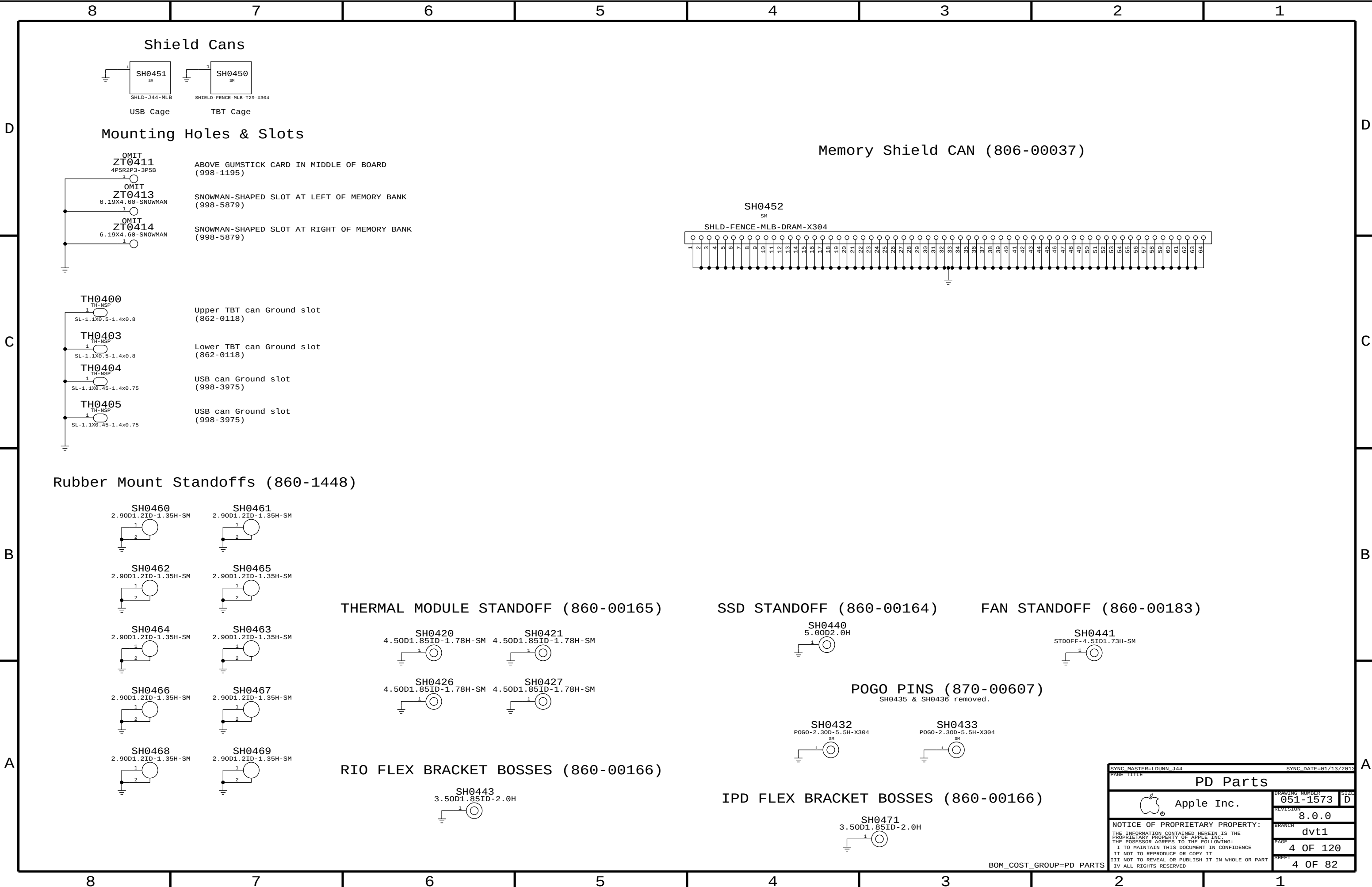
Main DRAM Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
333S0783	4	IC, SDRAM, 25nm 32Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	16G_HYNIX_1600
333S0784	4	IC, SDRAM, 25nm 32Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	16G_HYNIX_1866
333S0785	4	IC, SDRAM, 29nm 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	8G_HYNIX_1600
333S0786	4	IC, SDRAM, 29nm 16Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	8G_HYNIX_1866
333S0787	4	IC, SDRAM, 29nm 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	4G_HYNIX_1600
333S0788	4	IC, SDRAM, 29nm 8Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	4G_HYNIX_1866
333S0789	4	IC, SDRAM, 25nm 32Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	16G_ELPIDA_1600
333S0790	4	IC, SDRAM, 25nm 32Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	16G_ELPIDA_1866
333S0791	4	IC, SDRAM, 25nm 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	8G_ELPIDA_1600
333S0792	4	IC, SDRAM, 25nm 16Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	8G_ELPIDA_1866
333S0793	4	IC, SDRAM, 25nm 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	4G_ELPIDA_1600
333S0794	4	IC, SDRAM, 25nm 8Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	4G_ELPIDA_1866
333S00003	4	IC, SDRAM, 23nm 16Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	8G_SAMSUNG_1600
333S00004	4	IC, SDRAM, 23nm 16Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	8G_SAMSUNG_1866
333S00001	4	IC, SDRAM, 23nm 8Gb, LPDDR3-1600, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	4G_SAMSUNG_1600
333S00002	4	IC, SDRAM, 23nm 8Gb, LPDDR3-1866, 178P FBGA	U2300, U2400, U2500, U2600	CRITICAL	4G_SAMSUNG_1866

S2 DRAM Parts

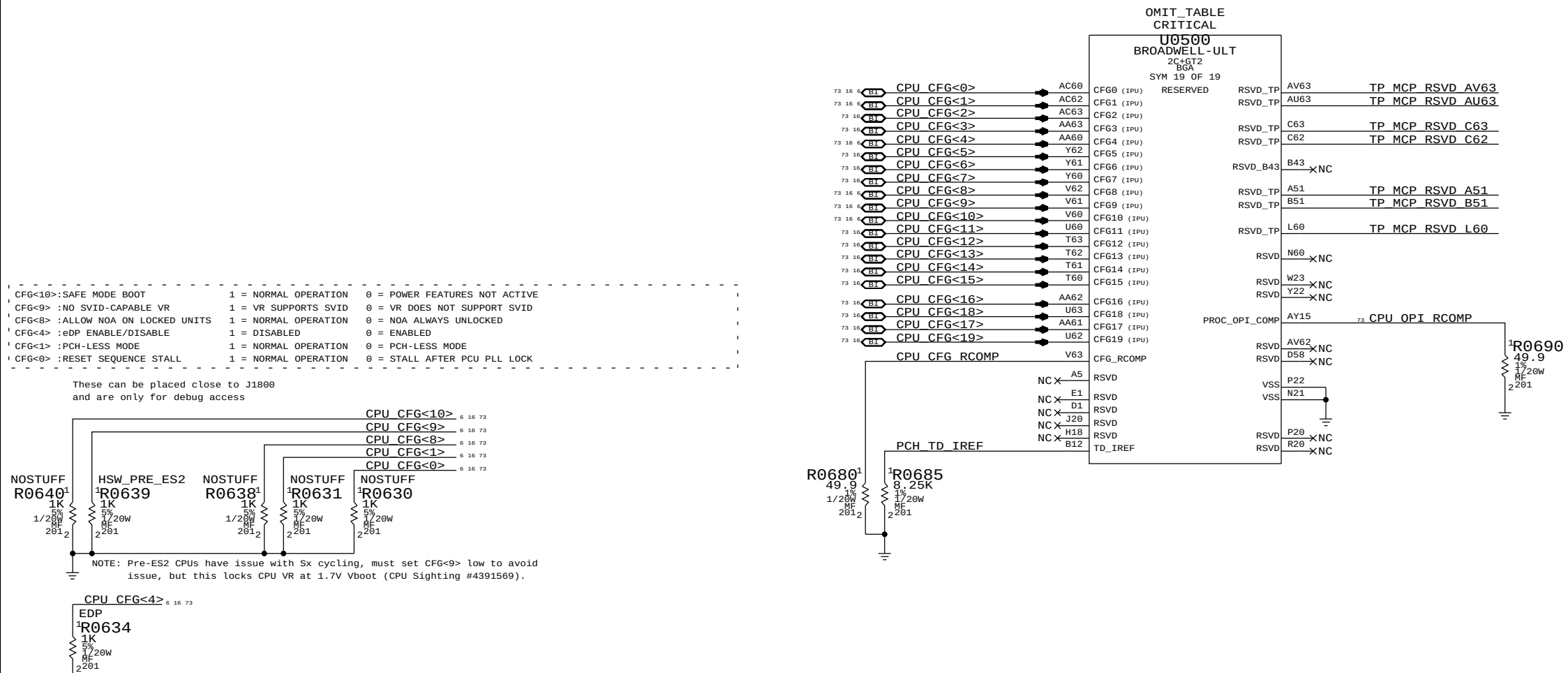
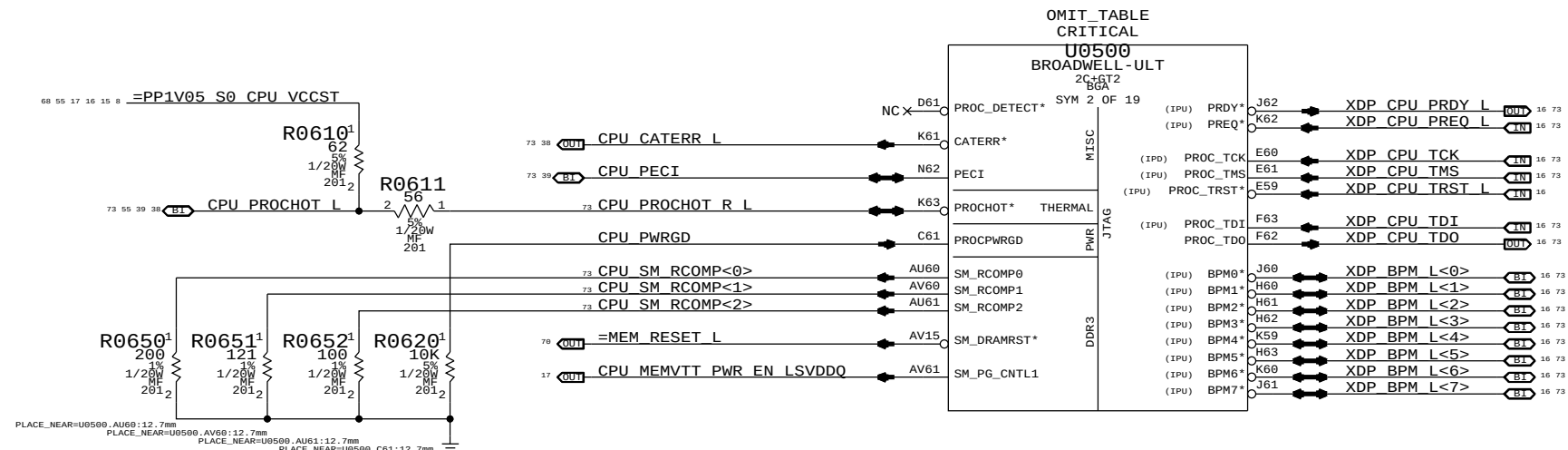
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333S0700	1	IC, SDRAM, 4GBIT. DDR3L - 1600, HUMA, 96B BGA	U4000		

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BOM Configuration			
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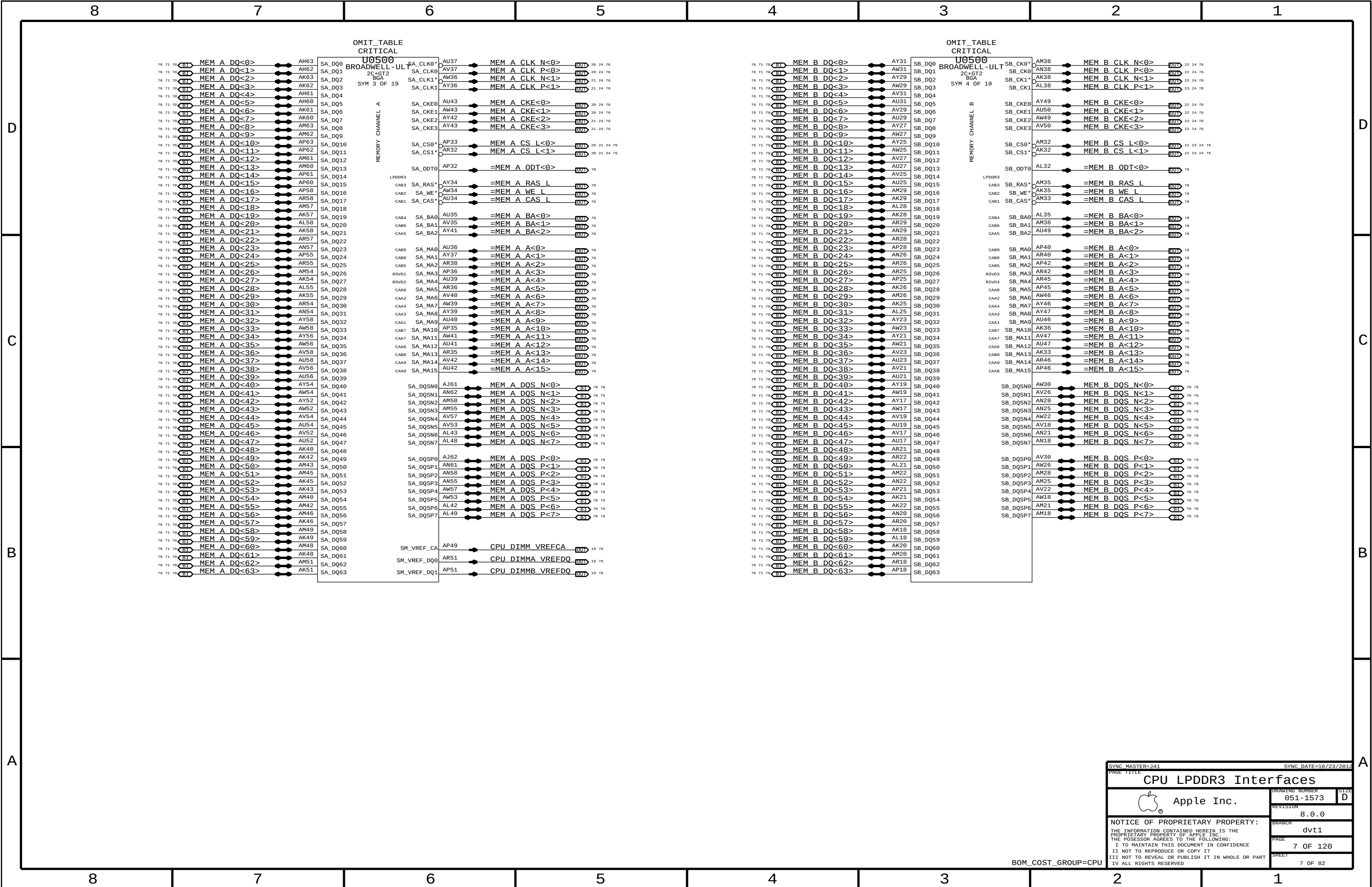
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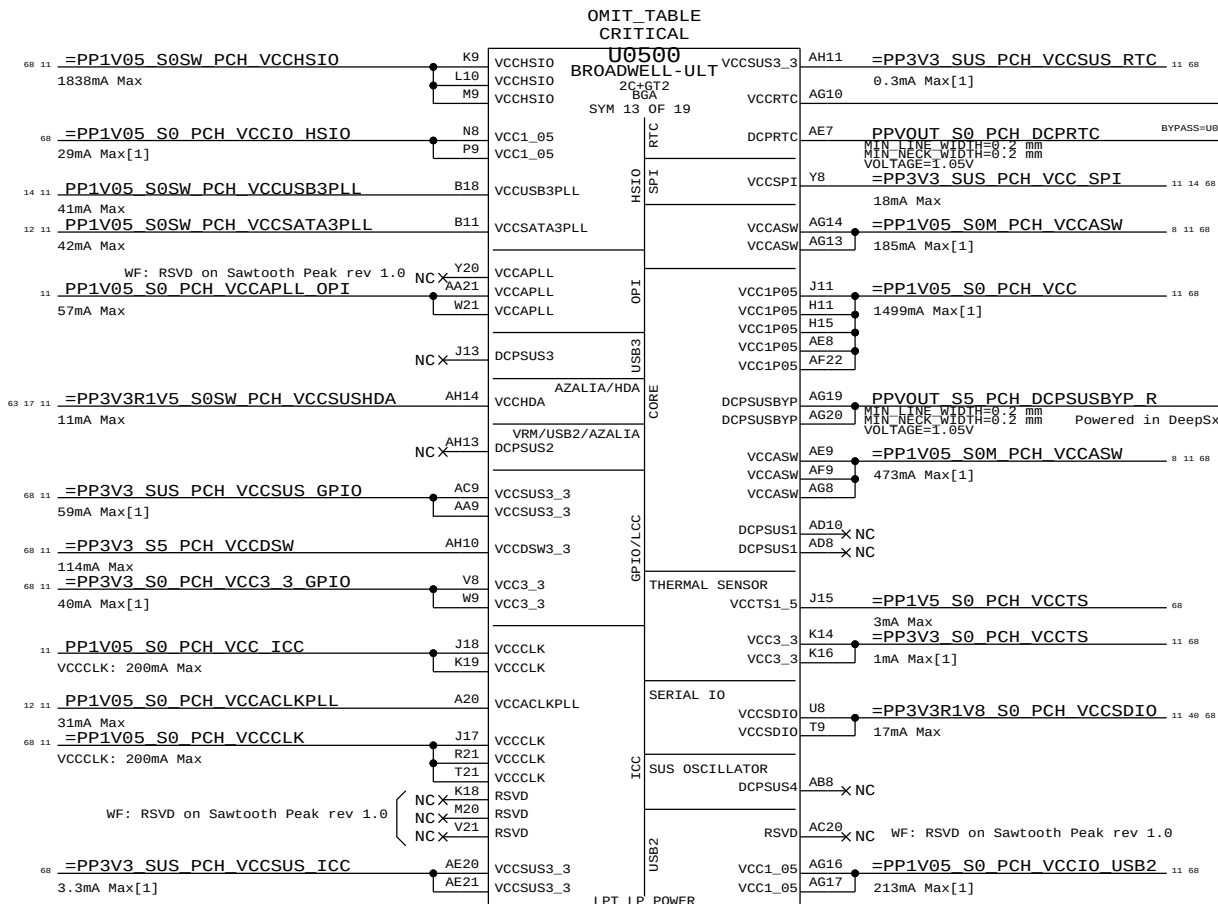
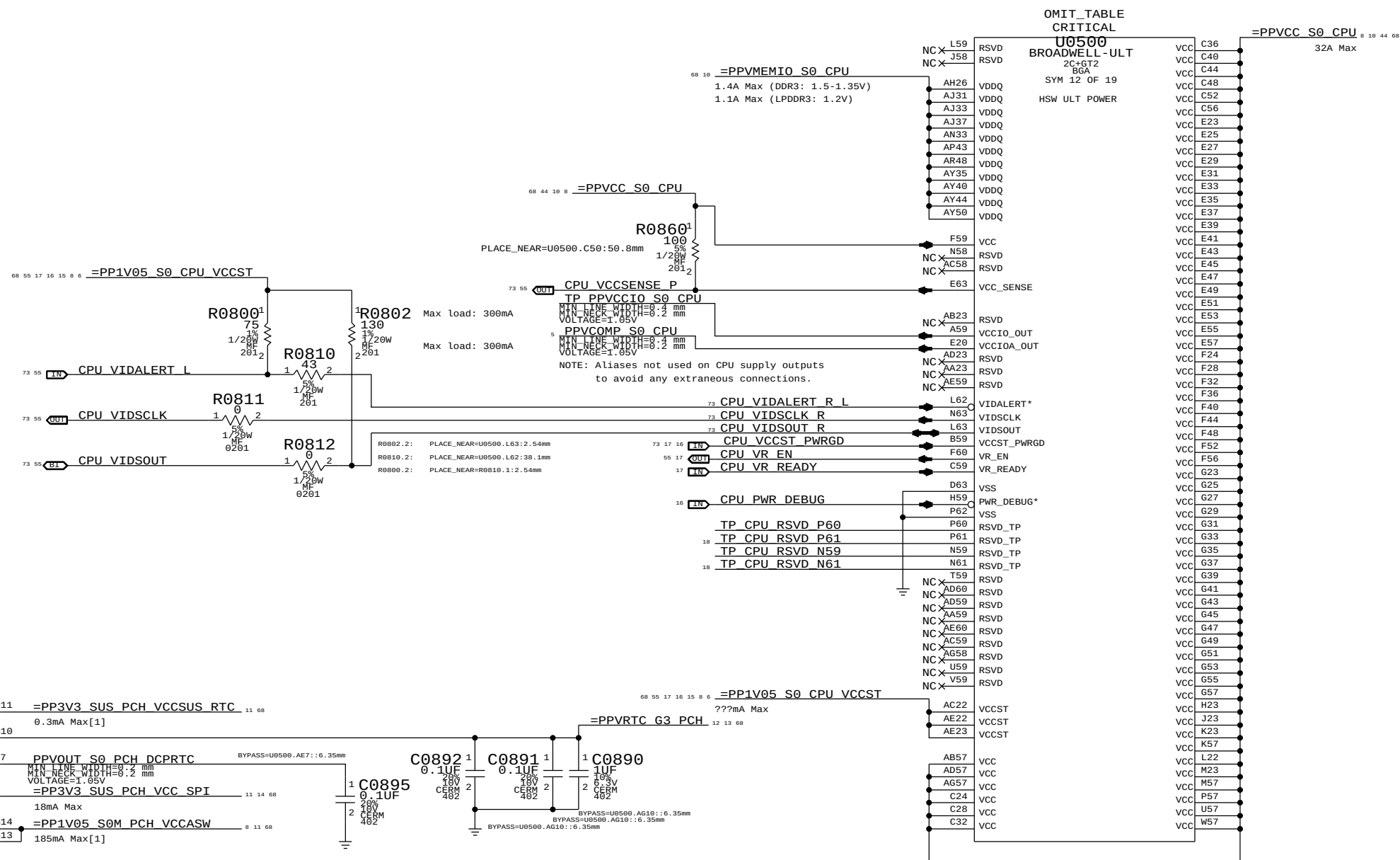


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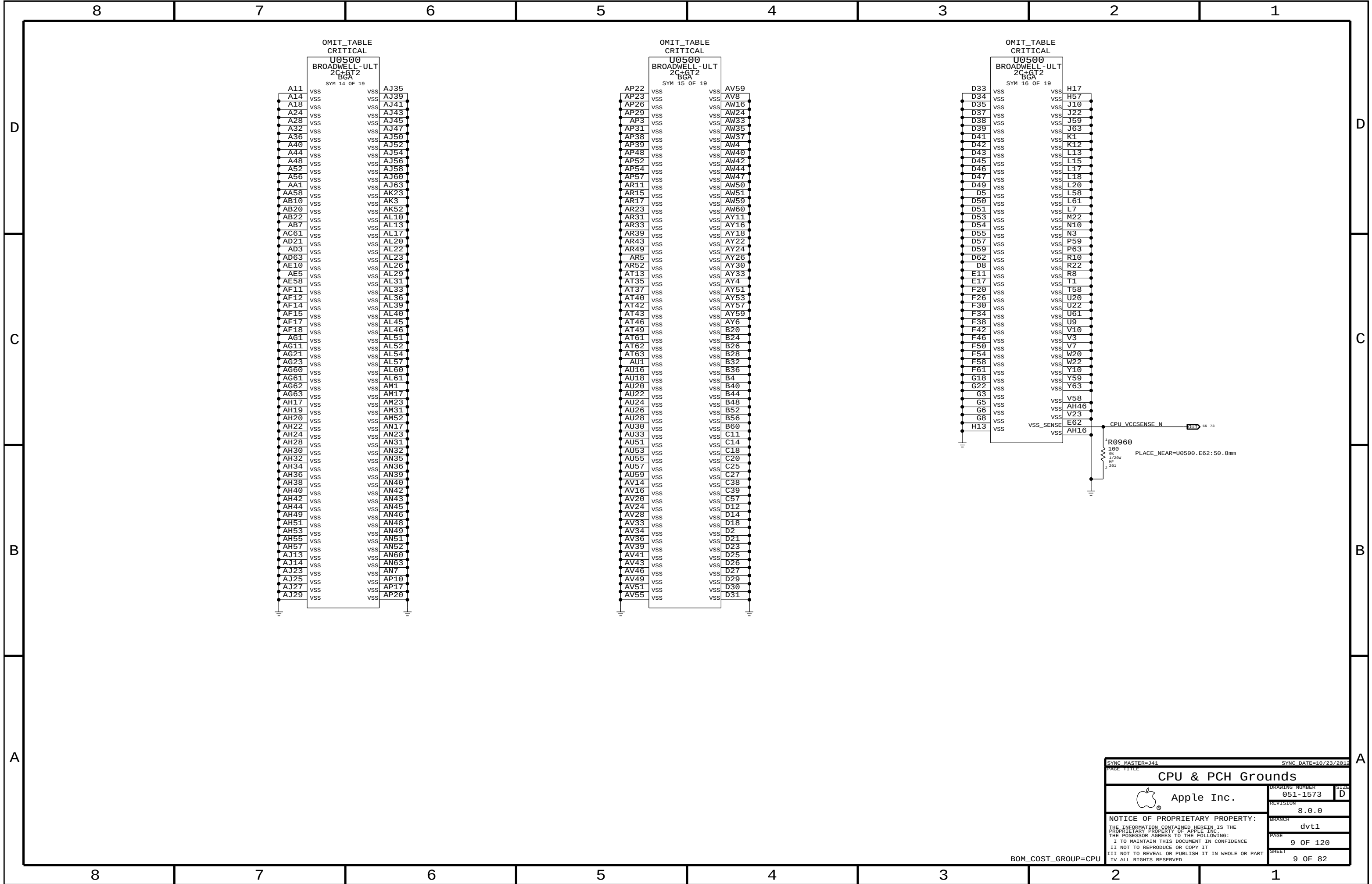
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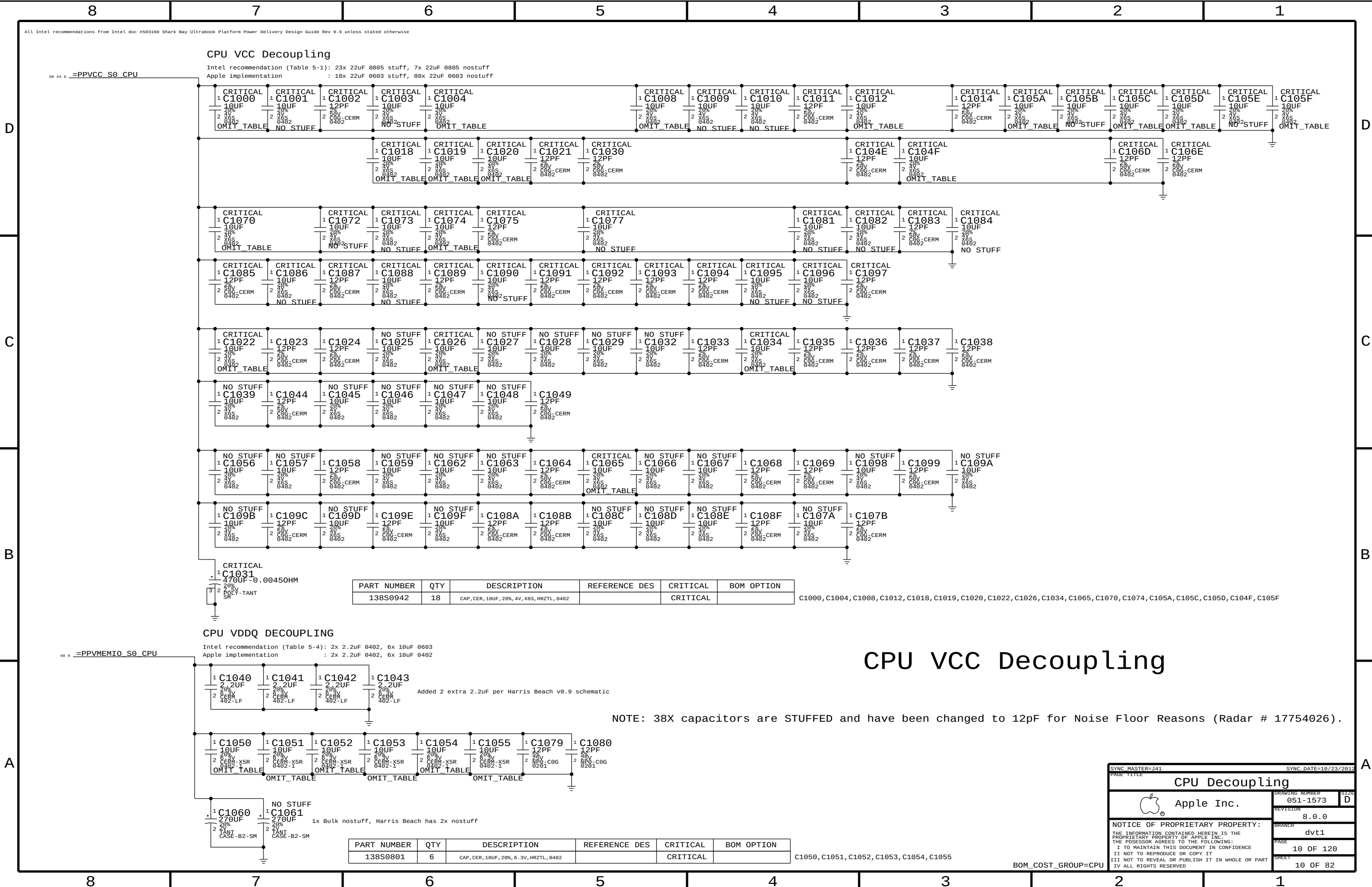


BDW-ULT current estimates from Broadwell Mobile ULT Processor EDS vol.1 Doc# 514405, Rev.: 0.9v1
Wildcat Point-LP current estimates from Wildcat Point-LP PCH EDS, Doc# 515621, Rev. 0.9
Note [1] current numbers from clarification email, from Srini, dated 9/10/2012 2:11pm.



SYMC MASTER-J41		SYMC DATE=10/23/2013	
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CPU & PCH Power			
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CPU VCC Decoupling

SYNC MASTER=141

SYNC DATE=10/23/2012

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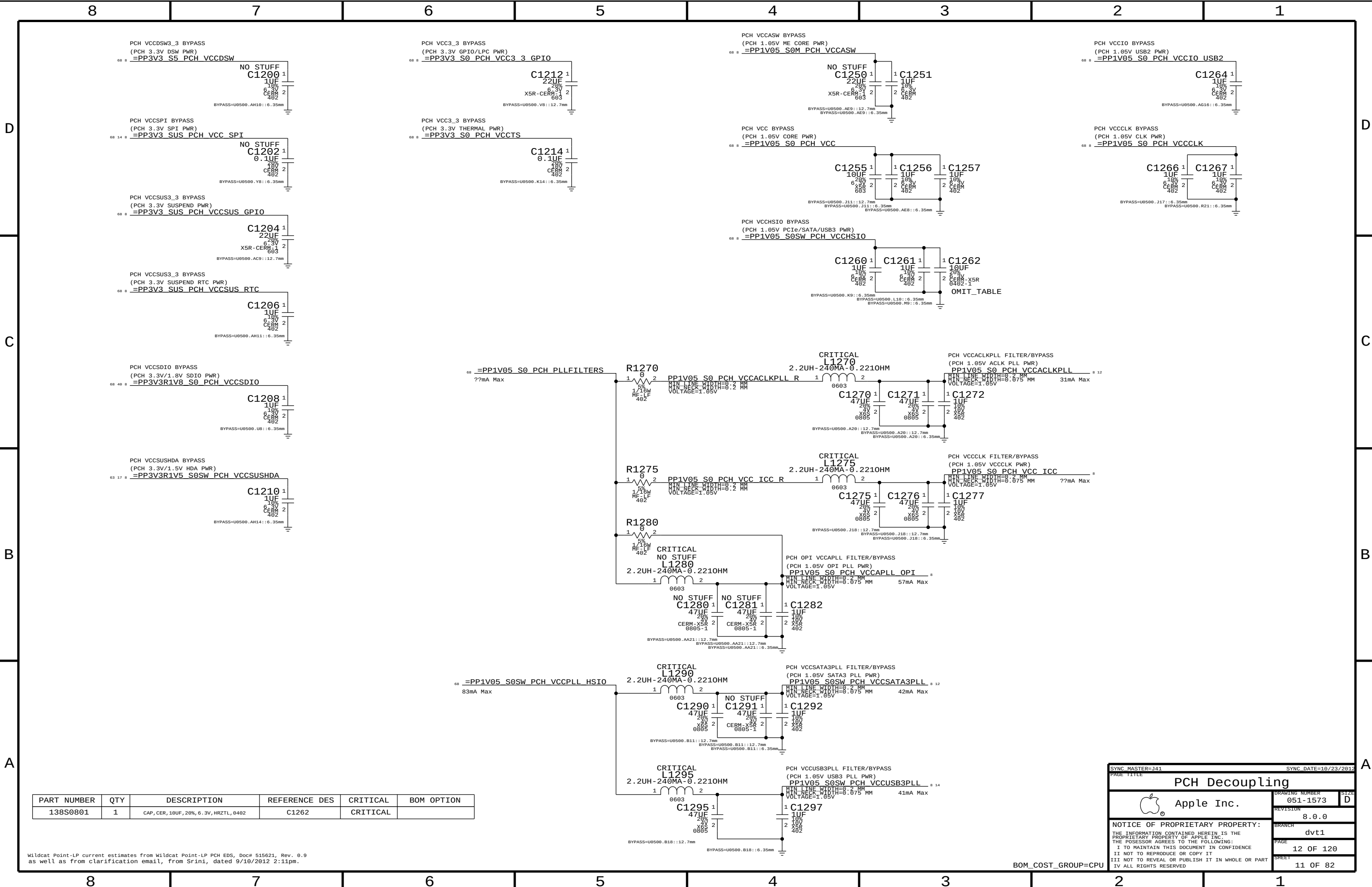
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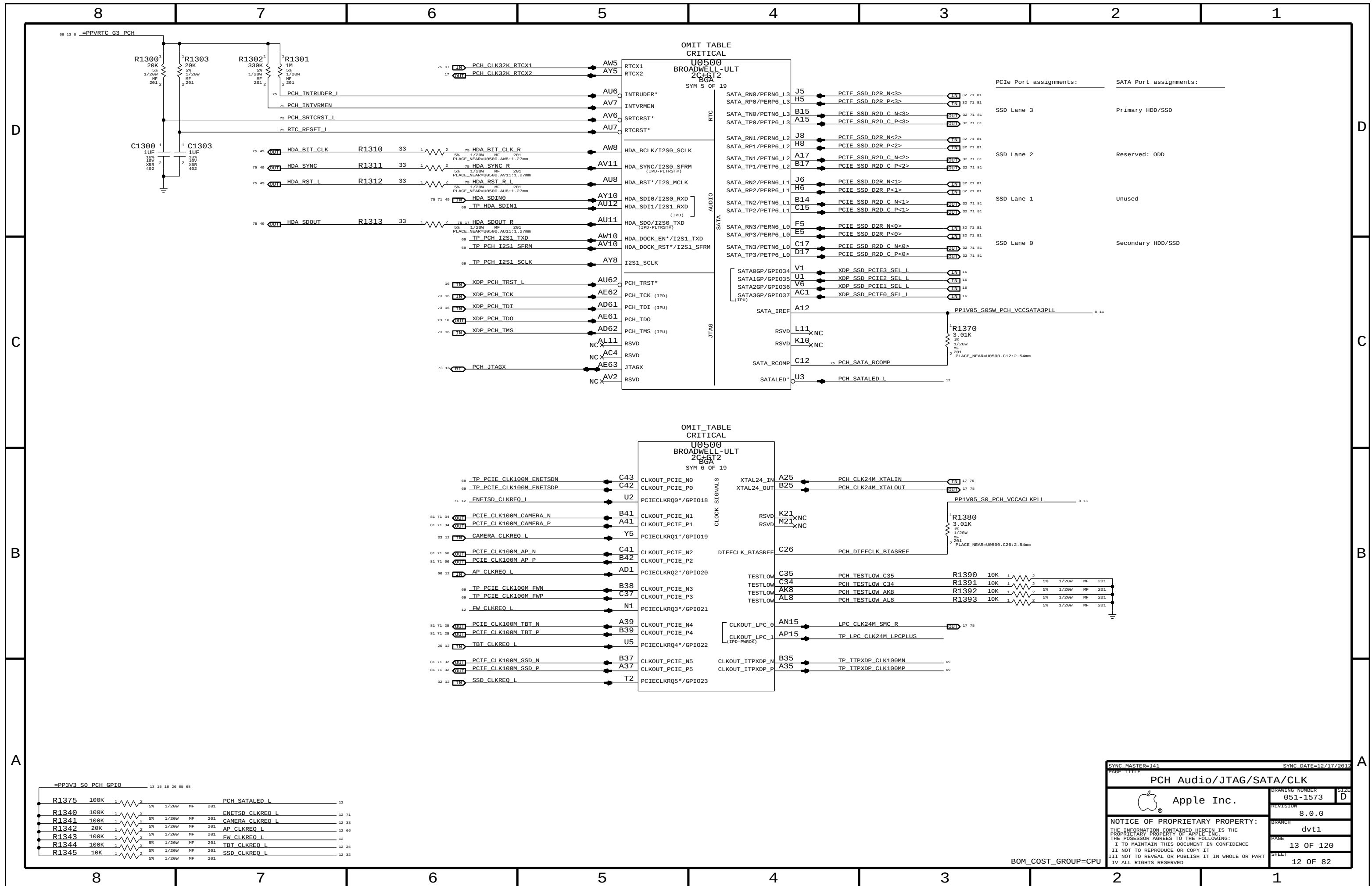


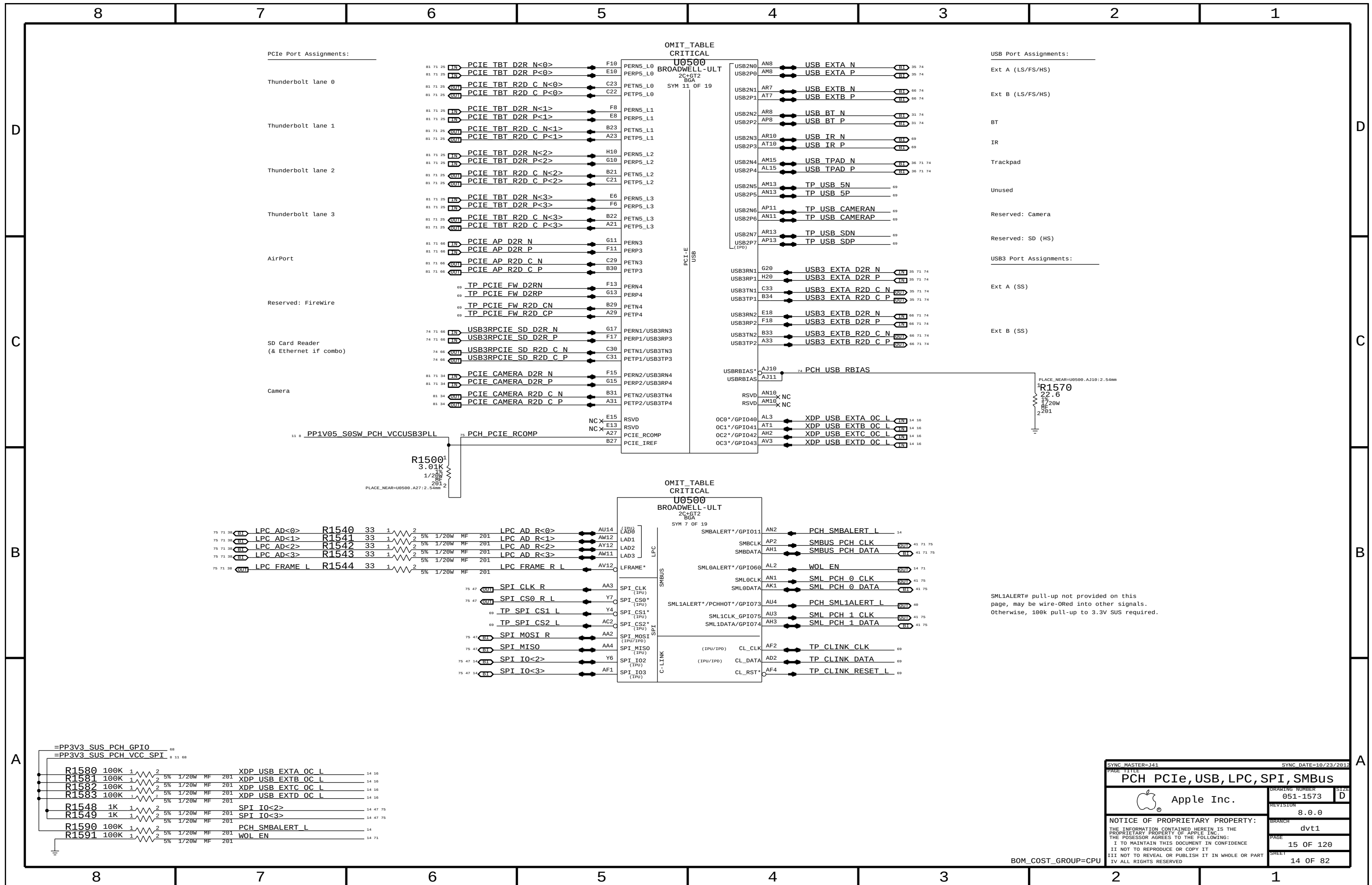
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138S0801	1	CAP, CER, 10UF, 20%, 6.3V, HRZTL, 0402	C1262	CRITICAL	

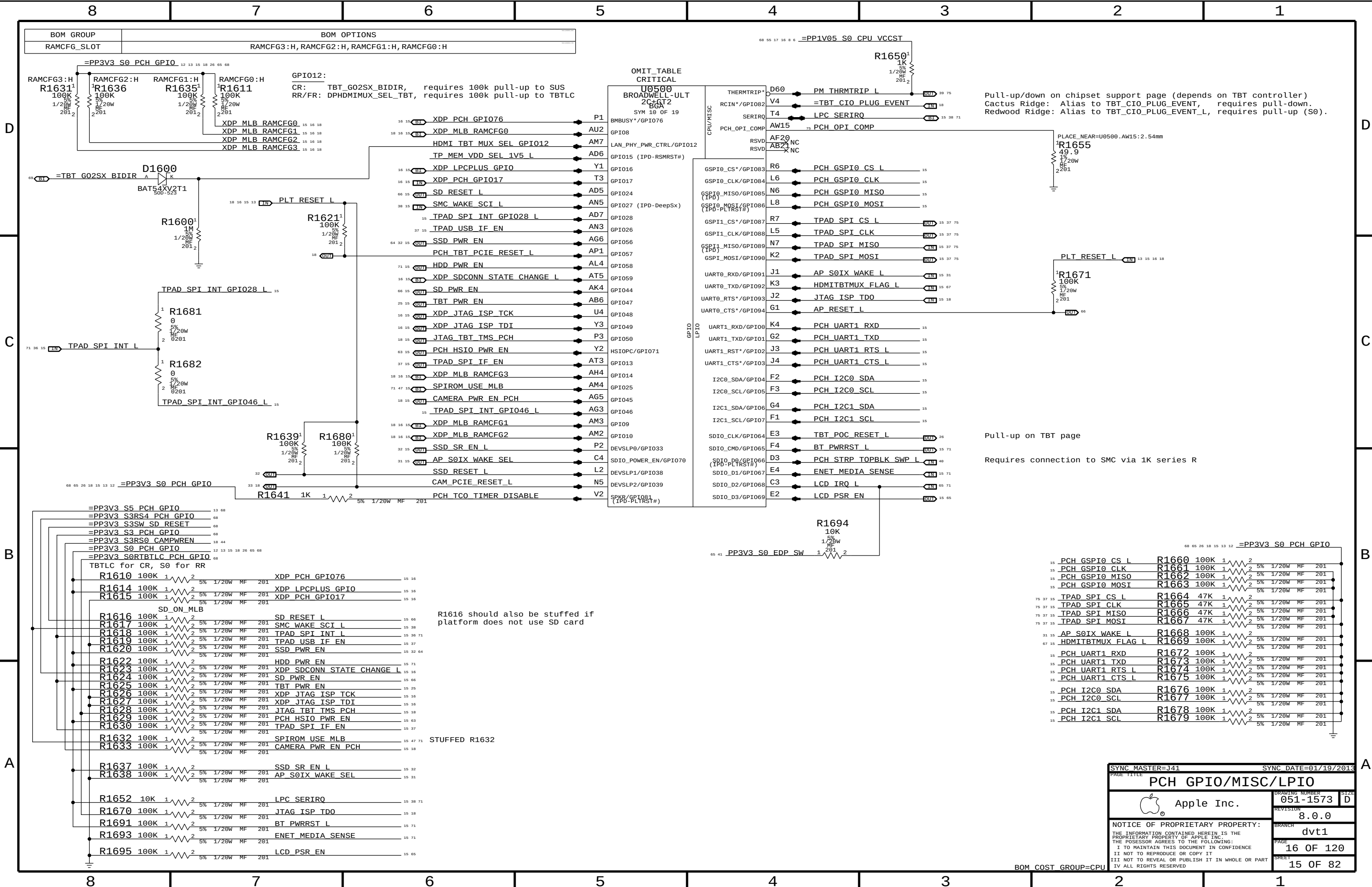
Wildcat Point-LP current estimates from Wildcat Point-LP PCH EDS, Doc# 515621, Rev. 0.9 as well as from clarification email, from Srini, dated 9/10/2012 2:11pm.

BOM_COST_GROUP=CPU

SYNC MASTER=J41		SYNC DATE=10/23/2012	
PAGE TITLE		PCH Decoupling	
		DRAWING NUMBER	051-1573
		REVISION	8.0.0
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		PAGE	12 OF 120
		SHEET	11 OF 82





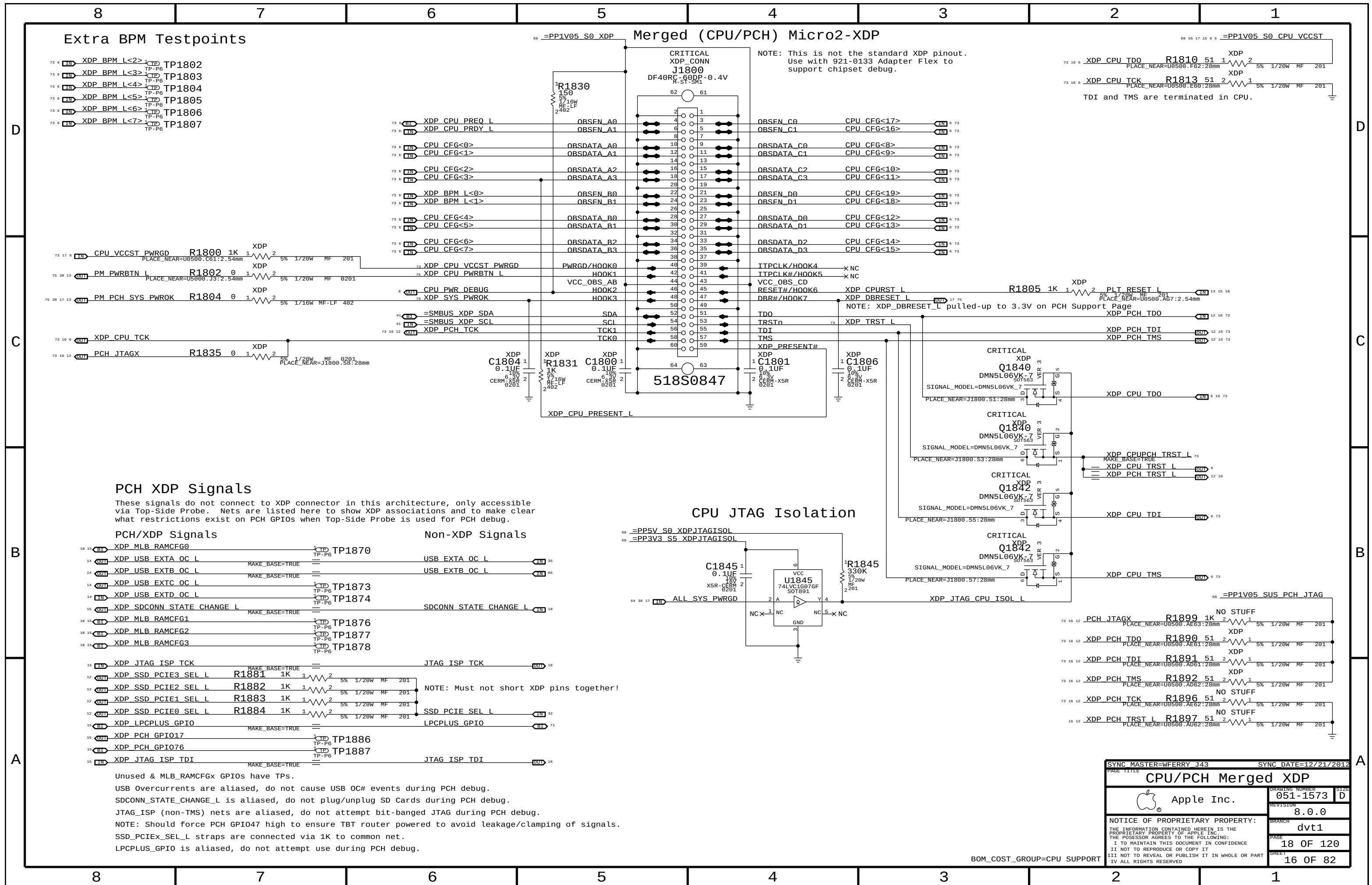


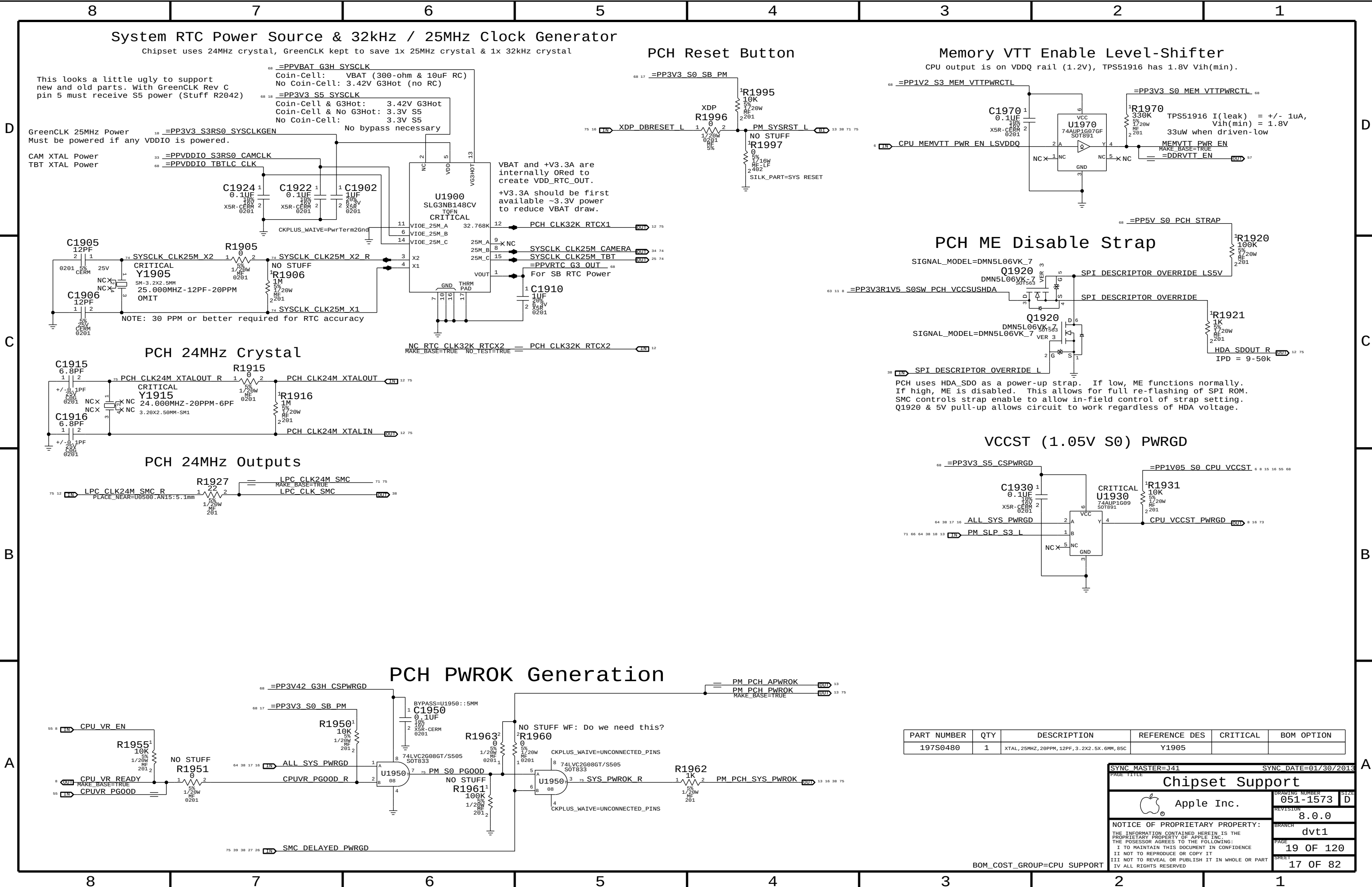
Pull-up/down on chipset support page (depends on TBT controller)
Cactus Ridge: Alias to TBT_CIO_PLUG_EVENT, requires pull-down.
Redwood Ridge: Alias to TBT_CIO_PLUG_EVENT_L, requires pull-up (S0).

Pull-up on TBT page
Requires connection to SMC via 1K series R

SYNC_MASTER=J41		SYNC_DATE=01/19/2013	
PAGE TITLE		PCH GPIO/MISC/LPIO	
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BOM COST GROUP=CPU





D

C

B

A

D

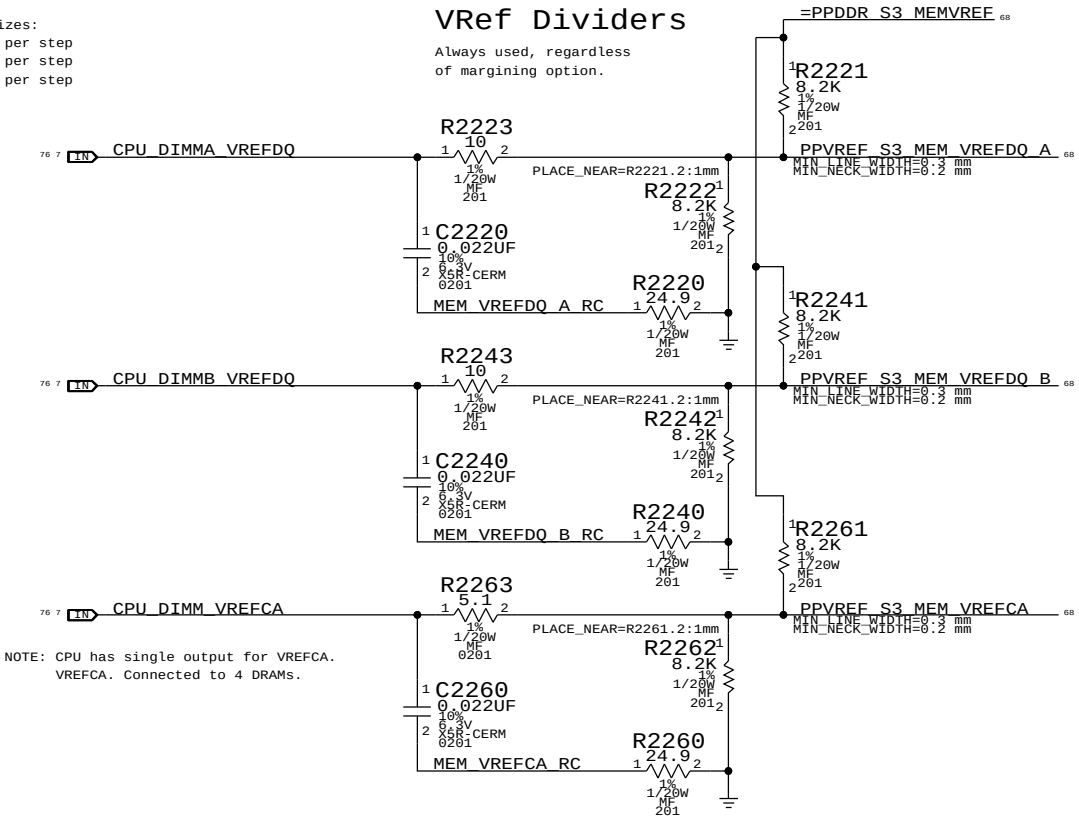
C

B

A

CPU-Based Margining

NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step
DDR3L (1.35V) 6.99mV per step
LPDDR3 (1.2V) ??mV per step



	MEM A VREF DQ		MEM B VREF DQ		MEM A VREF CA	MEM B VREF CA	MEM VREG	
DAC Channel:	A		B		C	C	D	
PCA9557D Pin:	1		2		3	4	5	
	LPDDR3 (1.2V)		DDR3L (1.35V)		LPDDR3 (1.2V)	DDR3L (1.35V)		
Nominal value	0.600V (DAC: 0x2E.5)		0.675V (DAC: 0x34)		1.200V (DAC: 0x5D)	1.343V (DAC: 0x68)		
Margin target:	0.300V - 0.900V (+/- 300mV)		0.337V - 1.013V (+/- 337.5mV)		0.800V - 1.600V (+/- 400mV)	0.972V - 1.714V (+/- 371mV)		
DAC range:	0.000V - 1.199V (0x00 - 0x5D)		0.000V - 1.354V (0x00 - 0x69)		0.000V - 2.397V (0x00 - 0xBA)	0.000V - 2.694V (0x00 - 0xD1)		
VRef current:	+73uA - -73uA (- = sourced)		+82uA - -82uA (- = sourced)		+21uA - -21uA (- = sourced)	+25uA - -25uA (- = sourced)		
DAC step size:	6.36mV / step @ output		6.36mV / step @ output		4.28mV / step @ output	3.53mV / step @ output		

NOTE: LPDDR3 assumes TPS51916 supply with 28.7k/57.6k divider
DDR3L assumes TPS51916 supply with 19.6k/57.6k divider

SYNC MASTER=YHARTANTO_344

SYNC DATE=01/02/2013

PAGE TITLE

LPDDR3 VREF Margining

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DRAWING NUMBER

051-1573

REVISION

8.0.0

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PAGE

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SHEET

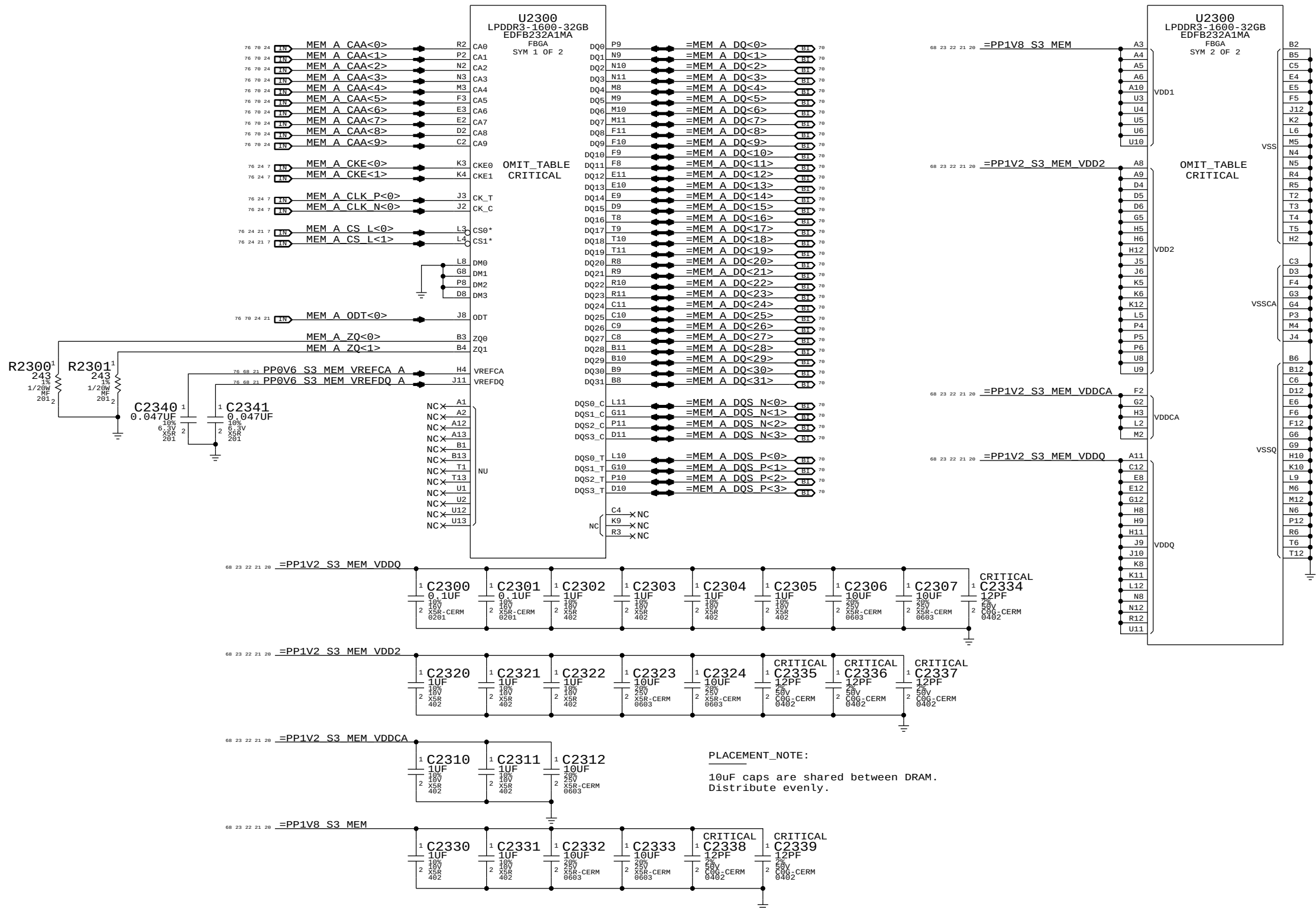
19 OF 82

SIZE

D

BOM_COST_GROUP=CPU_SUPPORT

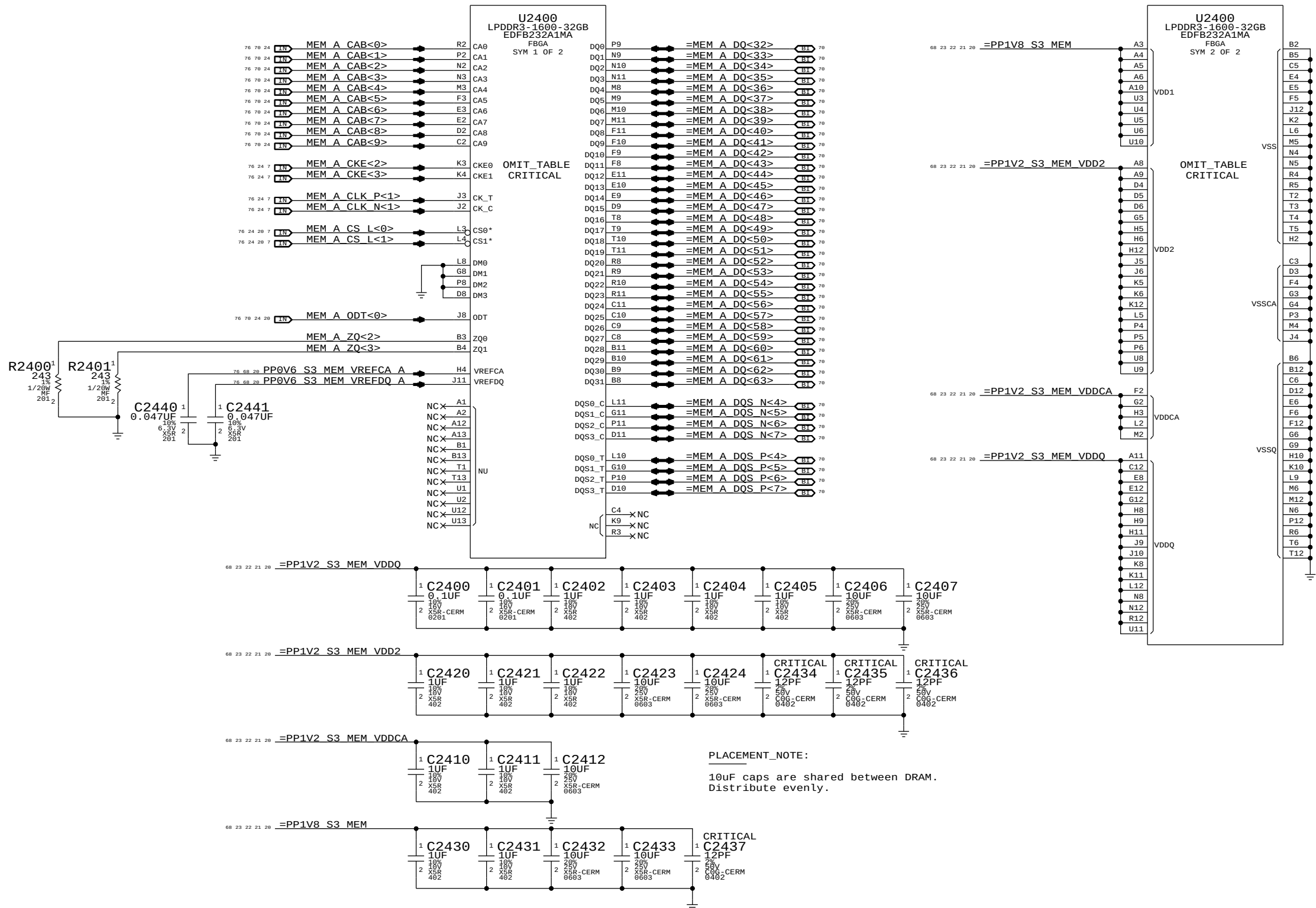
LPDDR3 CHANNEL A (0-31)



SYNC_MASTER=J41 MLB		SYNC_DATE=02/06/2013	
PAGE TITLE			
LPDDR3 DRAM Channel A (00-31)			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-1573		D
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		8.0.0	
		BRANCH	
		dvt1	
		PAGE	
		23 OF 120	
		SHEET	
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BOM_COST_GROUP=DRAM

LPDDR3 CHANNEL A (32-63)



SYNC MASTER=J41 MLB		SYNC DATE=02/06/2013	
PAGE TITLE			
LPDDR3 DRAM Channel A (32-63)			
 Apple Inc.	DRAWING NUMBER		SIZE
	051-1573		D
	REVISION		
		8.0.0	
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BOM_COST_GROUP=DRAM

D



B

A

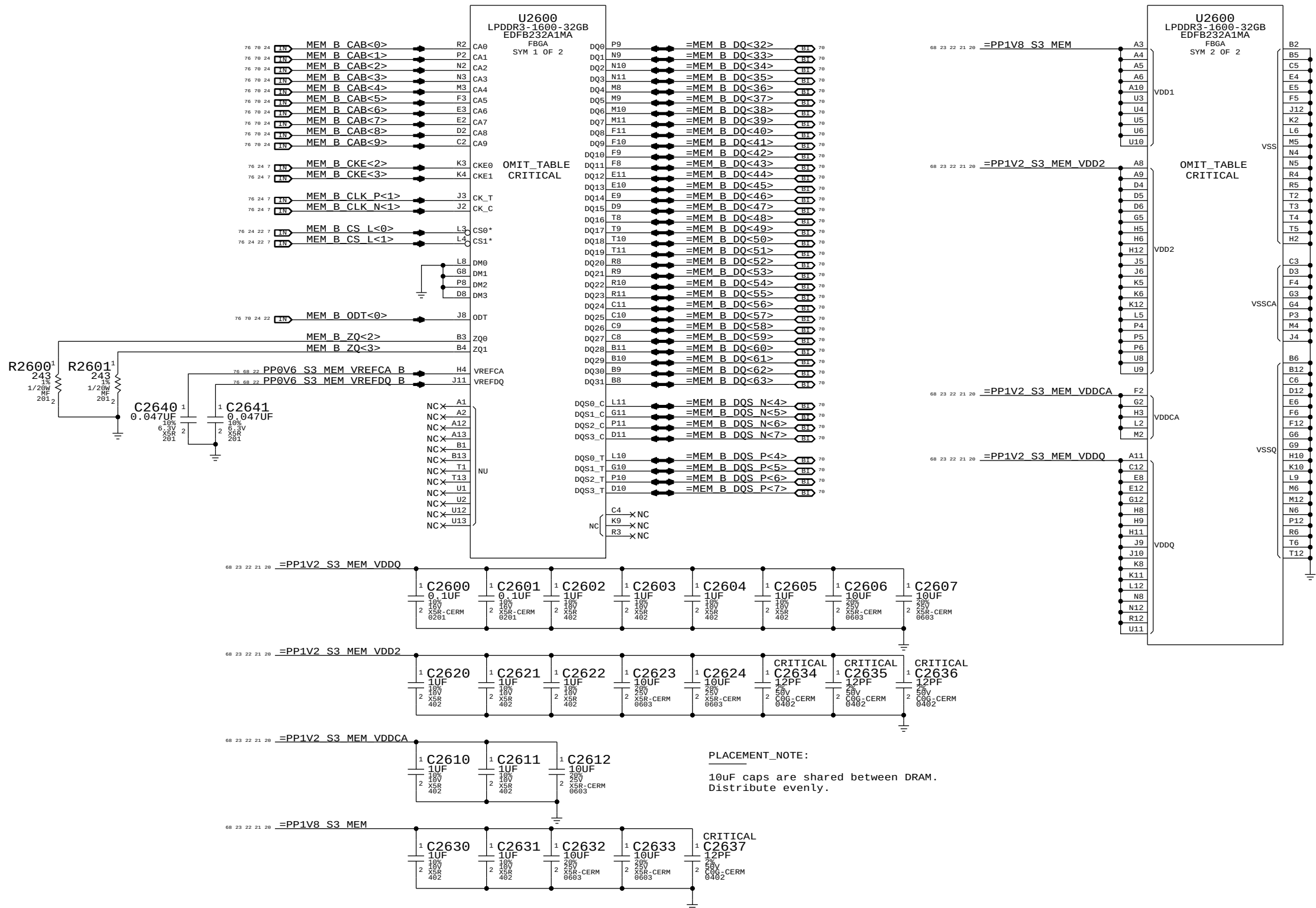
D

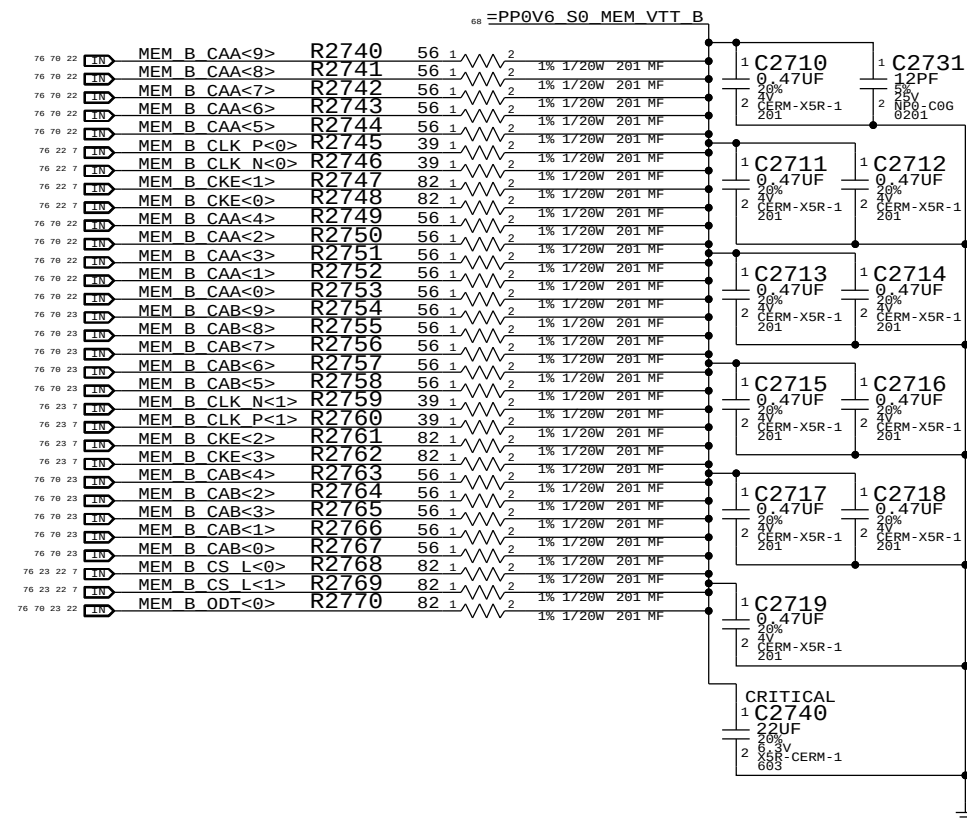
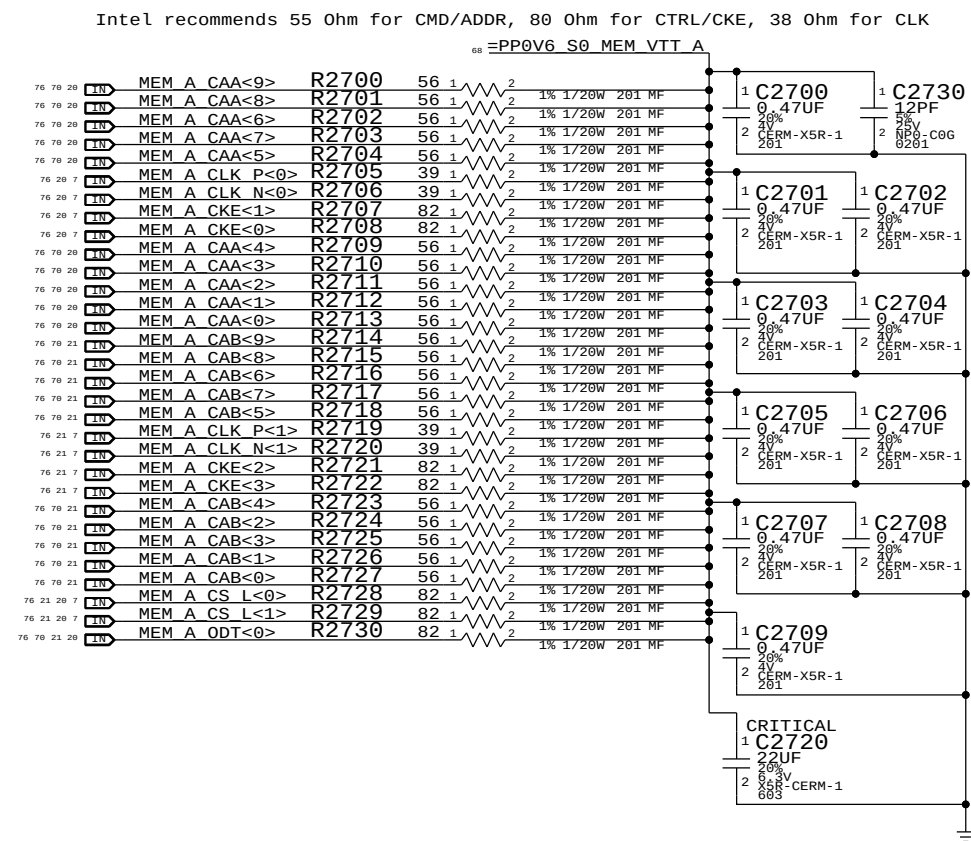
C

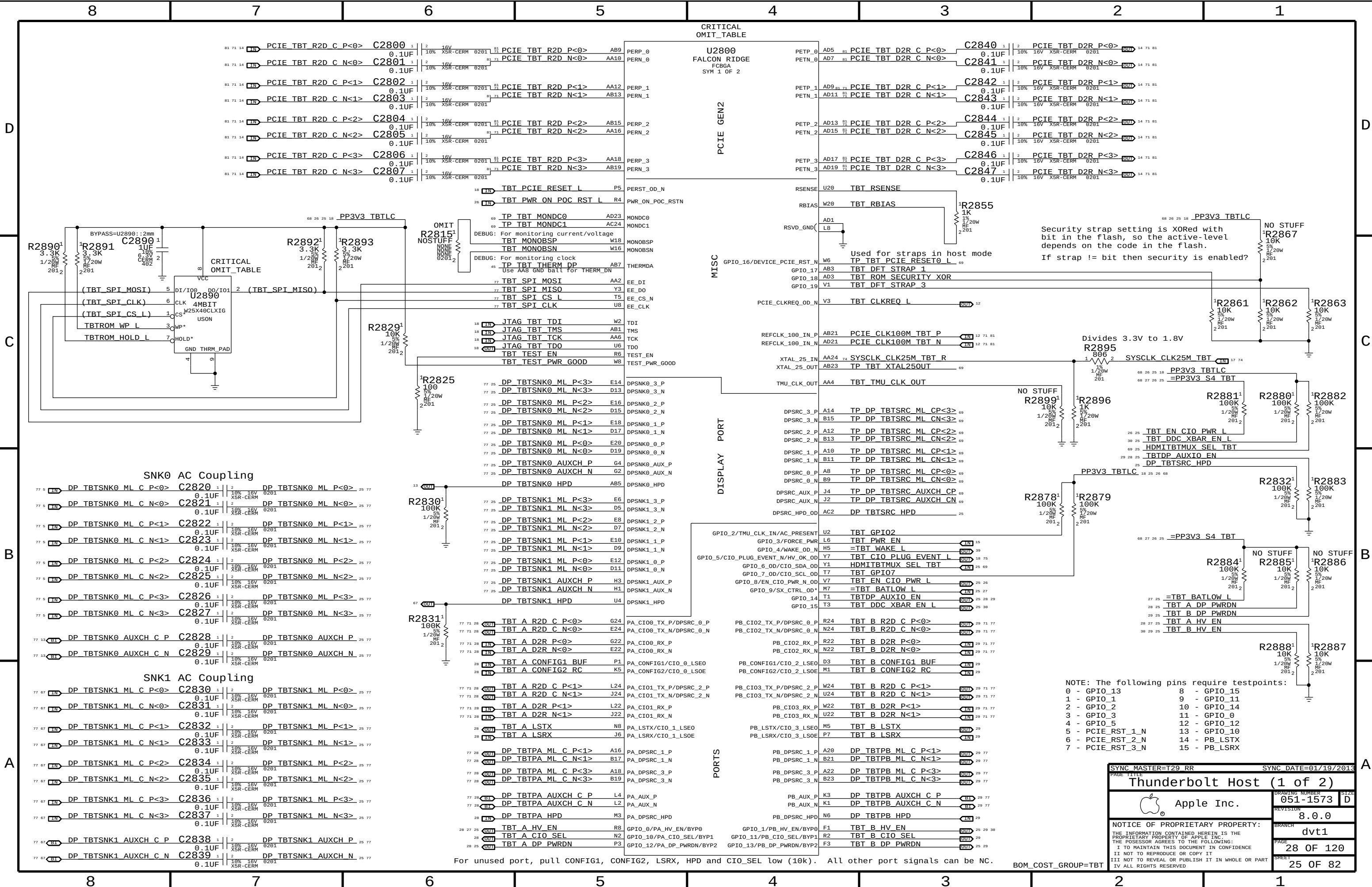
B

Δ

LPDDR3 CHANNEL B (32-63)







SYNC MASTER=T29 RR

SYNC DATE=01/19/2013

Thunderbolt Host (1 of 2)

 Apple Inc.

DRAWING NUMBER
051-1573

REVISION
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BRANCH
dvt1

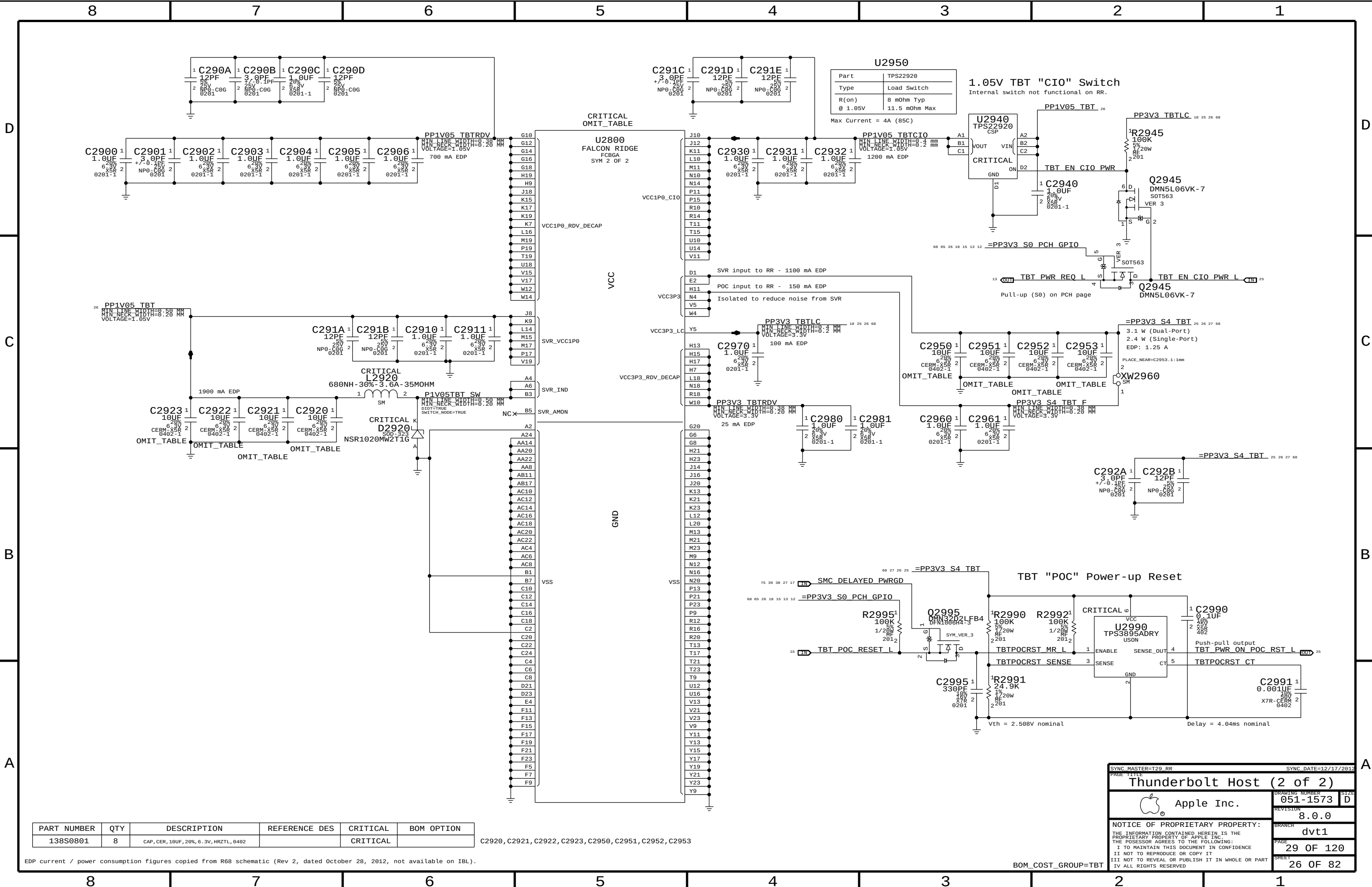
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For unused port, pull CONFIG1, CONFIG2, LSRX, HPD and CIO_SEL low (10k). All other port signals can be NC.

BOM_COST_GROUP=TBT



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0801	8	CAP, CER, 10UF, 20%, 6.3V, HRZTL, 0402		CRITICAL	

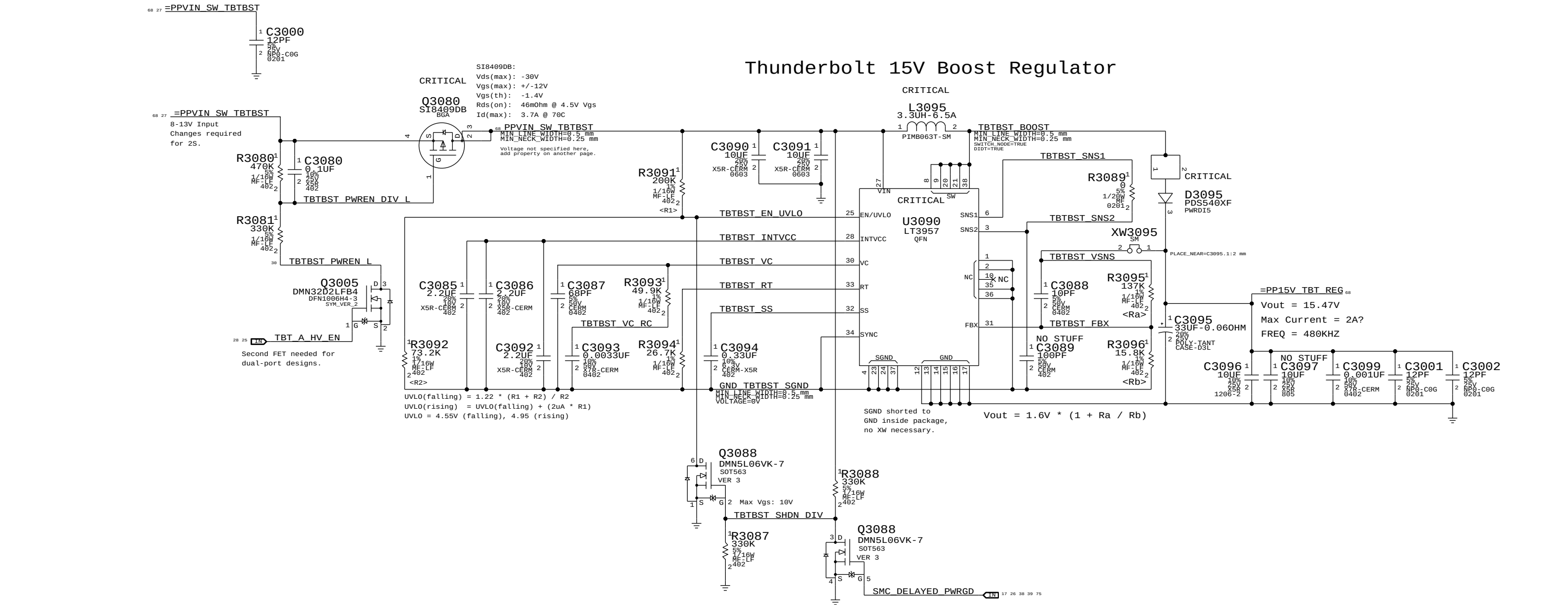
C2920, C2921, C2922, C2923, C2950, C2951, C2952, C2953

EDP current / power consumption figures copied from R68 schematic (Rev 2, dated October 28, 2012, not available on IBL).

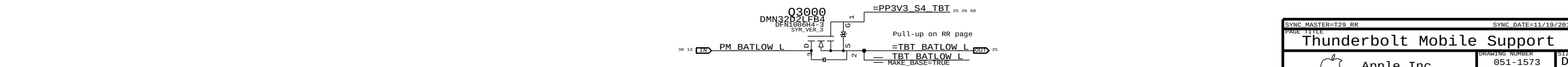
SYNC MASTER=T29_RR		SYNC DATE=12/17/2012	
PAGE TITLE		Thunderbolt Host (2 of 2)	
Apple Inc.		DRAWING NUMBER	051-1573
		REVISION	8.0.0
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IV ALL RIGHTS RESERVED			

BOM_COST_GROUP=TBT

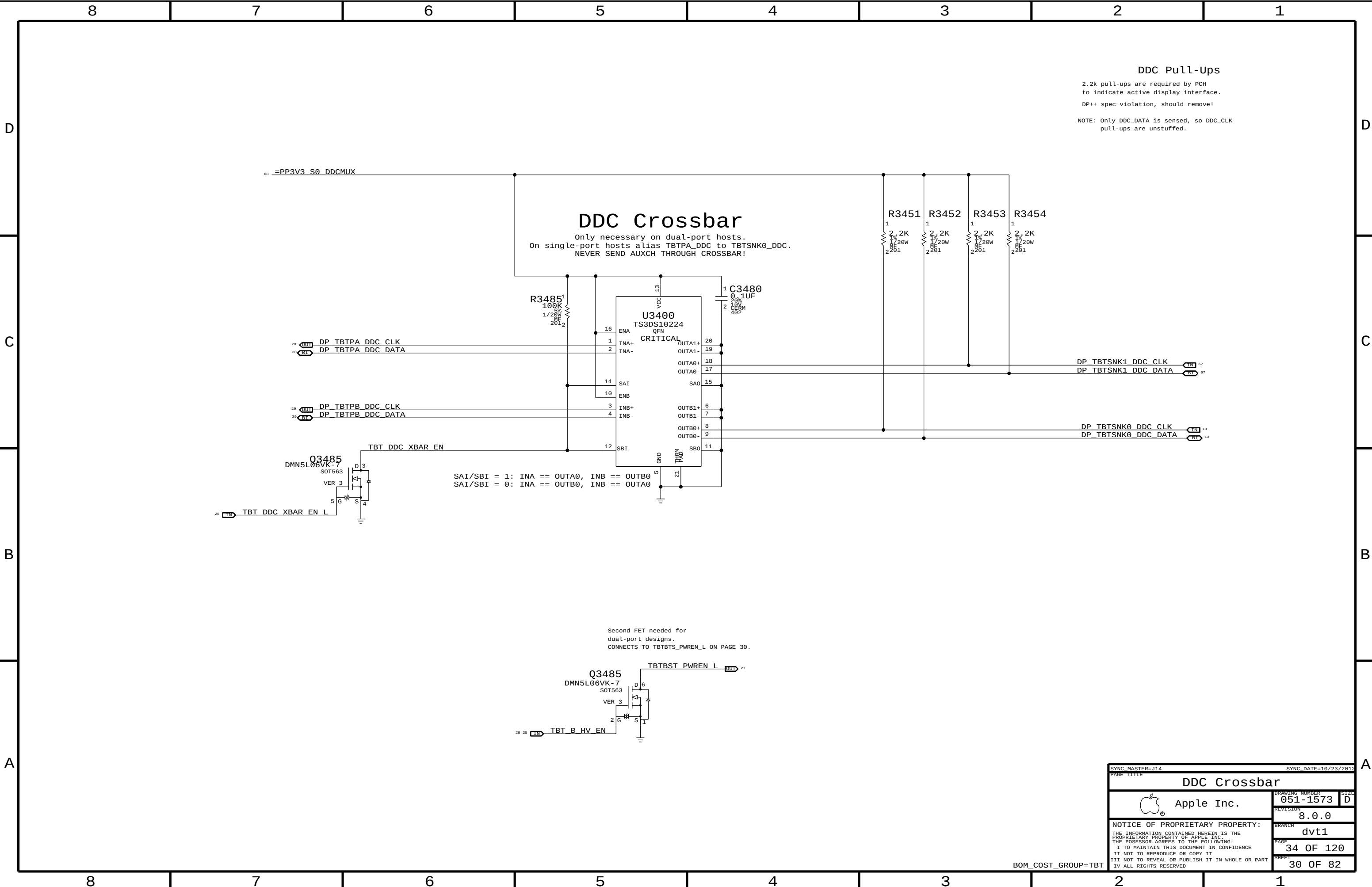
Power aliases required by this page: - =PPVIN_SW_TBTBST (8-13V Boost Input) - =PP15V_TBT_REG (15V Boost Output)
Signal aliases required by this page: (NONE)
BOM options provided by this page: (NONE)



BATLOW# Isolation



SYNC MASTER-T29 RR		SYNC DATE=11/19/2012	
PAGE 1		TITLE	
Thunderbolt Mobile Support			
		DRAWING NUMBER 051-1573	
Apple Inc.		SIZE D	
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DDC Pull-Ups

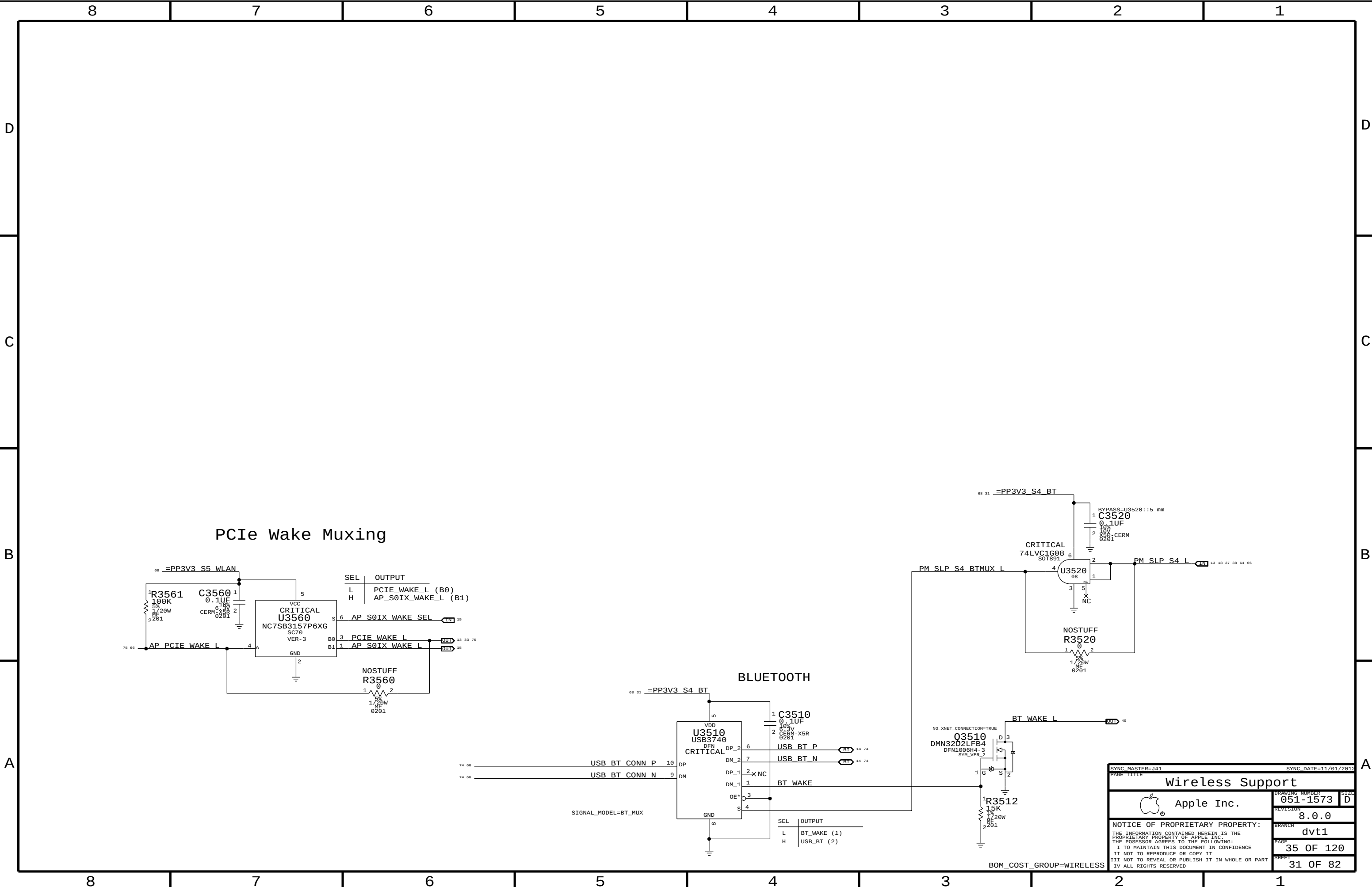
2.2k pull-ups are required by PCH to indicate active display interface.

DP++ spec violation, should remove!

NOTE: Only DDC_DATA is sensed, so DDC_CLK pull-ups are unstuffed.

SYNC MASTER=J14		SYNC DATE=10/23/2012	
PAGE TITLE			
DDC Crossbar			
		Apple Inc.	
DRAWING NUMBER		051-1573	SIZE D
REVISION		8.0.0	
BRANCH		dvt1	
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BOM_COST_GROUP=TBT



PCIe Wake Muxing

BLUETOOTH

SYNC MASTER=J41

SYNC DATE=11/01/2012

051-1573

8.0.0

dvt1

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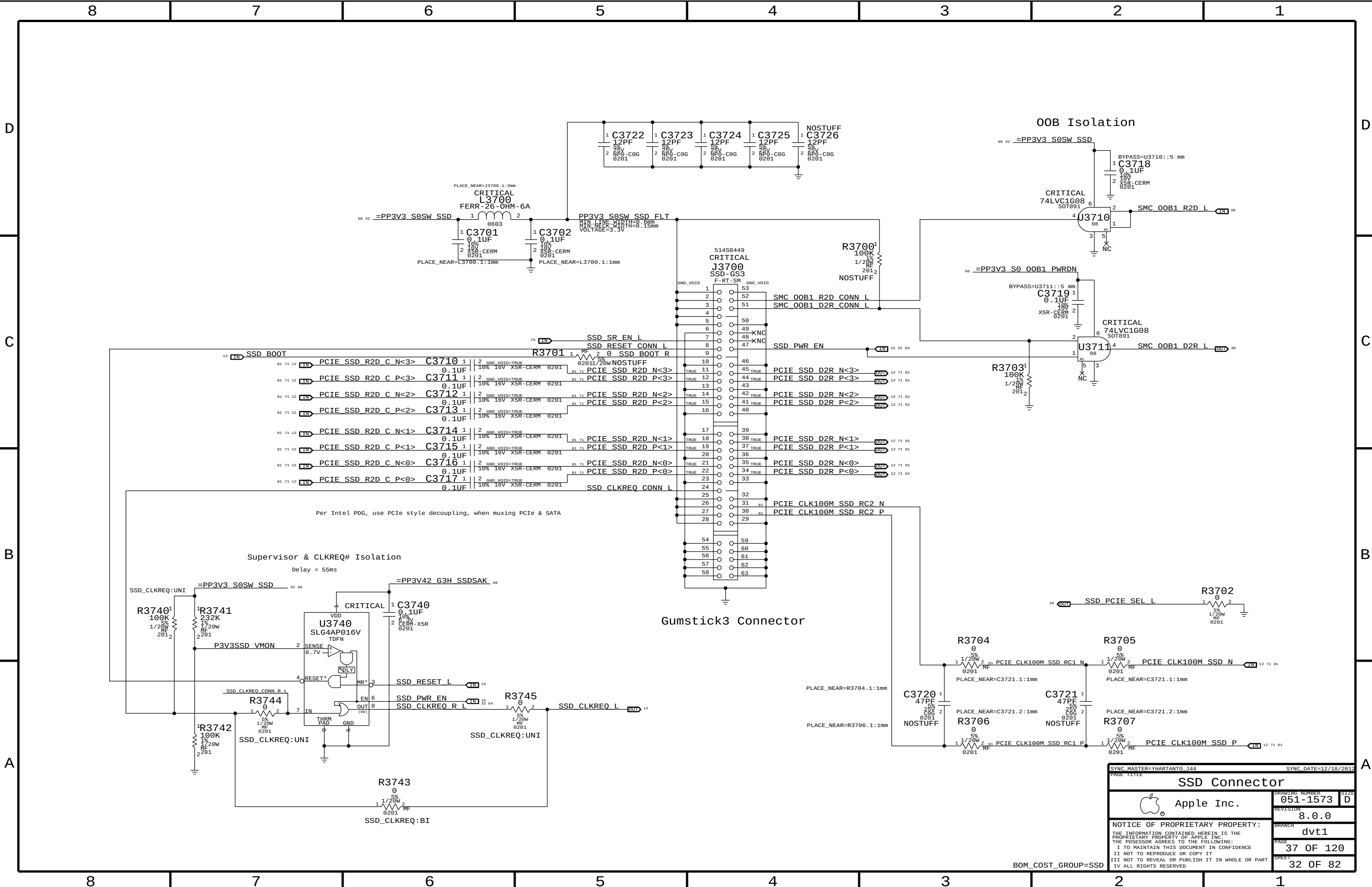
31 OF 82

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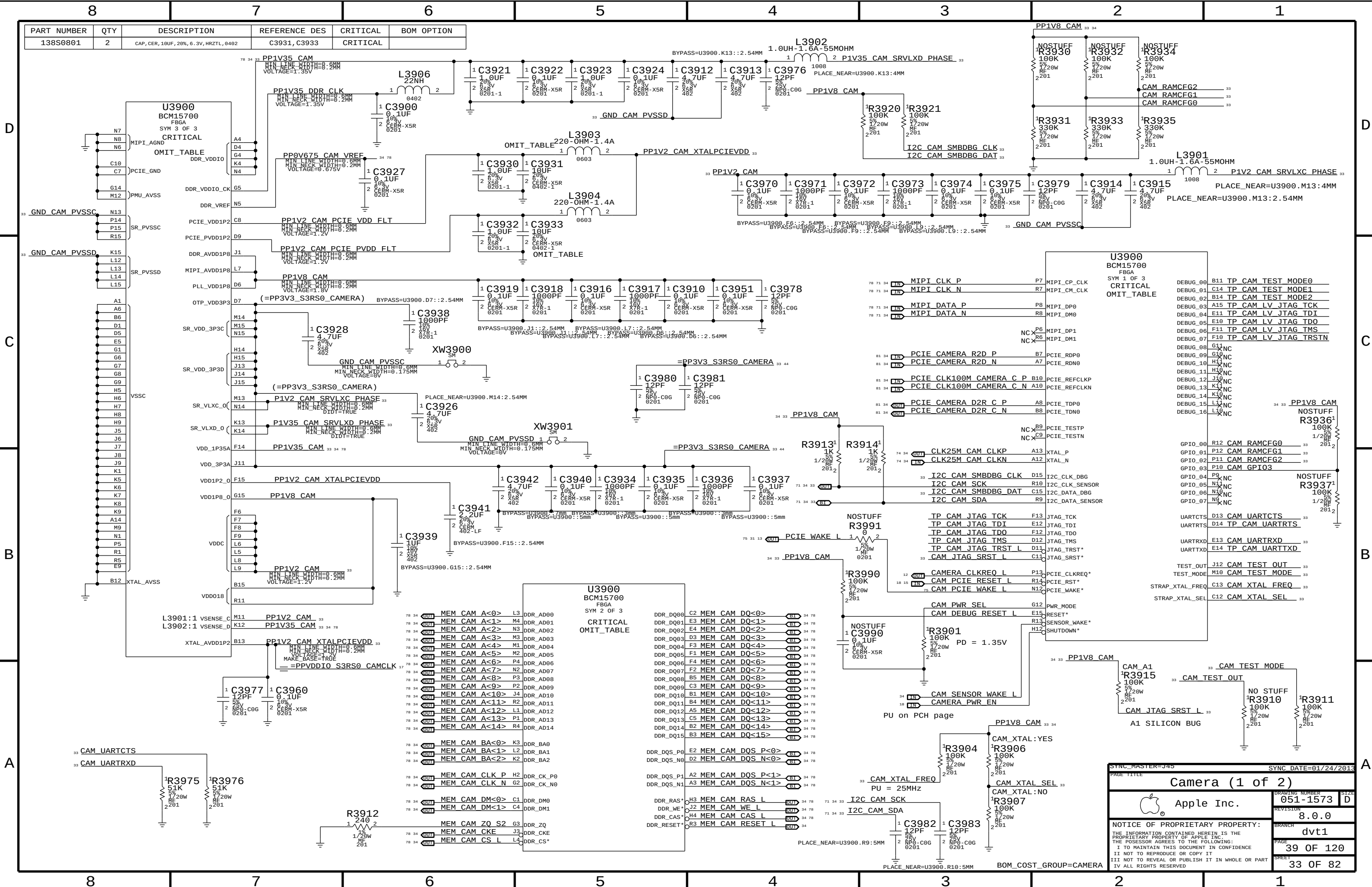
BOM_COST_GROUP=WIRELESS

Wireless Support



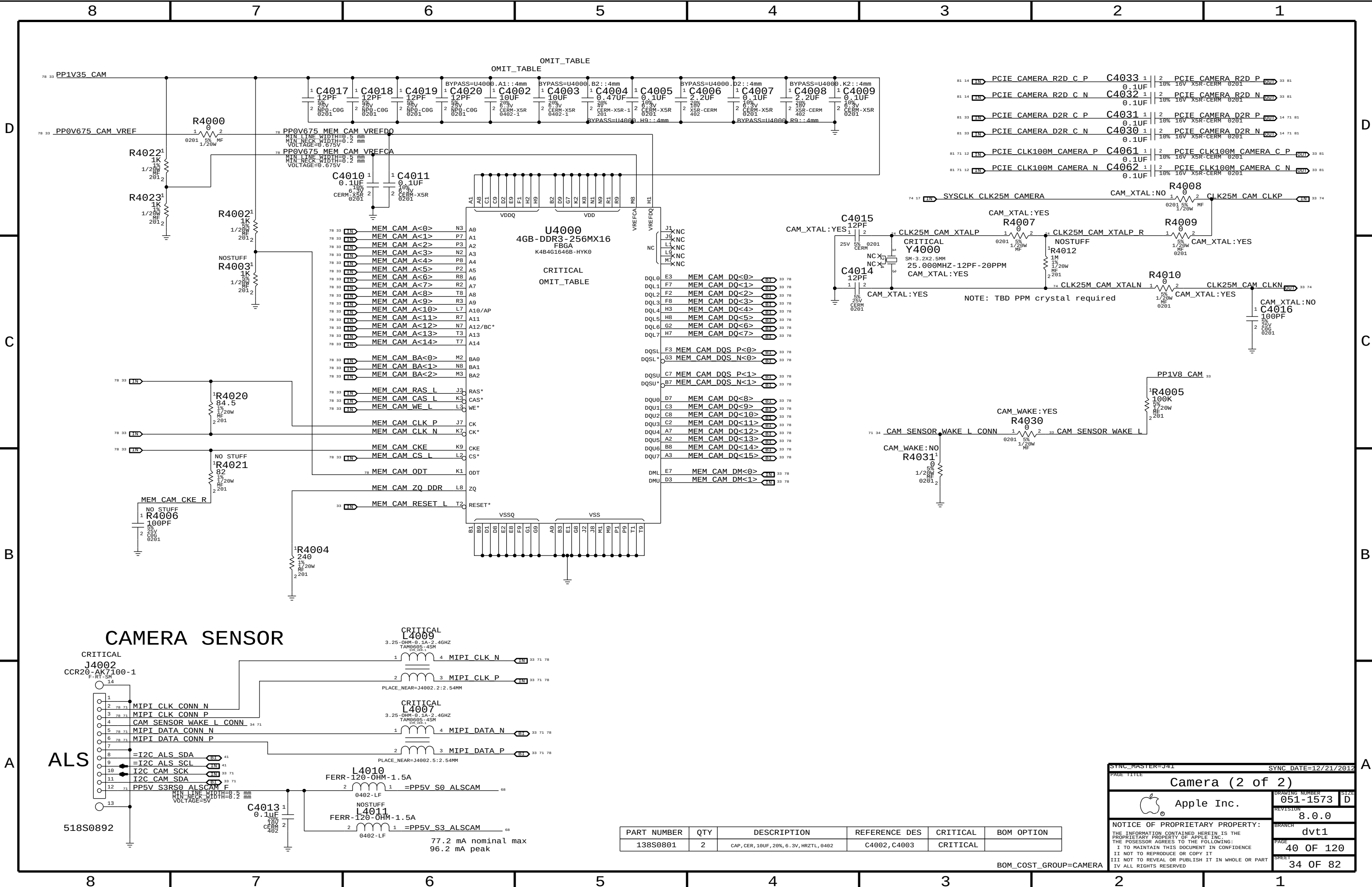
SYNC MASTER=YHARTANTO J44		SYNC DATE=12/18/2012	
PAGE TITLE		PAGE TITLE	
SSD Connector		DRAWING NUMBER	051-1573
Apple Inc.		REVISION	8.0.0
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		PAGE	37 OF 120
		SHEET	32 OF 82

BOM_COST_GROUP=SSD



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0801	2	CAP,CER,10UF,20%,6.3V,HRZTL,0402	C3931,C3933	CRITICAL	

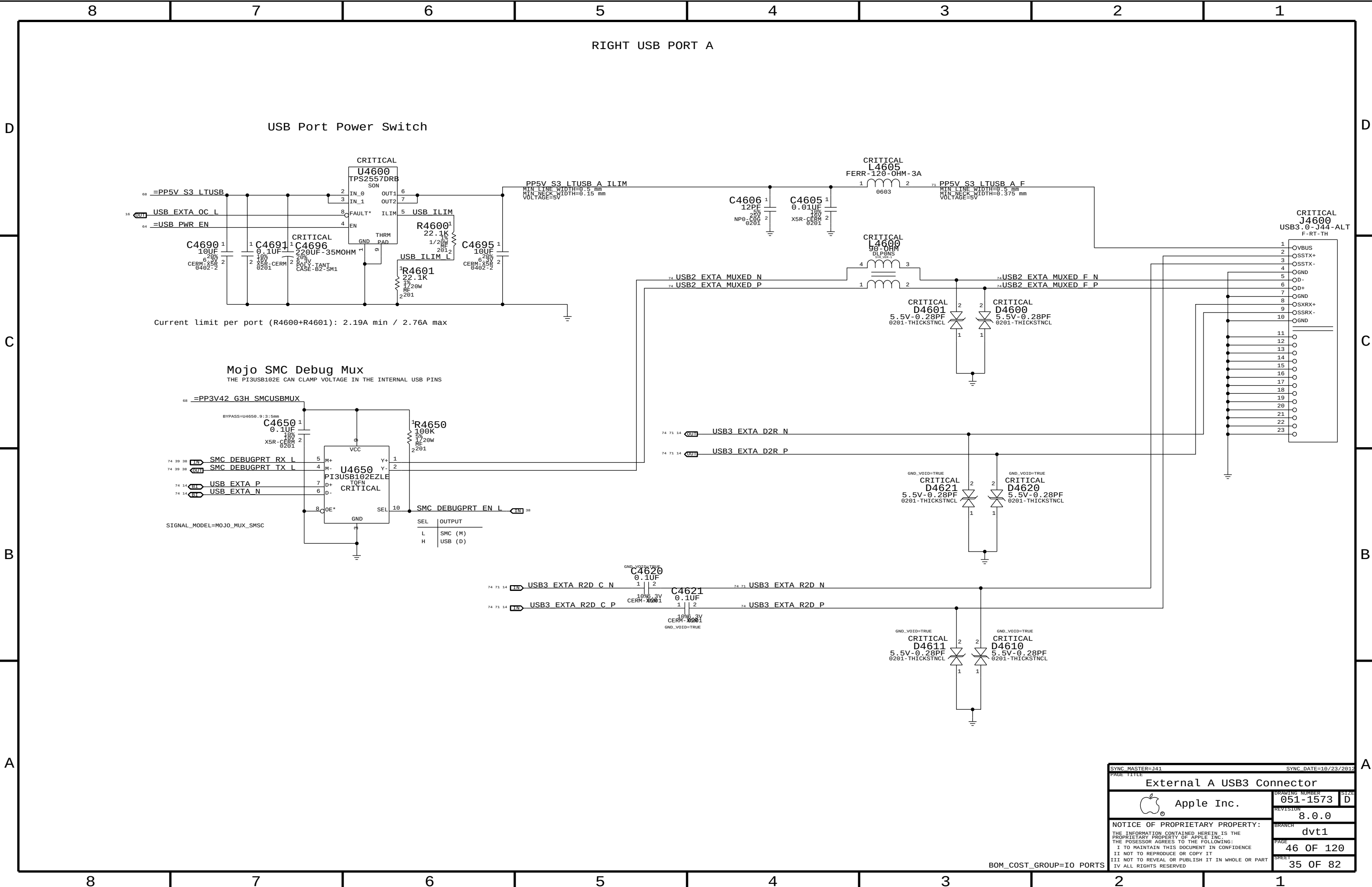
PAGE TITLE		SYNC DATE=01/24/2013	
Camera (1 of 2)		DRAWING NUMBER	051-1573
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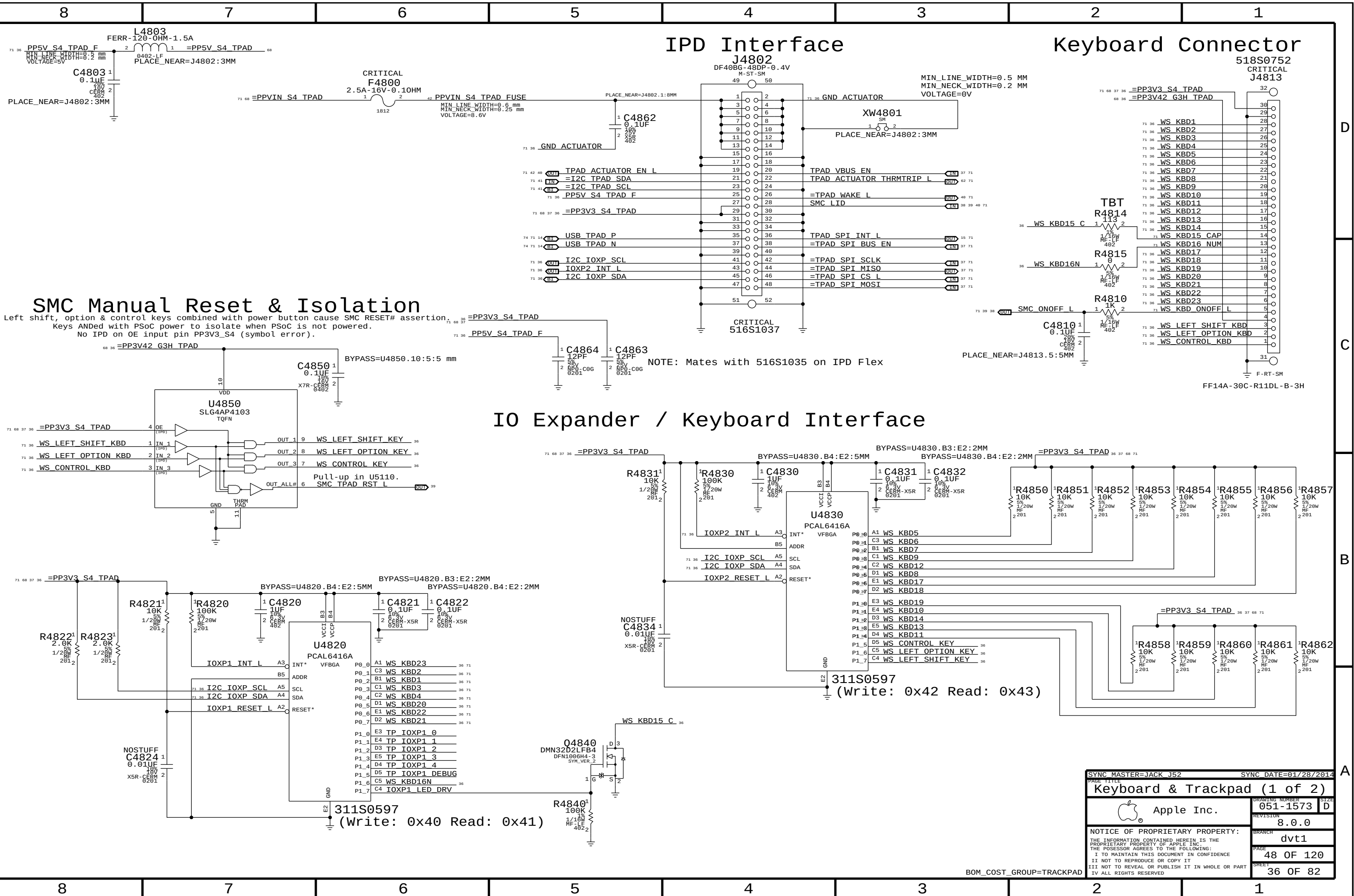
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
138S0801	2	CAP, CER, 10UF, 20%, 6.3V, HRZTL, 0402	C4002, C4003	CRITICAL	

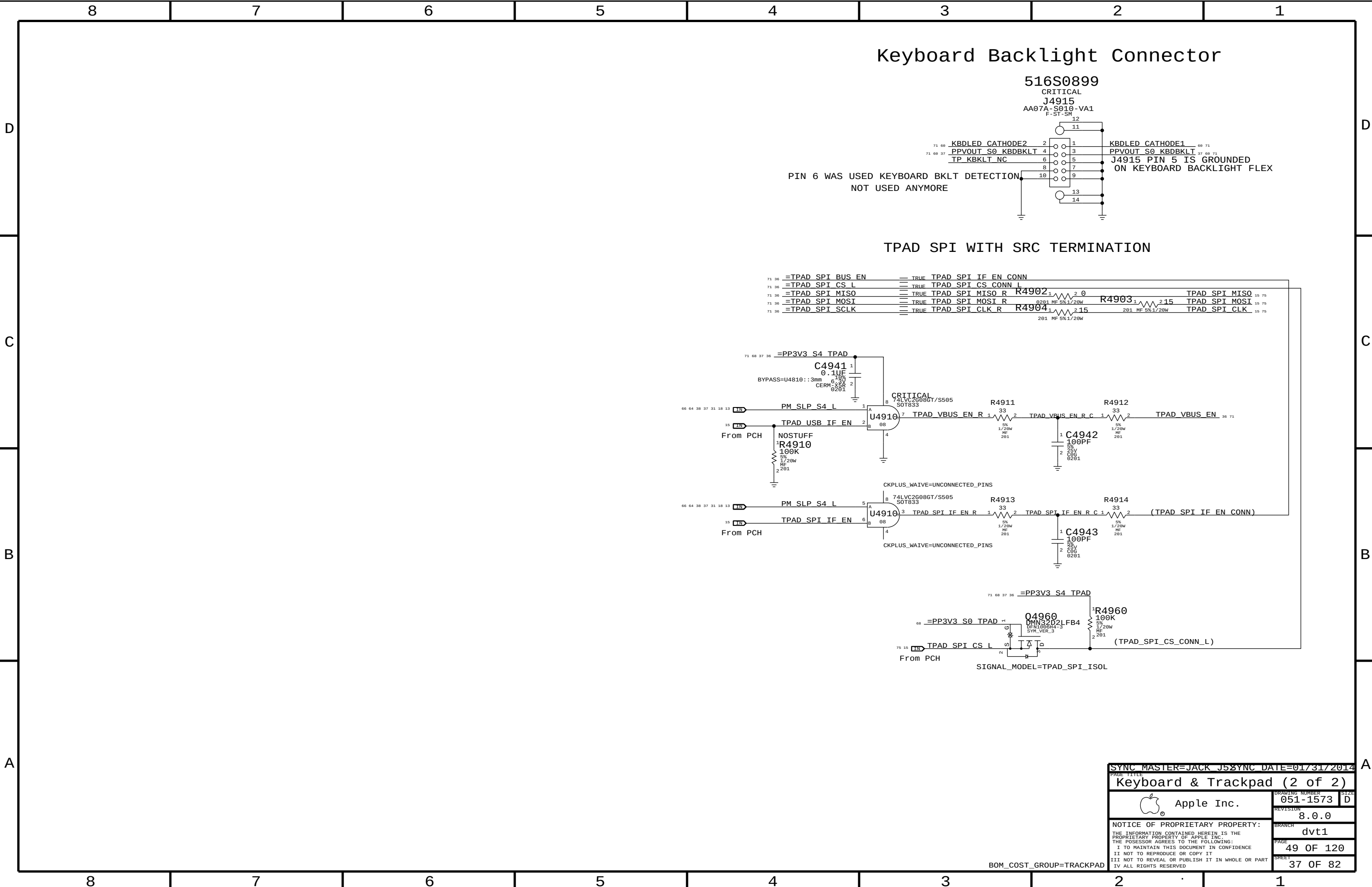
PAGE TITLE		SYNC DATE=12/21/2012	
Camera (2 of 2)		DRAWING NUMBER	051-1573
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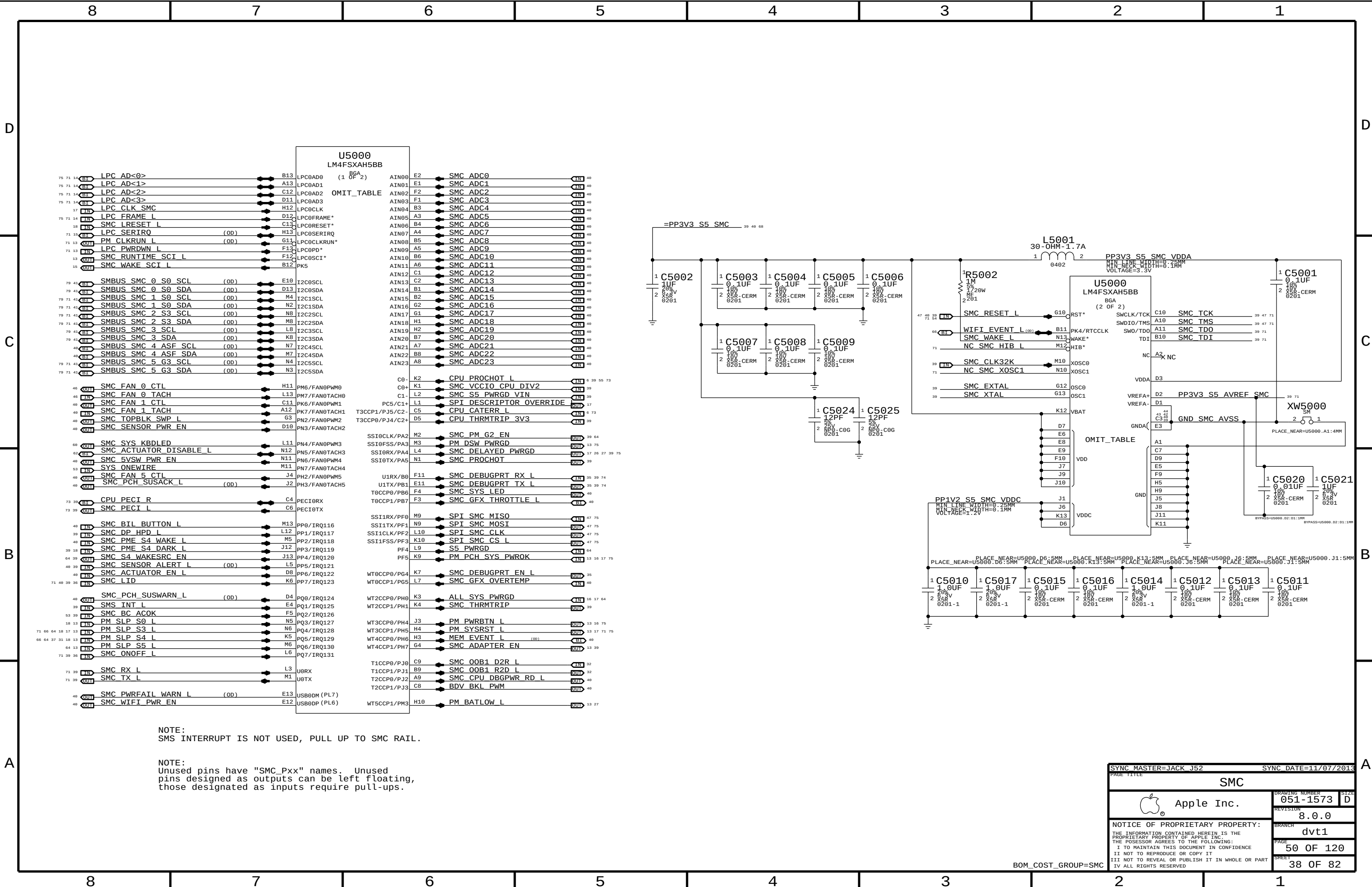
BOM_COST_GROUP=CAMERA



SYNC MASTER=J41		SYNC DATE=10/23/2012	
PAGE TITLE			
External A USB3 Connector			
 Apple Inc.		DRAWING NUMBER	051-1573
		SIZE	D
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		PAGE	46 OF 120
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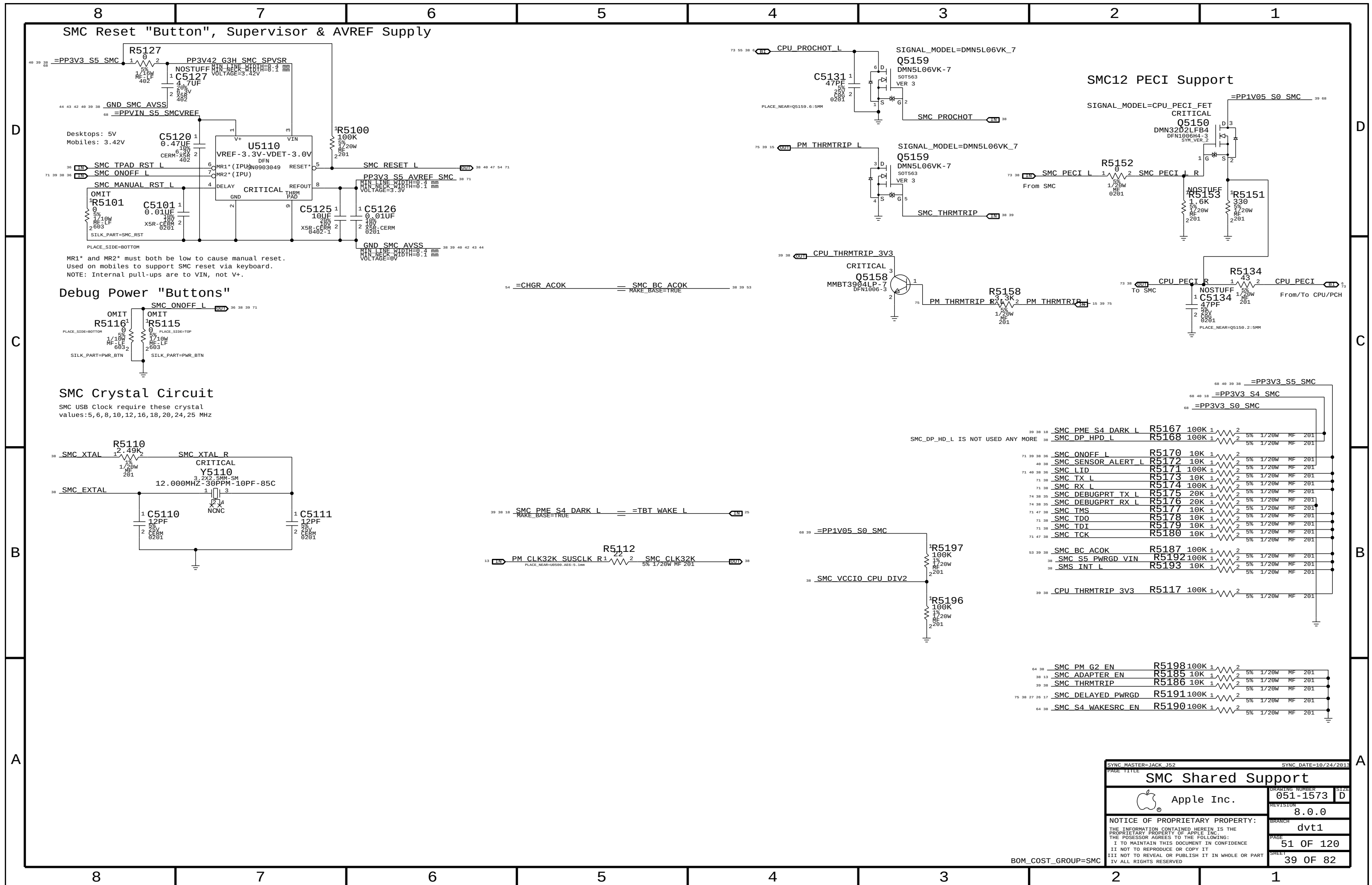


NOTE:
SMS INTERRUPT IS NOT USED, PULL UP TO SMC RAIL.

NOTE:
Unused pins have "SMC_Pxx" names. Unused pins designed as outputs can be left floating, those designated as inputs require pull-ups.

SYNC MASTER=JACK J52		SYNC DATE=11/07/2013	
PAGE TITLE			
SMC			
 Apple Inc.		DRAWING NUMBER	051-1573
		SIZE	D
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		BRANCH	dvt1
		PAGE	50 OF 120
		SHEET	38 OF 82

BOM_COST_GROUP=SMC



8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---



C



8	7	6	5	4	3	2	1
---	---	---	---	---	---	---	---

```

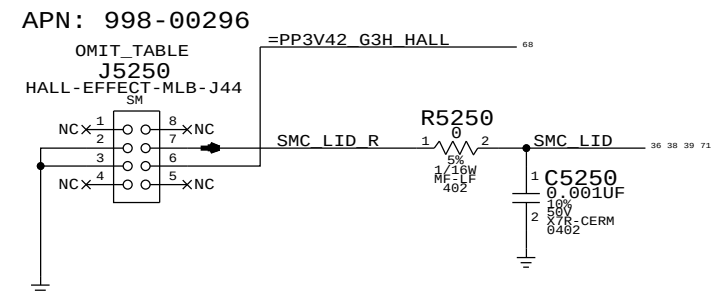
66 39 38 ==PP3V3 S5 SMC
SMC CPU_DBGPWR_RD_L == SMC BOARDID
MAKE_BASE=TRUE
R5232
10K
5%
1/20W
0805
2281
SMCBOARDID:16
R5233
10K
5%
1/20W
0805
2281
SMCBOARDID:8

```

S4 SMC Wake Sources

The diagram illustrates the S4 SMC Wake Sources. It features two input lines, TPAD WAKE L (pin 36) and BT WAKE L (pin 31), which are connected to a common node. This node is connected to a resistor R5282 (100K, 5%, 20W, 201) which is connected to a power source =PP3V3 S4 SMC (pins 18, 39, 40, 68). The output of the resistor is connected to SMC PME S4 WAKE L (pin 38) with the attribute MAKE_BASE=TRUE. A second section shows TPAD ACTUATOR_EN_L (pin 36) connected to a resistor R5297 (100K, 5%, 20W, 201) which is connected to a power source =PP3V3 S4 SMC (pins 18, 39, 40, 68). The output of the resistor is connected to SMC ACTUATOR_EN_L (pin 3) with the attribute MAKE_BASE=TRUE.

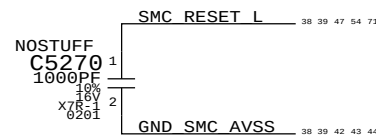
1 C5271
220PF
10%
25V
X7R-CERM
201



Specify one of these BOM GROUPS.

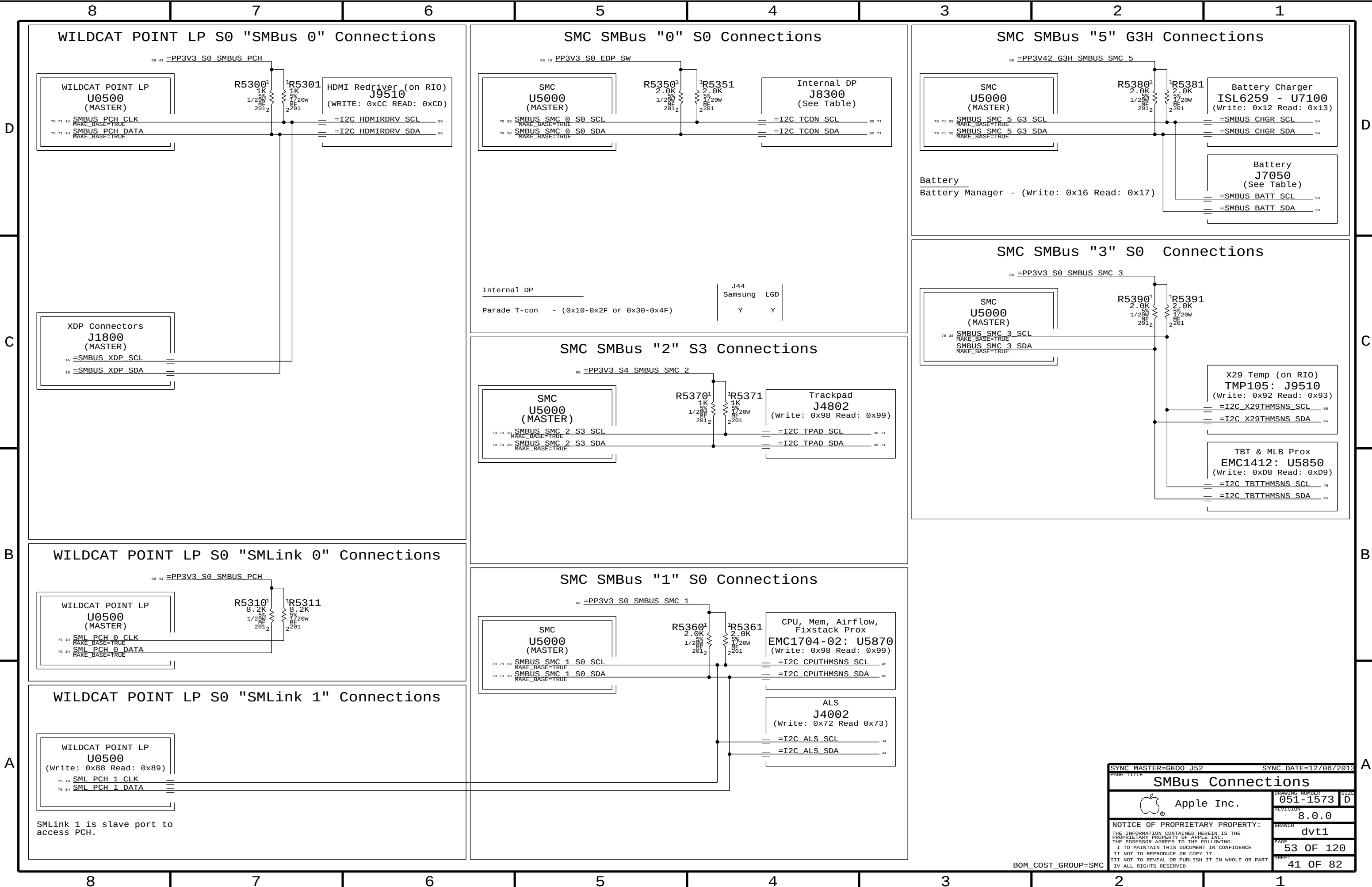
Specify one of these BOM GROUPS.

Requires EMC1412-1 or EMC1412-2 instead of EMC1412-A, new APN needs to be created.



PAGE TITLE		SMC Project Support	
 Apple Inc.	DRAWING NUMBER	051-1573	SIZE D
	REVISION 8.0.0		
	BRANCH dvt1		
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PAGE		52 OF 120	
SHEET		40 OF 82	

BOM_COST_GROUP=SMC



D

C



B

A



B

A



B

A



A

A

A



D

C



B

B



B

B



B

B



B

B

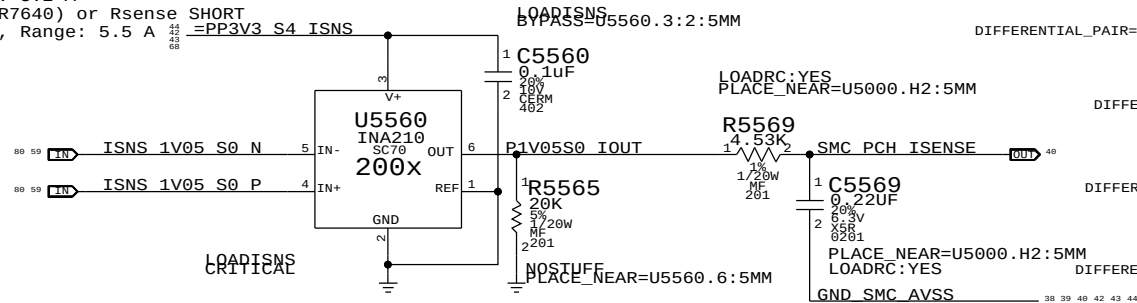


B

B

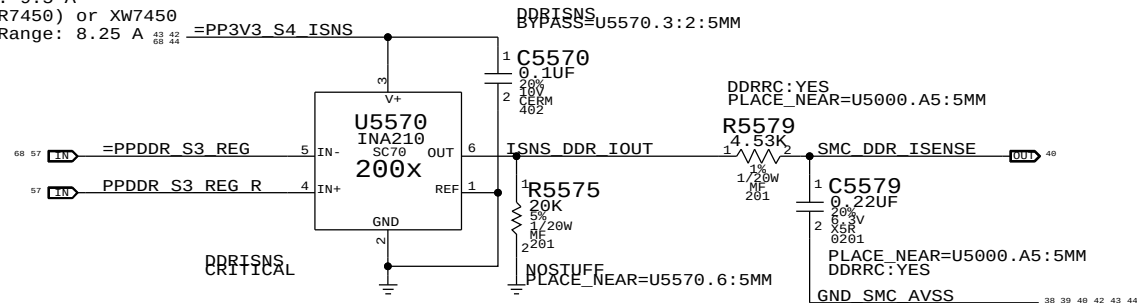
PCH 1.05V Current Sense (IC1C)

Gain: 200x, EDP: 5.2 A
Rsense: 0.003 (R7640) or Rsense SHORT
Vsense: 15.6 mV, Range: 5.5 A ⁴⁴₄₂ =PP3V3 S4 ISNS
SMC ADC: 19 ⁴³₆₀



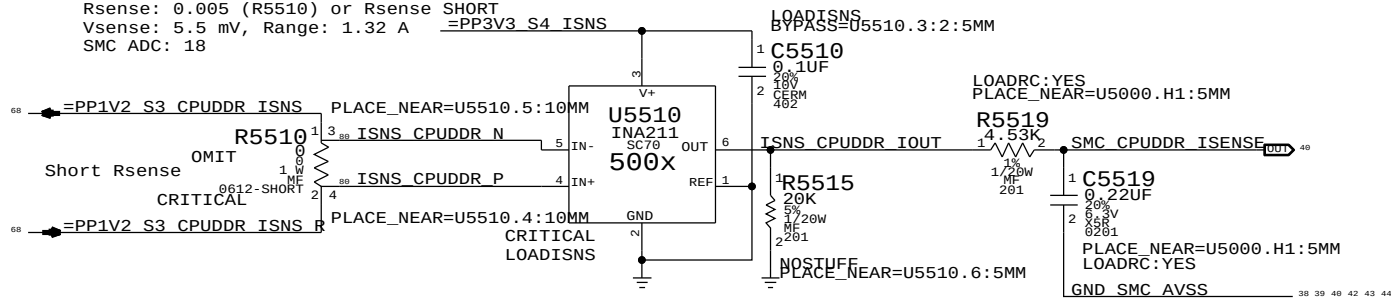
DDR 1.2V S3 (CPU & Memory) Current Sense (IM0C)

Gain: 200x, EDP: 9.5 A
Rsense: 0.002 (R7450) or XW7450
Vsense: 19 mV, Range: 8.25 A
SMC ADC: 09



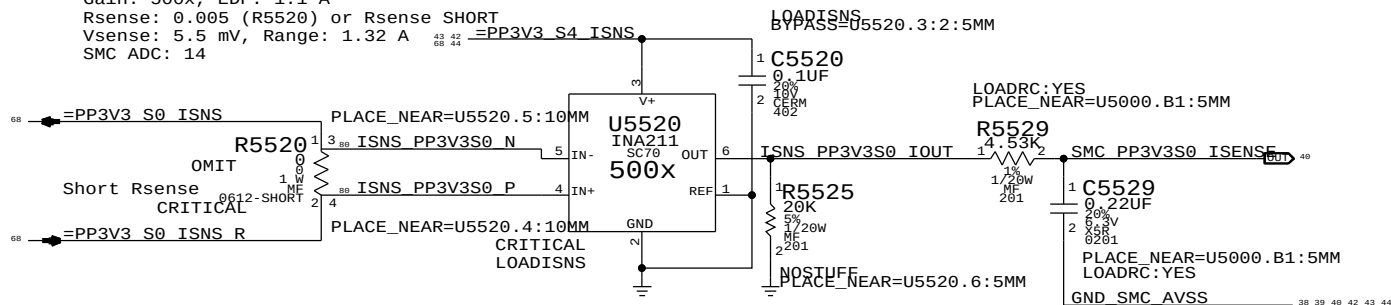
CPU DDR 1.2V S3 (CPU Only) Current Sense (IM1C)

Gain: 500x, EDP: 1.1 A
Rsense: 0.005 (R5510) or Rsense SHORT
Vsense: 5.5 mV, Range: 1.32 A =PP3V3 S4 ISNS
SMC ADC: 18



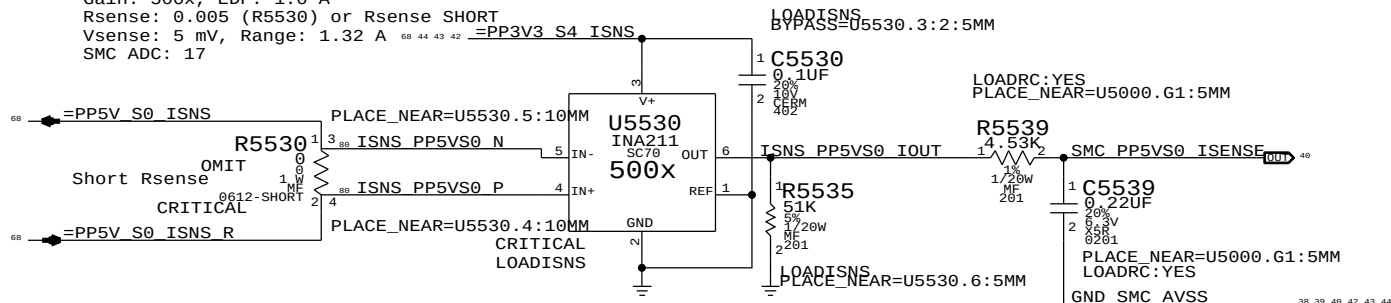
3.3V S0 Rail Current Sense (IR3C)

Gain: 500x, EDP: 1.1 A
Rsense: 0.005 (R5520) or Rsense SHORT
Vsense: 5.5 mV, Range: 1.32 A ^{43 42}_{68 44} =PP3V3 S4 IS
SMC ADC: 14



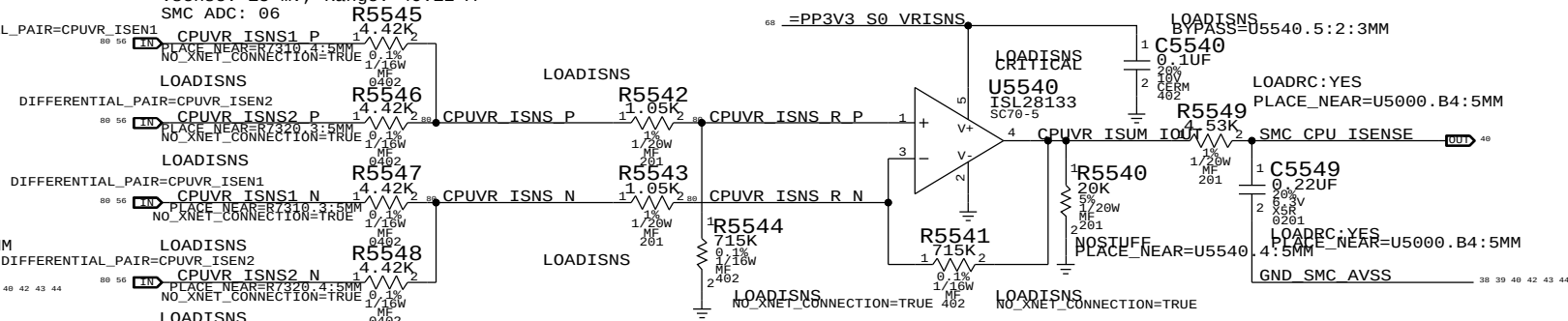
5V S0 Rail Current Sense (IR5C)

Gain: 500x, EDP: 1.0 A
Rsense: 0.005 (R5530) or Rsense SHORT
Vsense: 5 mV, Range: 1.32 A 68 44 43 42 =PP3V3 S4 IS
SMC ADC: 17



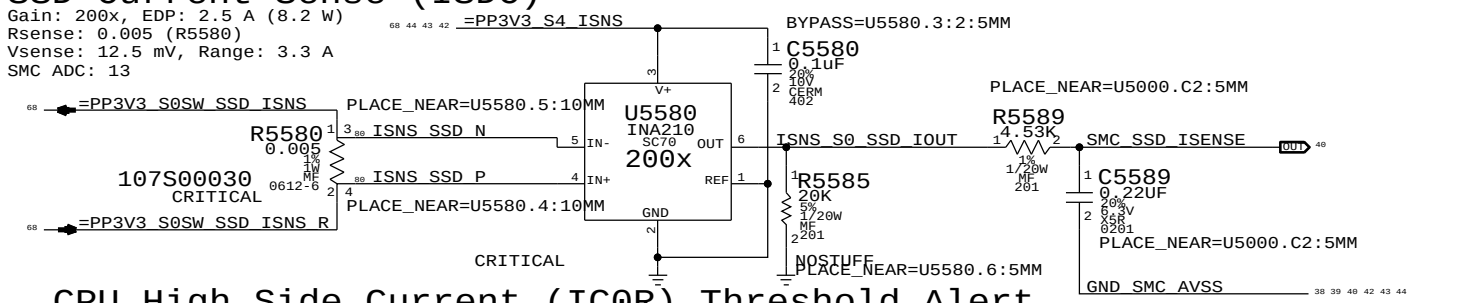
CPU Fixed Current Sense (IC0C)

Gain: 219.33x, EDP: 40 A
Rsense: 2x of 0.00075 (R7310, R7320), Rsum: 0.000375
Vsense: 15 mV, Range: 40.12 A
SMC ADC: 06 R5545



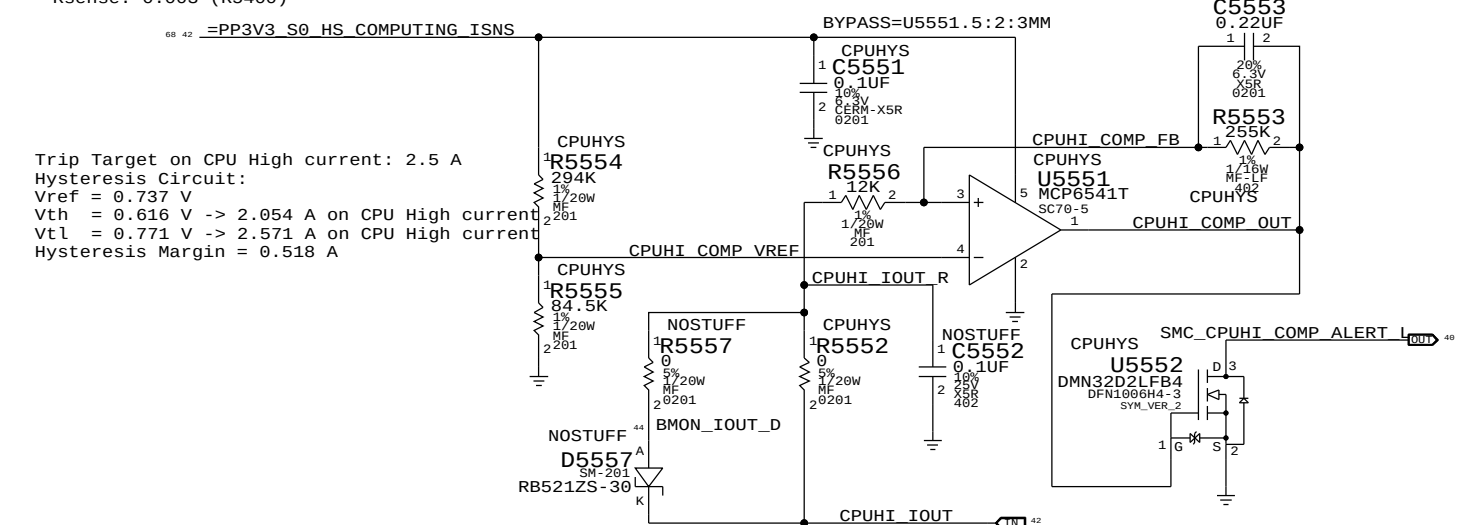
SSD Current Sense (ISDC)

Gain: 200x, EDP: 2.5 A (8.2 W)
Rsense: 0.005 (R5580)
Vsense: 12.5 mV, Range: 3.3 A
SMC ADC: 13



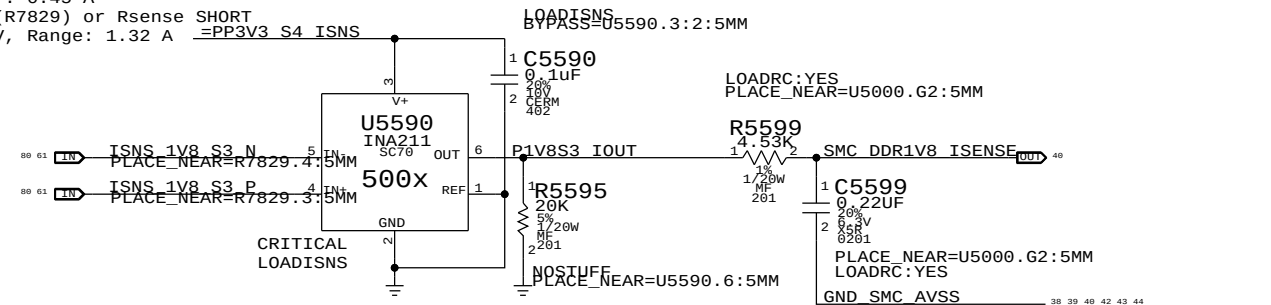
CPU High Side Current (ICOR) Threshold Alert

Gain: 100x
Rsense: 0.003 (R5400)



DDR 1.8V Current Sense (IM2C)

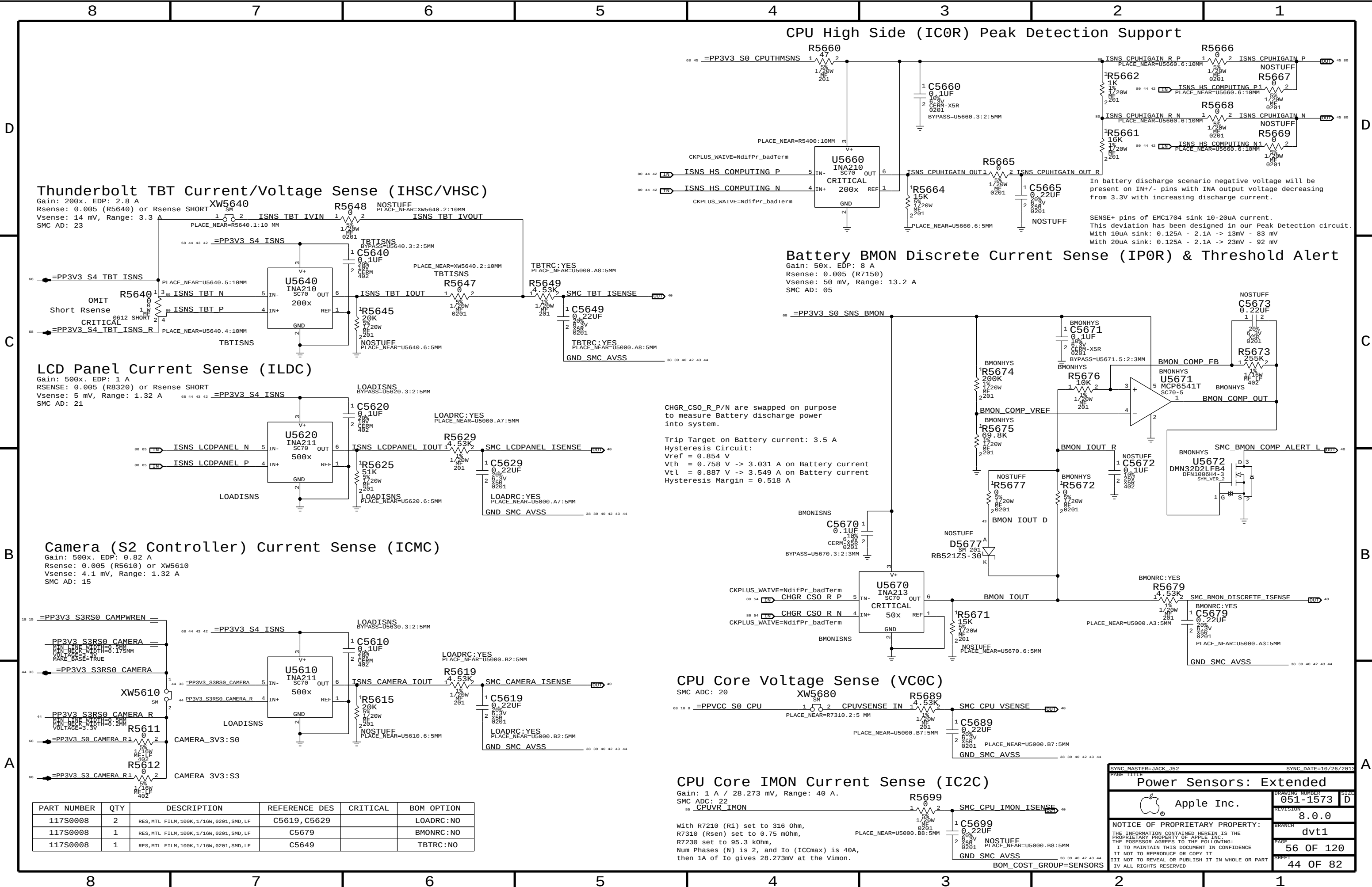
Gain: 500x, EDP: 0.45 A
Rsense: 0.005 (R7829) or Rsense SHORT
Vsense: 2.25 mV, Range: 1.32 A =PP3V3 S4 ISNS
SMC ADC: 12



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	3	RES,MTL FLIM,100K, 1/16W,0201,SMD, LF	C5569, C5519, C5599		LOADRC:NO
117S0008	3	RES,MTL FLIM,100K, 1/16W,0201,SMD, LF	C5529, C5539, C5549		LOADRC:NO
117S0008	1	RES,MTL FLIM,100K, 1/16W,0201,SMD, LF	C5579		DDRRC:NO

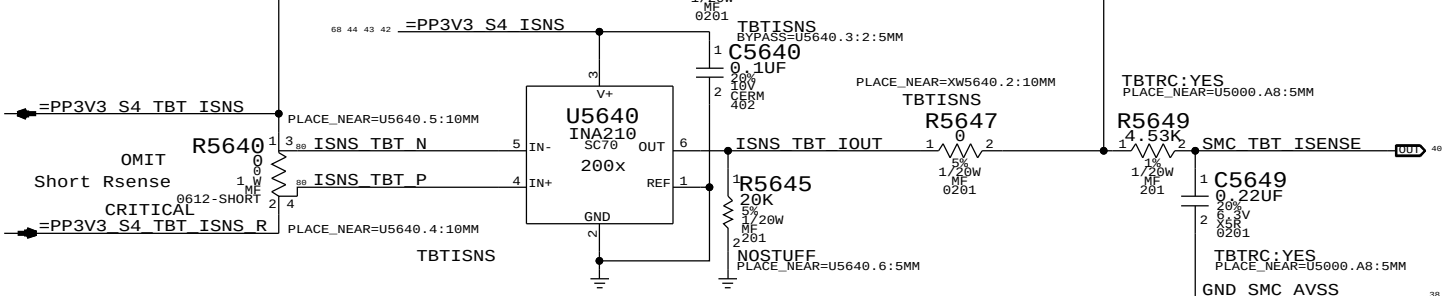
SYNC MASTER=JACK J52		SYNC DATE=12/06/2013	
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Power Sensors: Load Side			
 Apple Inc.		DRAWING NUMBER	SIZE
		051-1573	D
		REVISION	
		8.0.0	
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		PAGE	
		55 OF 120	
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		43 OF 82	

BOM_COST_GROUP=SENSORS



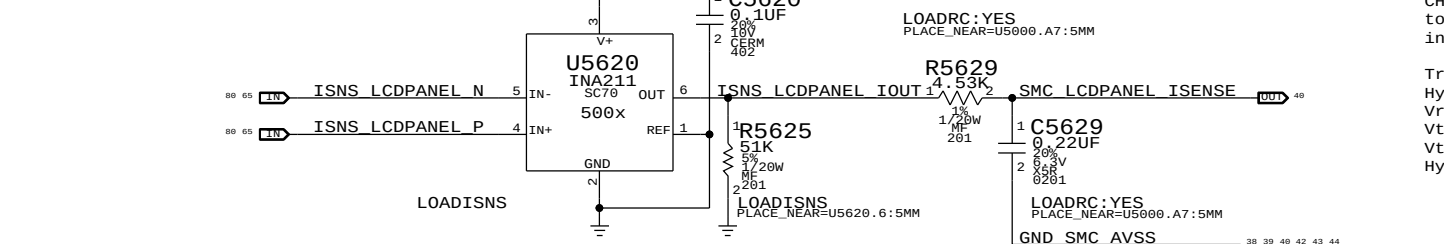
Thunderbolt TBT Current/Voltage Sense (IHSC/VHSC)

Gain: 200x. EDP: 2.8 A
Rsense: 0.005 (R5640) or Rsense SHORT
Vsense: 14 mV, Range: 3.3
SMC AD: 23



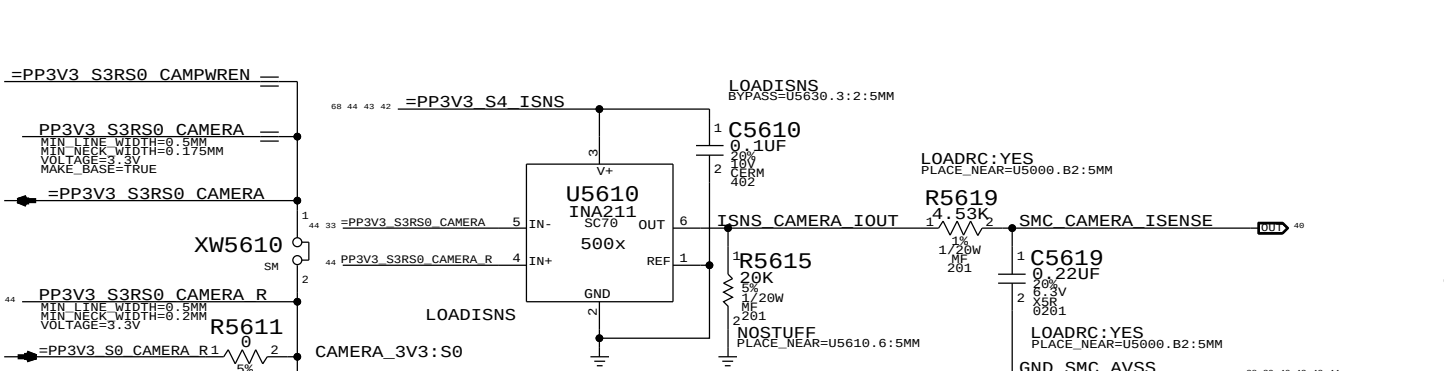
LCD Panel Current Sense (ILDC)

Gain: 500x. EDP: 1 A
RSENSE: 0.005 (R8320) or Rsense SHORT
Vsense: 5 mV, Range: 1.32 A
SMC AD: 21



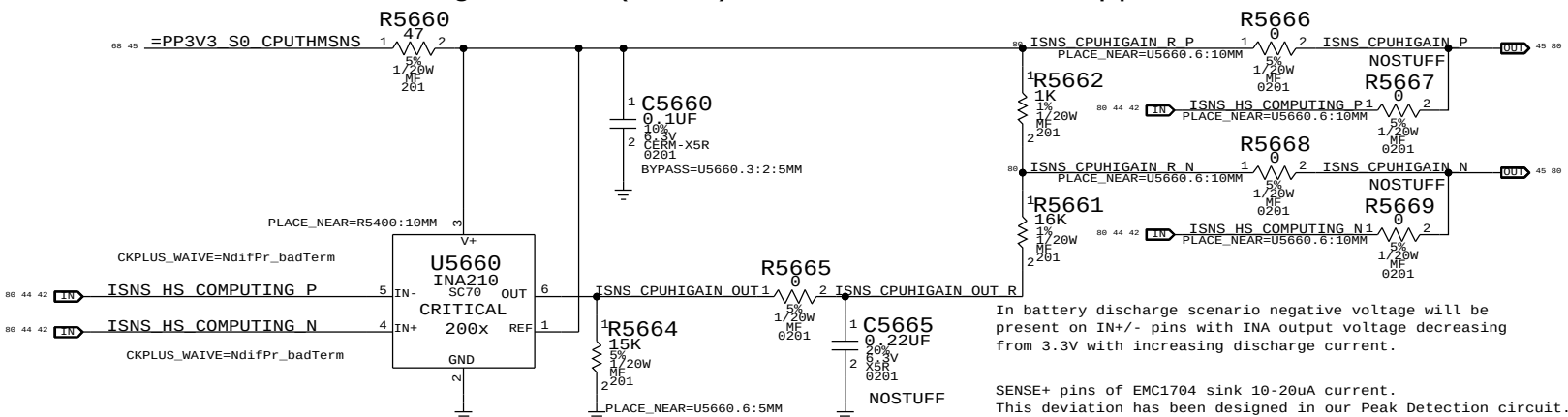
Camera (S2 Controller) Current Sense (ICMC)

Gain: 500x. EDP: 0.82 A
Rsense: 0.005 (R5610) or XW5610
Vsense: 4.1 mV, Range: 1.32 A
SMC AD: 15



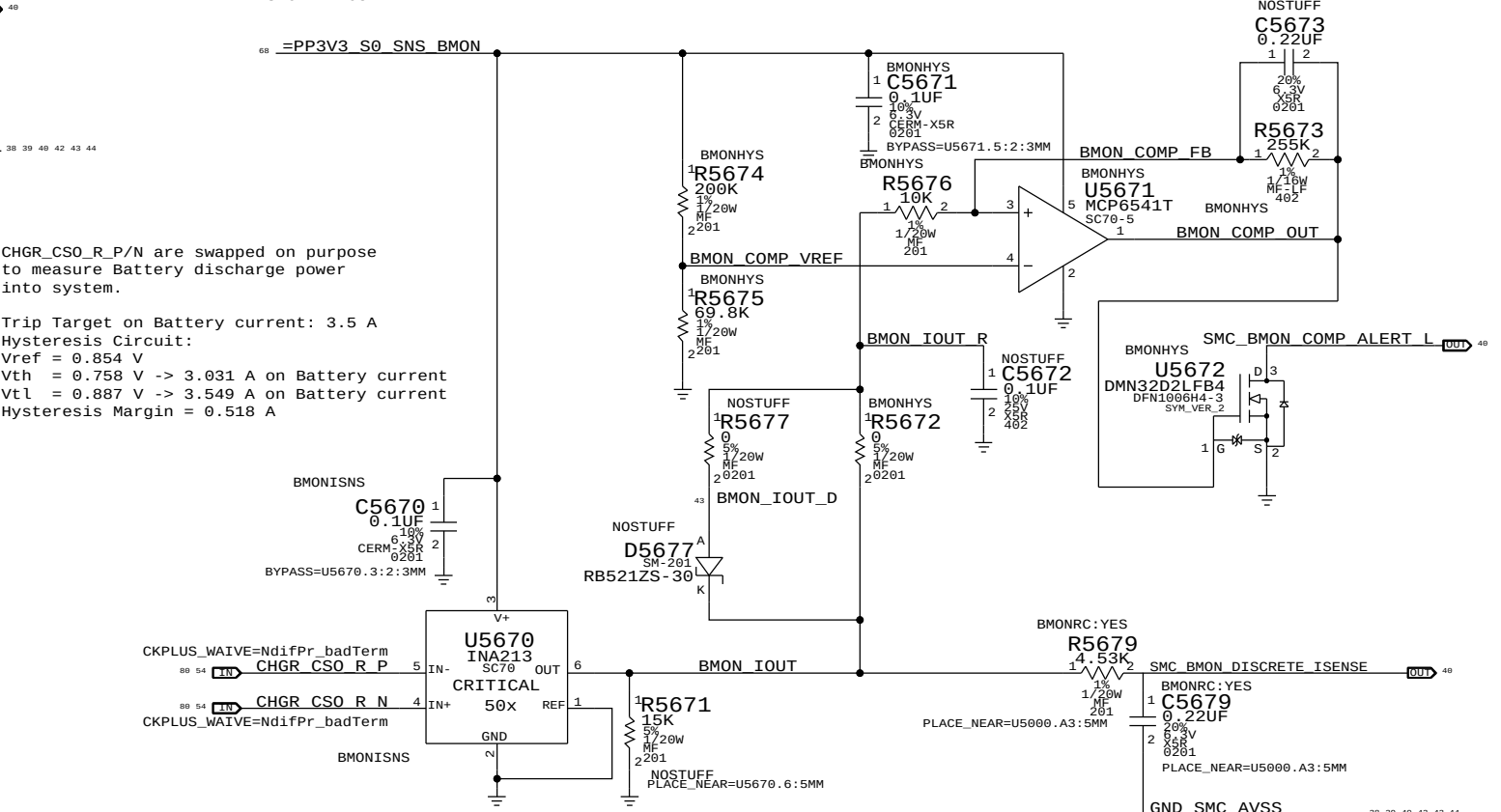
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0008	2	RES, MTL FILM, 100K, 1/16W, 0201, SMD, LF	C5619, C5629		LOADRC:NO
117S0008	1	RES, MTL FILM, 100K, 1/16W, 0201, SMD, LF	C5679		BMONRC:NO
117S0008	1	RES, MTL FILM, 100K, 1/16W, 0201, SMD, LF	C5649		TBTRC:NO

CPU High Side (IC0R) Peak Detection Support



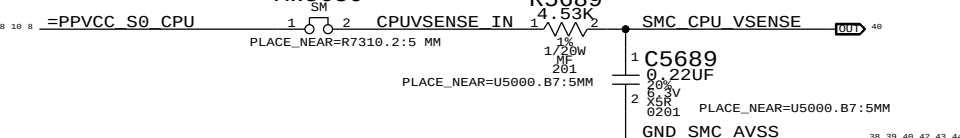
Battery BMON Discrete Current Sense (IP0R) & Threshold Alert

Gain: 50x. EDP: 8 A
Rsense: 0.005 (R7150)
Vsense: 50 mV, Range: 13.2 A
SMC AD: 05



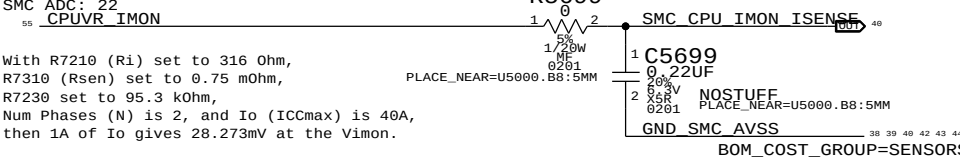
CPU Core Voltage Sense (VC0C)

SMC ADC: 20



CPU Core IMON Current Sense (IC2C)

Gain: 1 A / 28.273 mV, Range: 40 A.
SMC ADC: 22



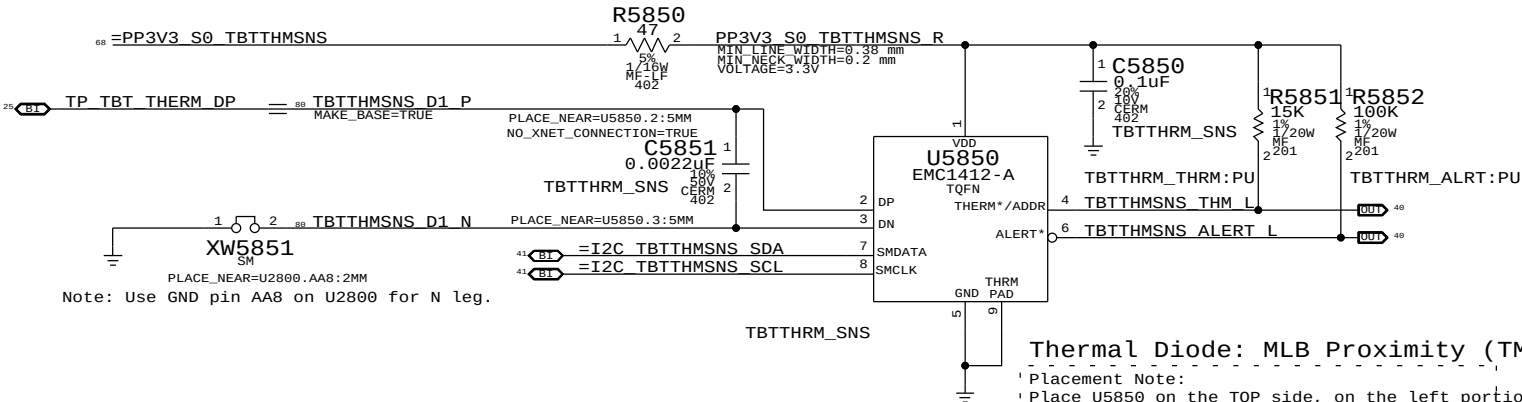
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PAGE TITLE		Power Sensors: Extended	
Apple Inc.		DRAWING NUMBER	051-1573
		REVISION	8.0.0
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Thermal Sensor A:
Thunderbolt Die, MLB Proximity

I2C Write: 0xD8, I2C Read: 0xD9

Thermal Diode: TBT Die (THSP)

Placement Note:
The P leg connects to THERMDA pin of the TBT chip, the N leg connect to pin AA8.



U5850 I2C Address:
By setting R5851 to 15k, I2C address for U5850 is 0xD8/0xD9.

Thermal Diode: MLB Proximity (TMLB)

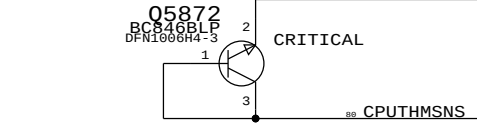
Placement Note:
Place U5850 on the TOP side, on the left portion of the board, 1" to the right of USB connector.

Thermal Sensor B & CPU High Peak Detection:
CPU Proximity, Memory Proximity, Airflow, Fin Stack Proximity

I2C Write: 0x98, I2C Read: 0x99

Thermal Diode: Airflow (TA0P)

Placement Note:
Place Q5871, Airflow thermal indicator, above the SSD, on the BOTTOM side.



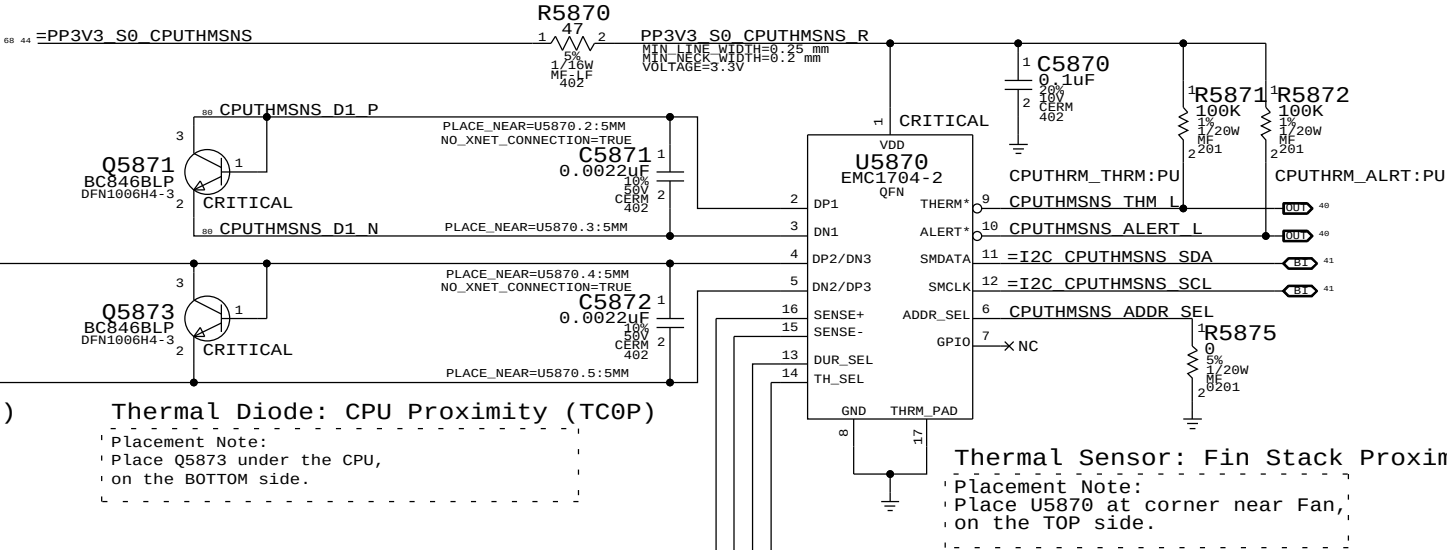
Thermal Diode: Memory Proximity (TM0P)

Placement Note:
Place Q5872 between two rows of Memory devices, between channel A and B, on the BOTTOM side.



Thermal Diode: CPU Proximity (TC0P)

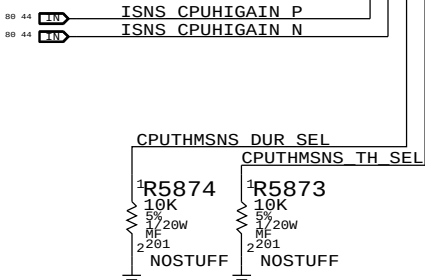
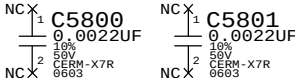
Placement Note:
Place Q5873 under the CPU, on the BOTTOM side.



Thermal Sensor: Fin Stack Proximity (Th1H)

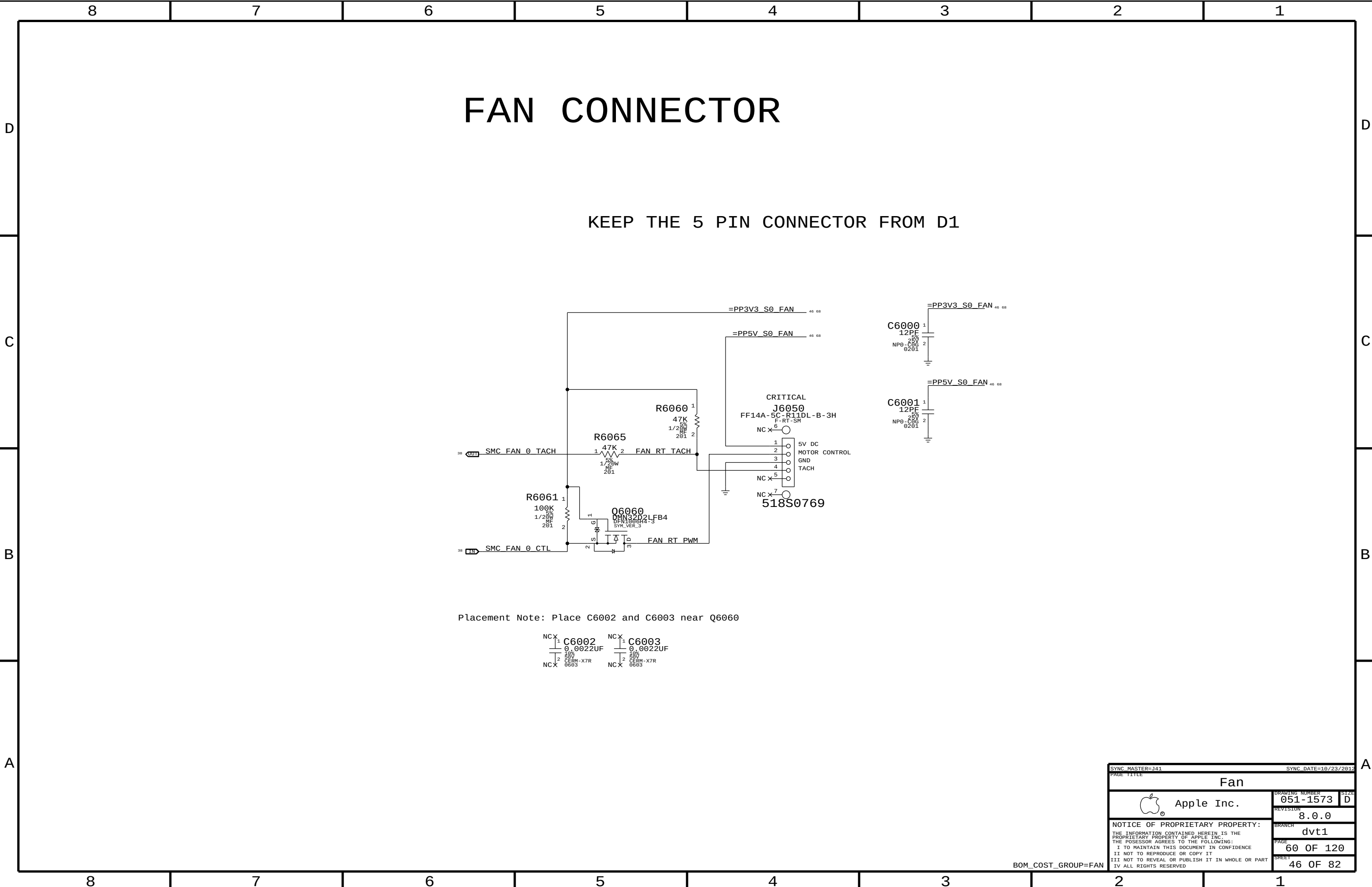
Placement Note:
Place U5870 at corner near Fan, on the TOP side.

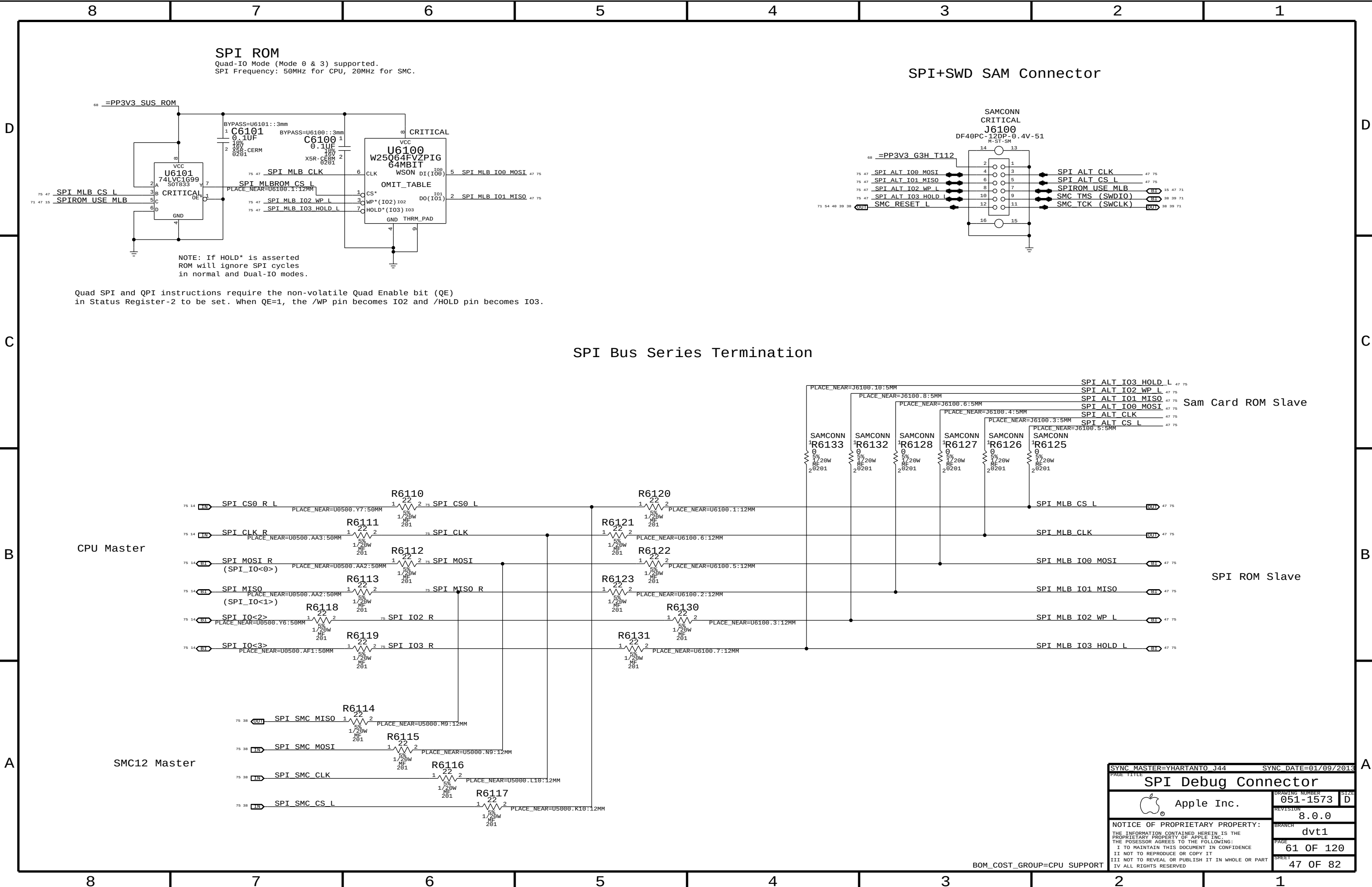
Placement Note: Place C5800 and C5801 near Q5871.



SYNC MASTER=YHARTANTO J44		SYNC DATE=01/07/2013	
PAGE TITLE		Thermal Sensors	
Apple Inc.		DRAWING NUMBER	051-1573
		REVISION	8.0.0
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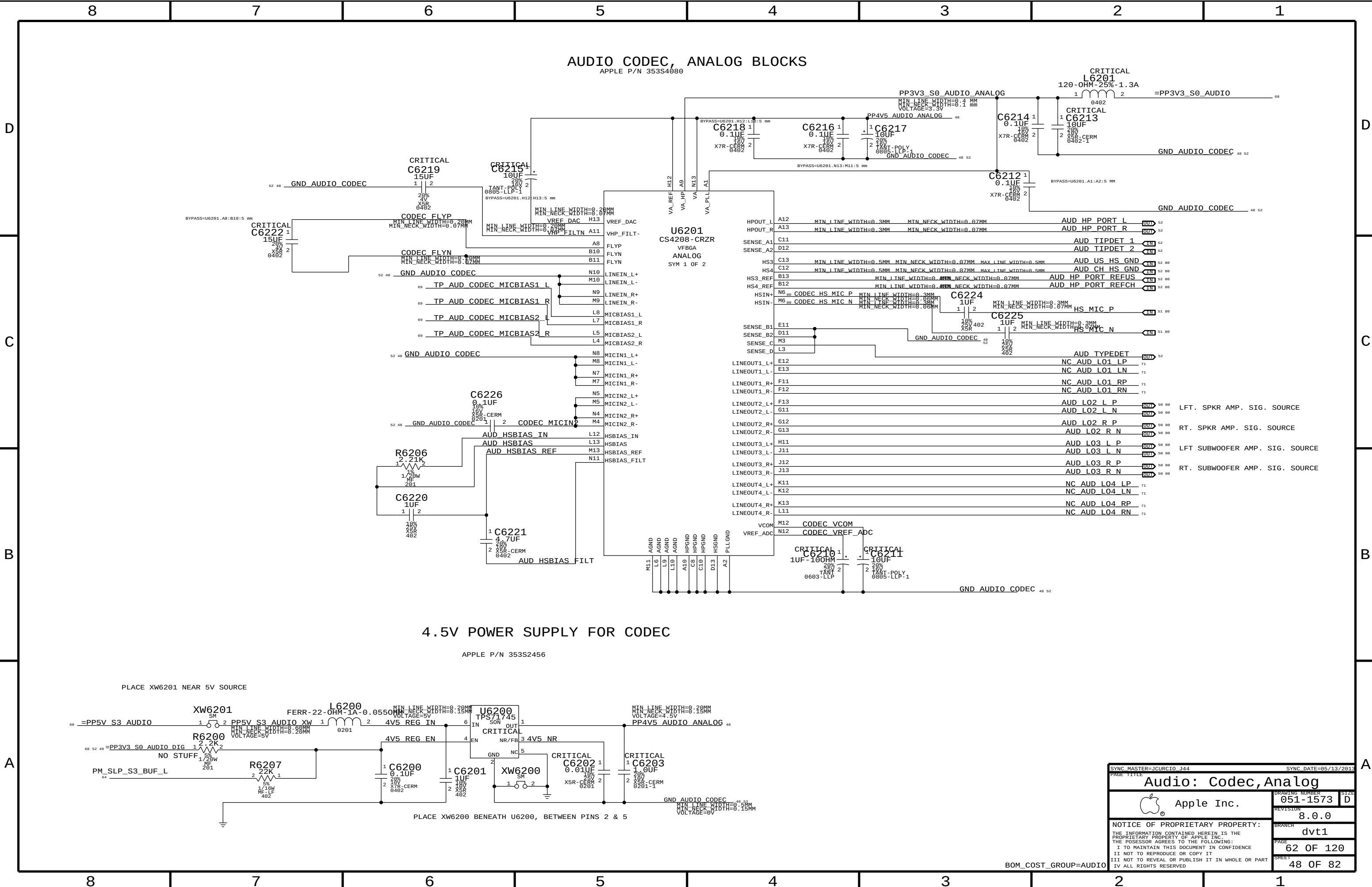
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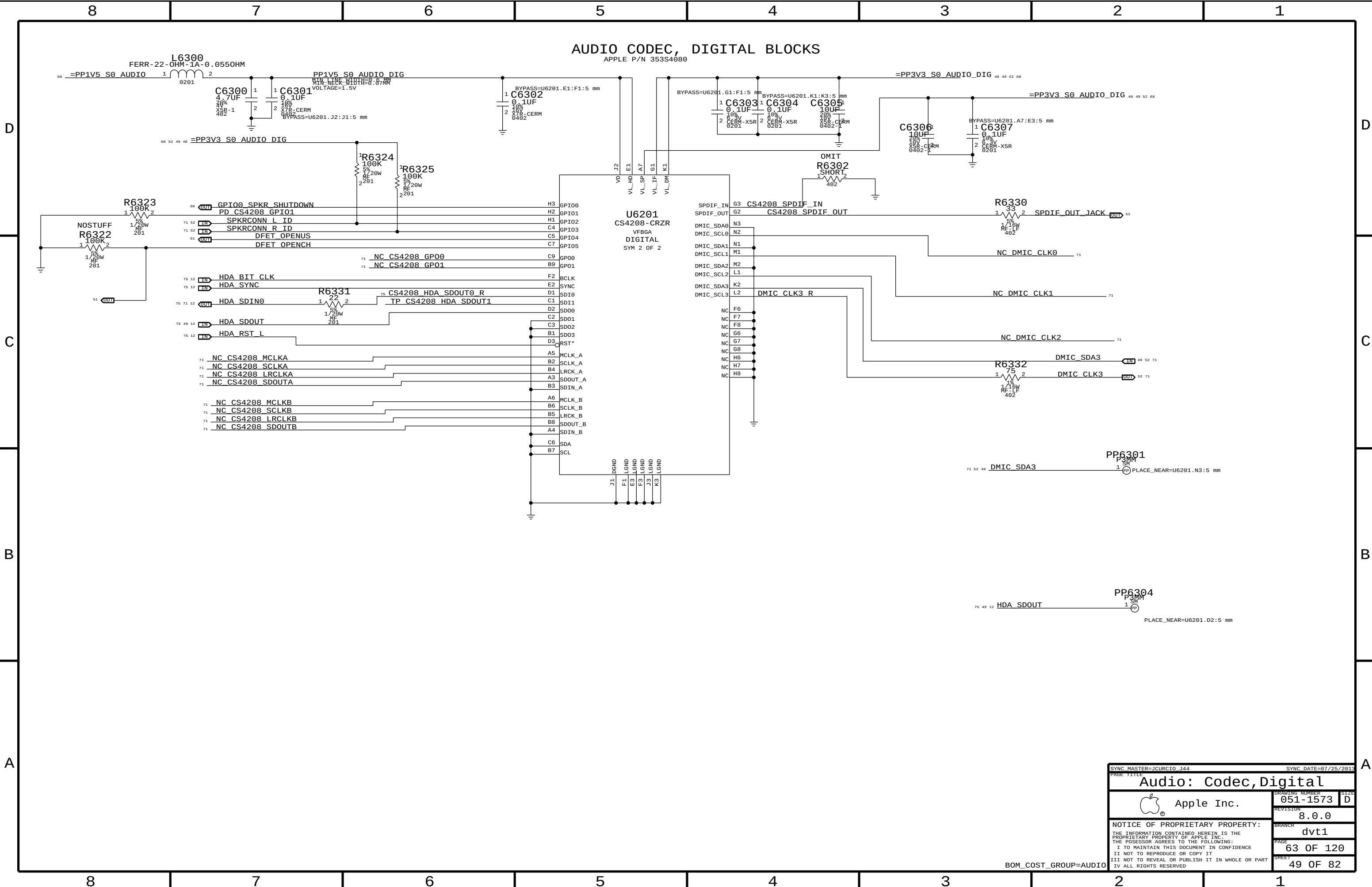




PAGE TITLE		SYNC DATE=01/09/2013	
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SBI Debug Connector		DRAWING NUMBER	051-1573
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BOM_COST_GROUP=CPU_SUPPORT





8

7

6

5

4

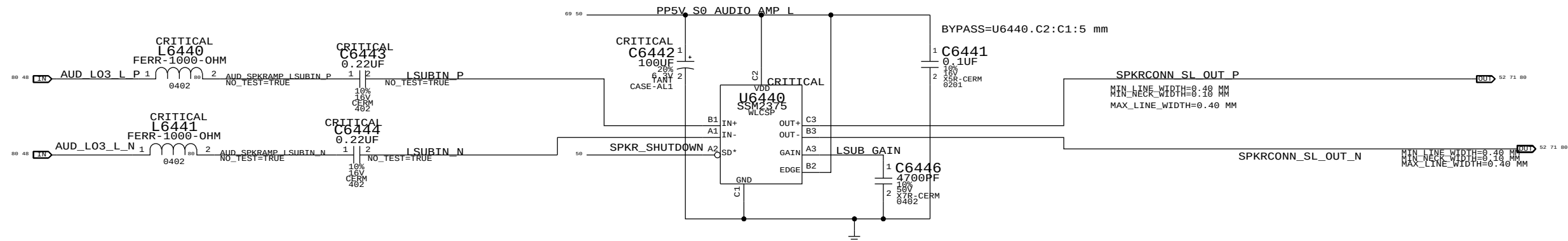
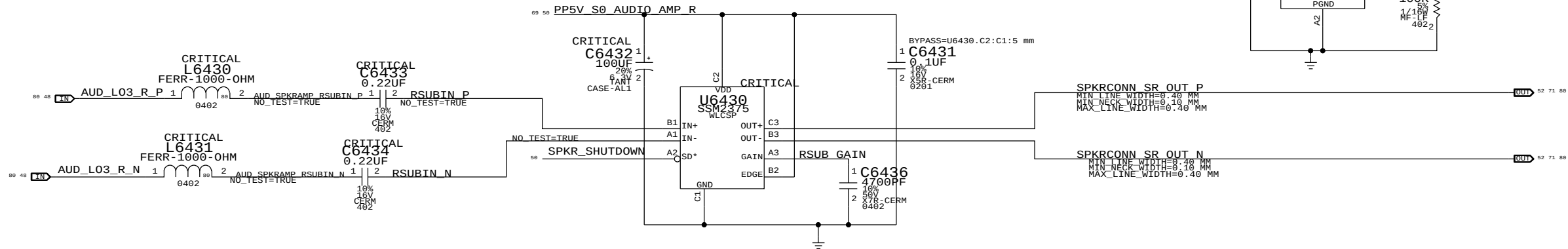
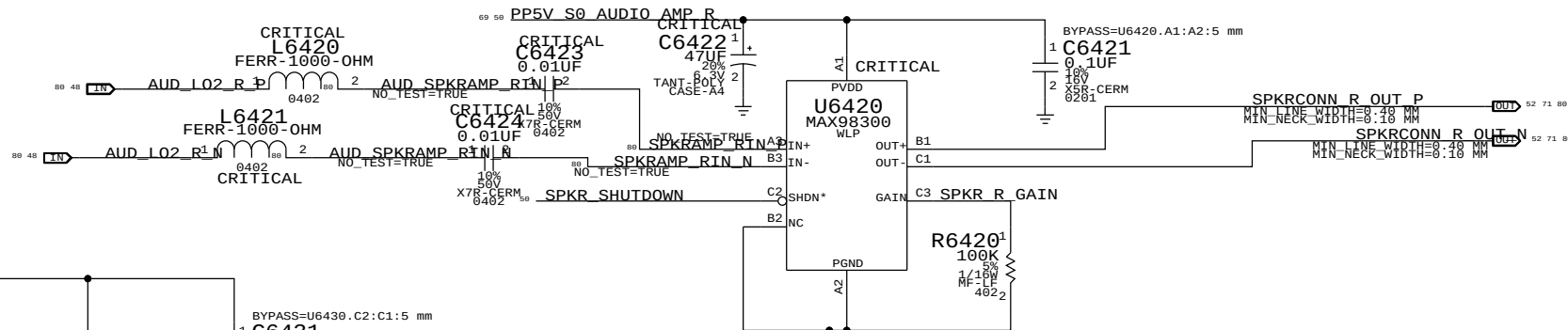
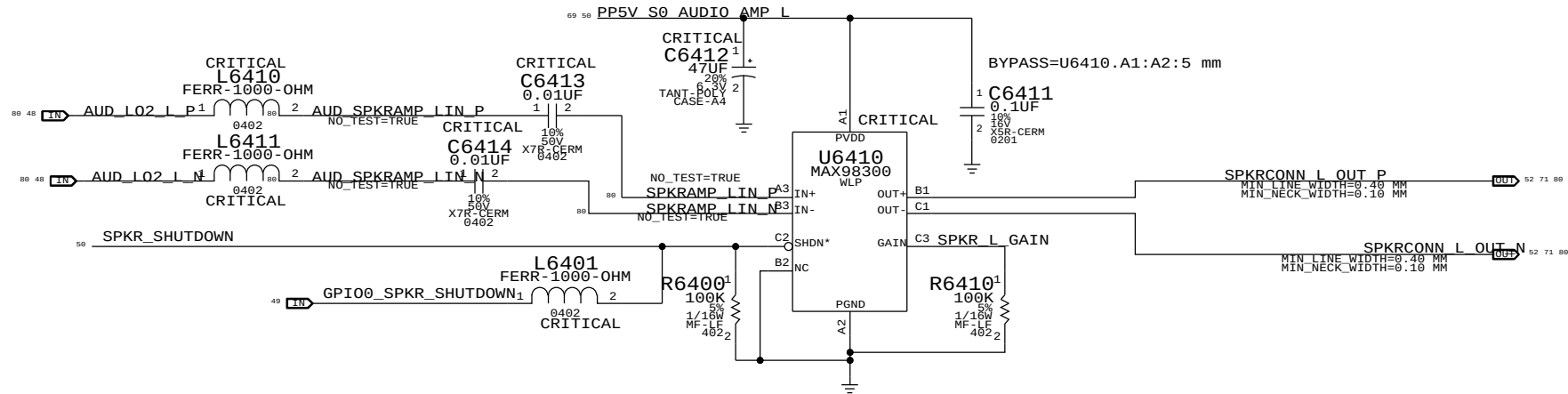
3

2

1

4X MONO SPEAKER AMPLIFIERS (MAX98300 & SSM2375)

APN: 353S2888 & 353S2958
GAIN = +3 DB
1ST ORDER FC (L&R) = NOM 569 HZ
1ST ORDER FC (SUB) = NOM 9 HZ

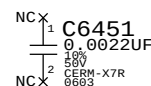
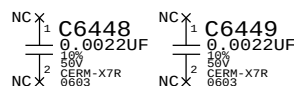


Placement Note: Place C6447 and C6452 near U6420

Placement Note: Place C6448 and C6449 near U6430

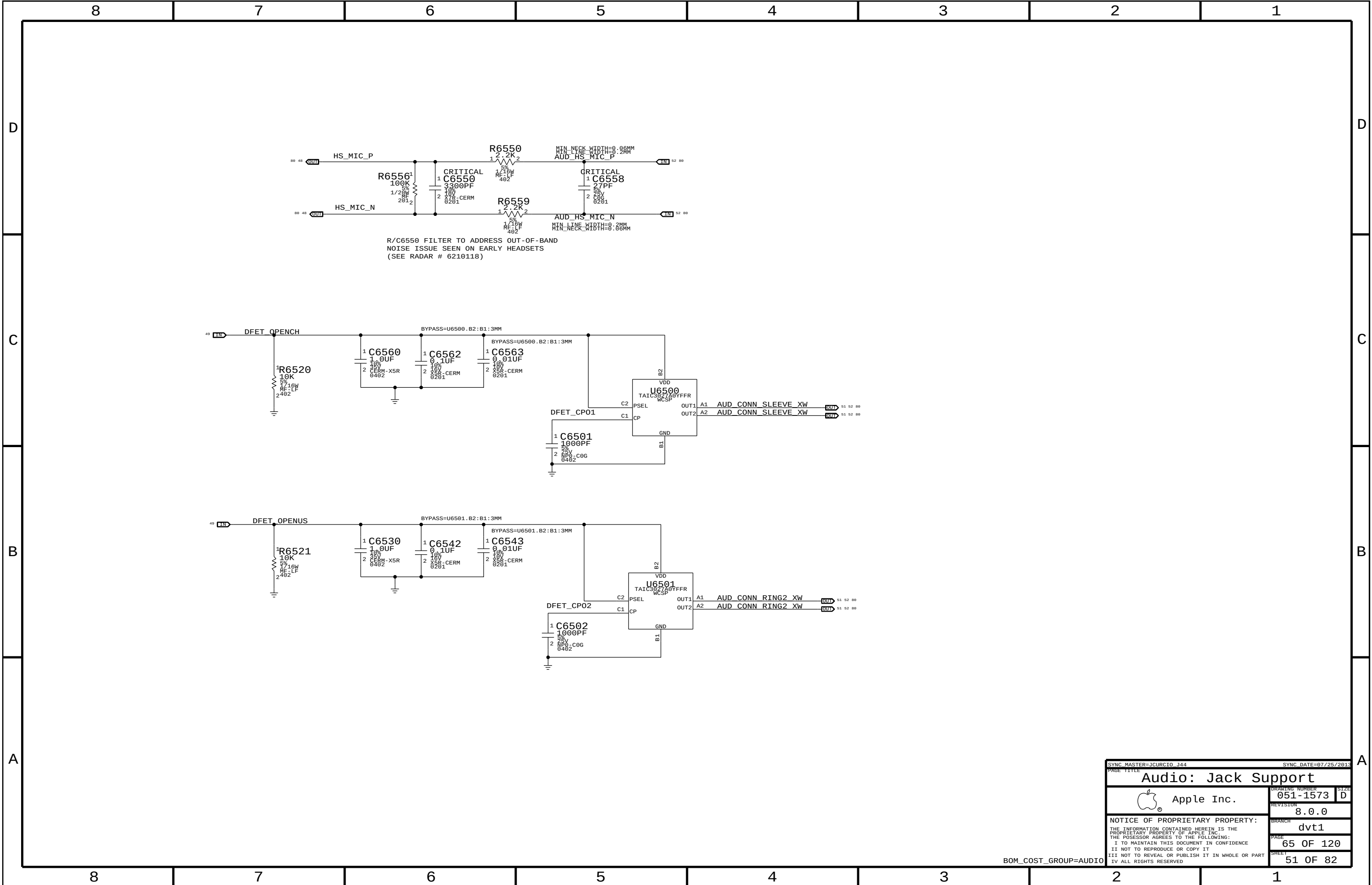
Placement Note: Place C6450 near U6410

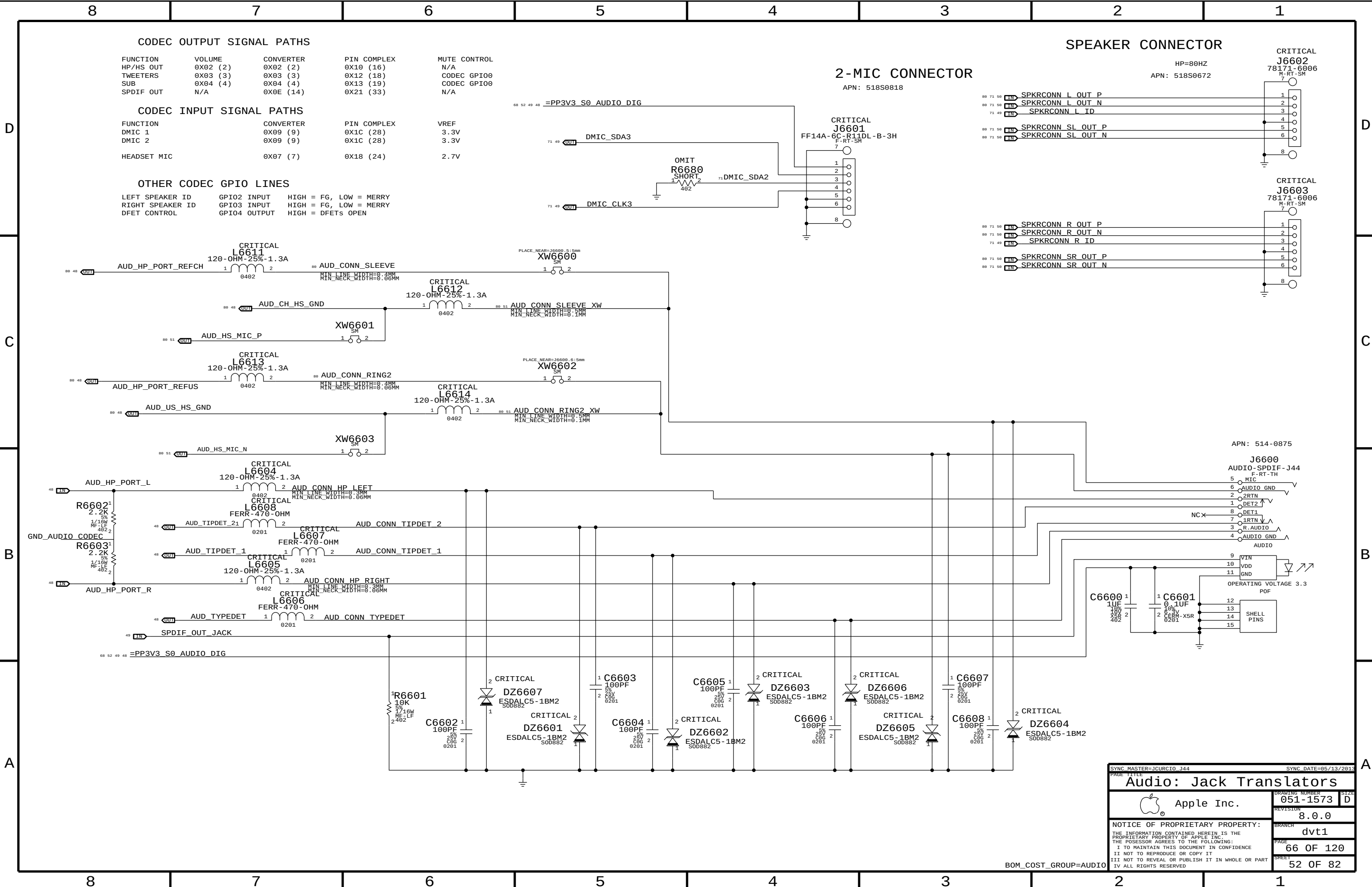
Placement Note: Place C6451 near U6440



BOM_COST_GROUP=AUDIO

SYNC MASTER=DIRK 344		SYNC DATE=01/09/2013	
PAGE TITLE			
Audio: Speaker Amps			
 Apple Inc.		DRAWING NUMBER	051-1573
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CODEC OUTPUT SIGNAL PATHS

FUNCTION	VOLUME	CONVERTER	PIN COMPLEX	MUTE CONTROL
HP/HS OUT	0X02 (2)	0X02 (2)	0X10 (16)	N/A
TWEETERS	0X03 (3)	0X03 (3)	0X12 (18)	CODEC GPIO0
SUB	0X04 (4)	0X04 (4)	0X13 (19)	CODEC GPIO0
SPDIF OUT	N/A	0X0E (14)	0X21 (33)	N/A

CODEC INPUT SIGNAL PATHS

FUNCTION	CONVERTER	PIN COMPLEX	VREF
DMIC 1	0X09 (9)	0X1C (28)	3.3V
DMIC 2	0X09 (9)	0X1C (28)	3.3V
HEADSET MIC	0X07 (7)	0X18 (24)	2.7V

OTHER CODEC GPIO LINES

LEFT SPEAKER ID	GPIO2 INPUT	HIGH = FG, LOW = MERRY
RIGHT SPEAKER ID	GPIO3 INPUT	HIGH = FG, LOW = MERRY
DFET CONTROL	GPIO4 OUTPUT	HIGH = DFETS OPEN

2-MIC CONNECTOR

APN: 518S0818

SPEAKER CONNECTOR

HP=80HZ
APN: 518S0672

CRITICAL
J6602
78171-6006
M-RT-SM

CRITICAL
J6603
78171-6006
M-RT-SM

APN: 514-0875

J6600
AUDIO-SPDIF-J44
F-RT-TH

OPERATING VOLTAGE 3.3
POF

SYNC MASTER=ICURC10 J44

SYNC DATE=05/13/2013

Audio: Jack Translators

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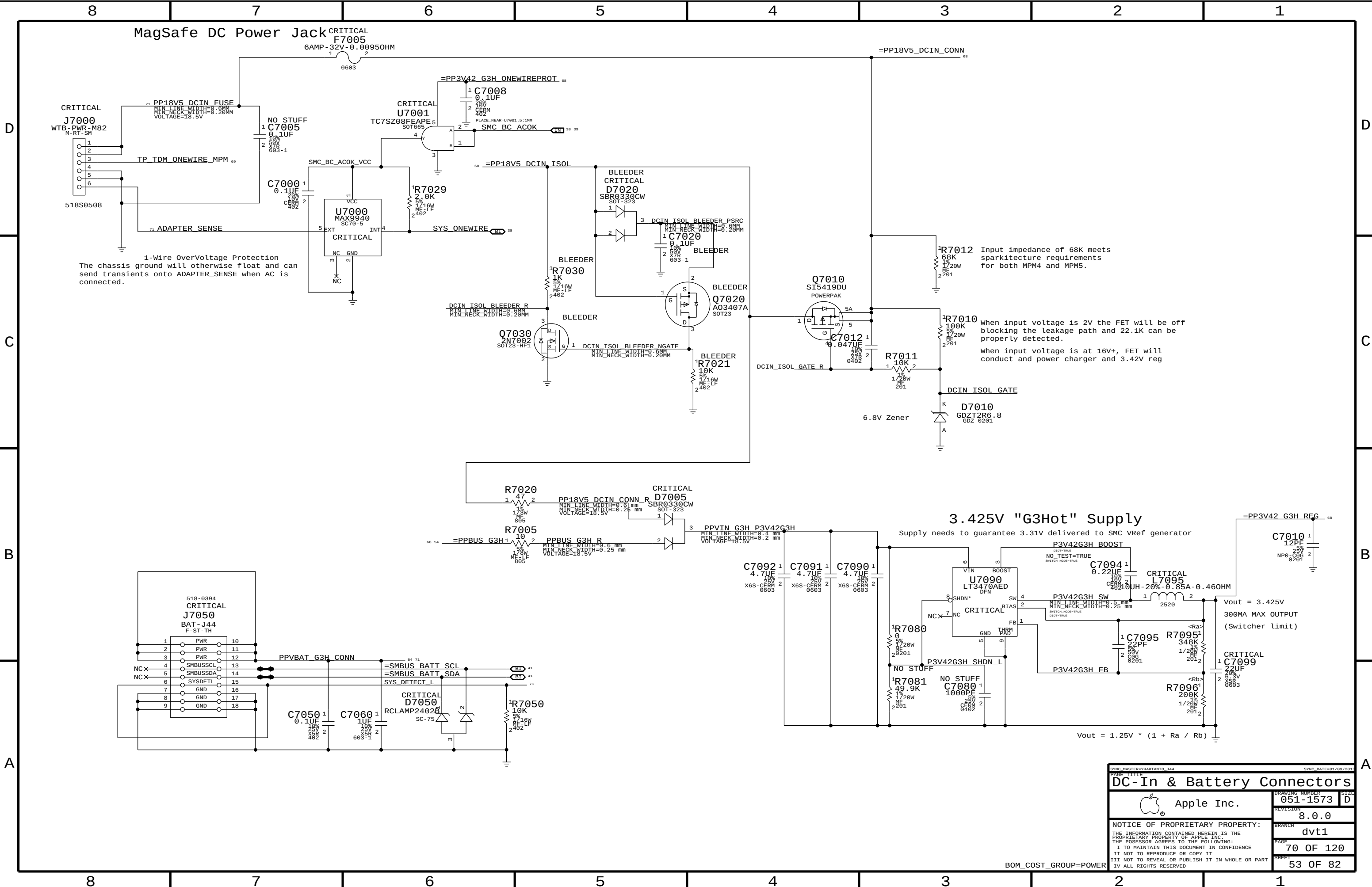
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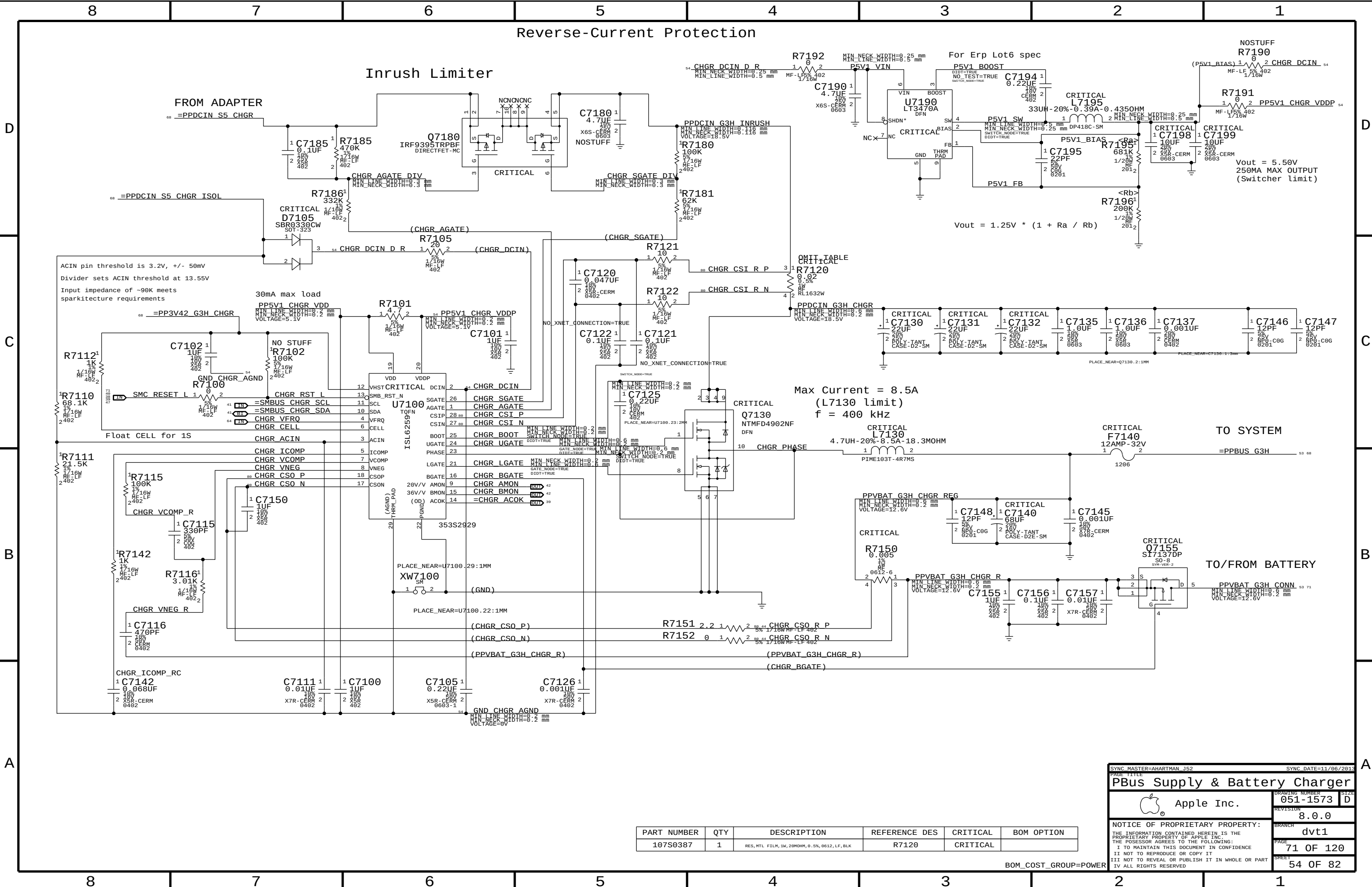
SHEET
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BOM_COST_GROUP=AUDIO



DC-In & Battery Connectors		DRAWING NUMBER	051-1573	SIZE	D		
Apple Inc.		REVISION	8.0.0				
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BOM_COST_GROUP=POWER



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
107S0387	1	RES, MTL FILM, 1W, 20MOHM, 0.5%, 0612, LF, BLK	R7120	CRITICAL	

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PAGE

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SYNC MASTER=AHARTMAN_352

SYNC DATE=11/06/2013

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051-1573

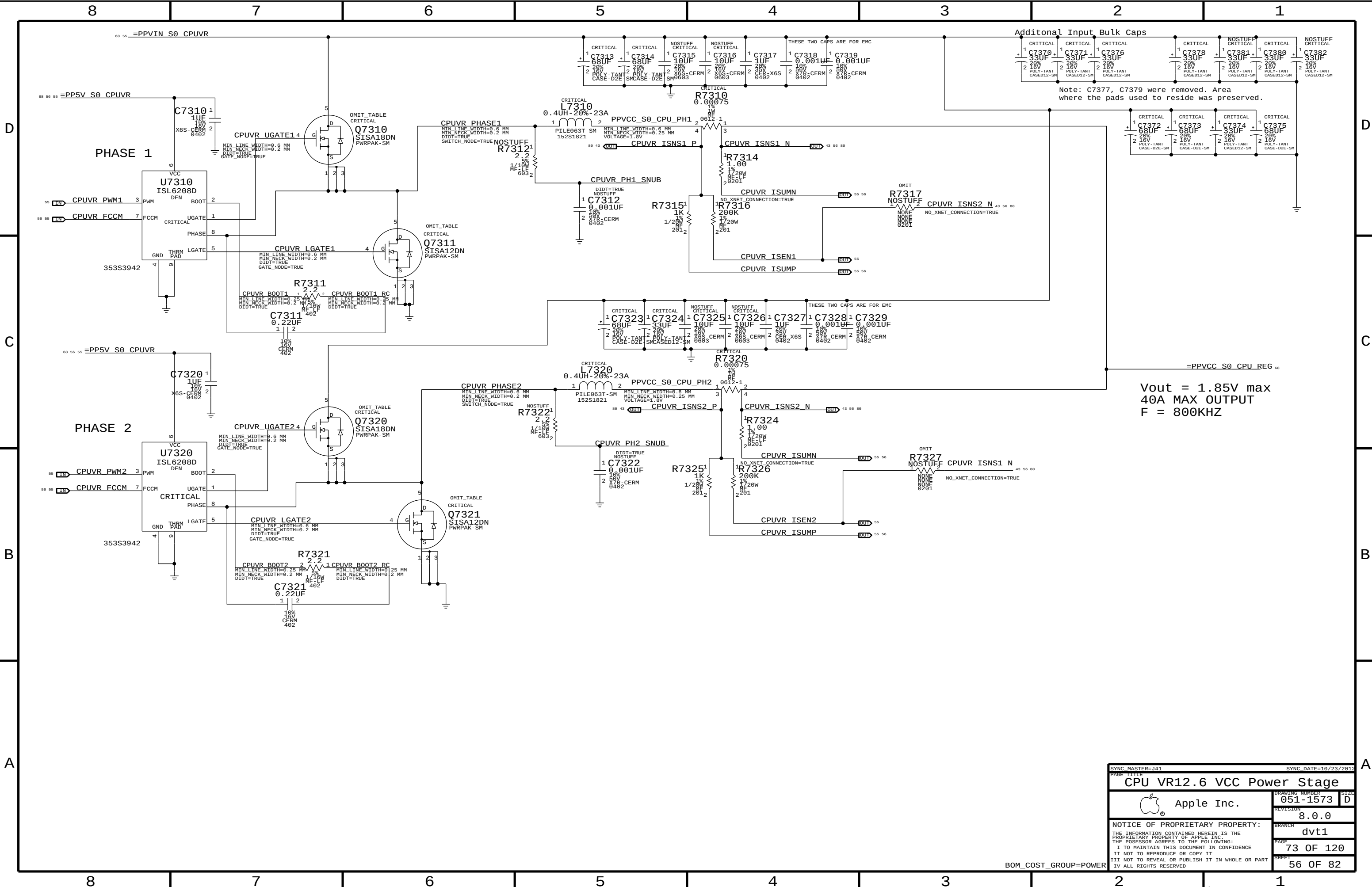
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dvt1

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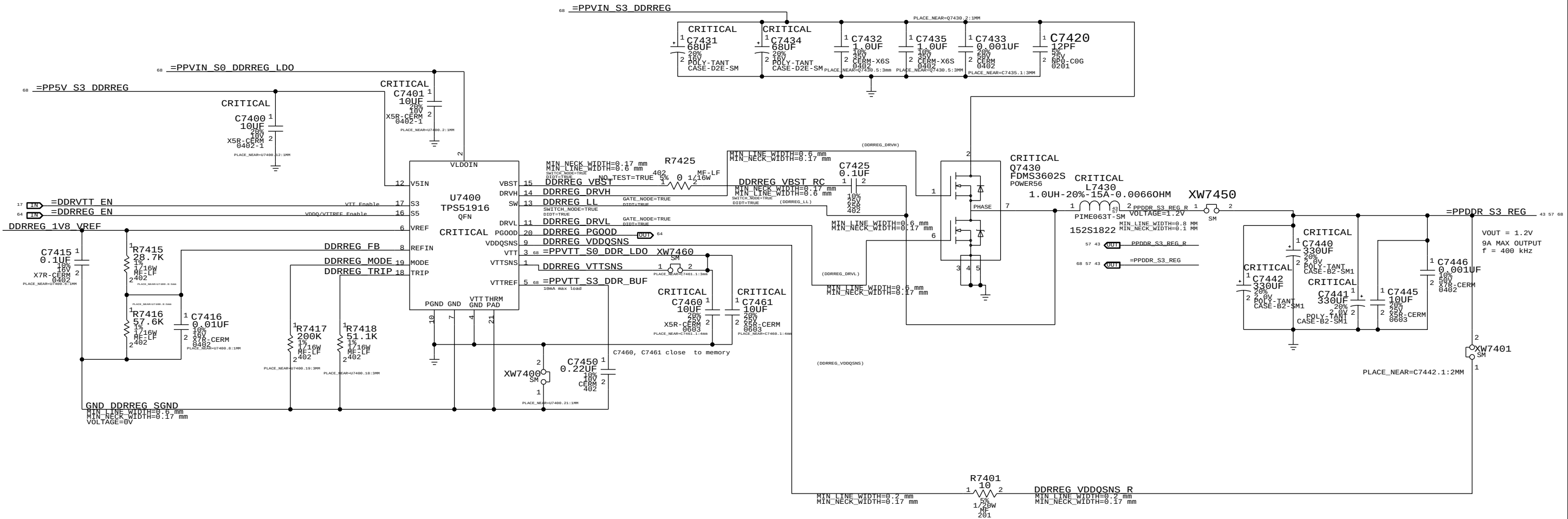
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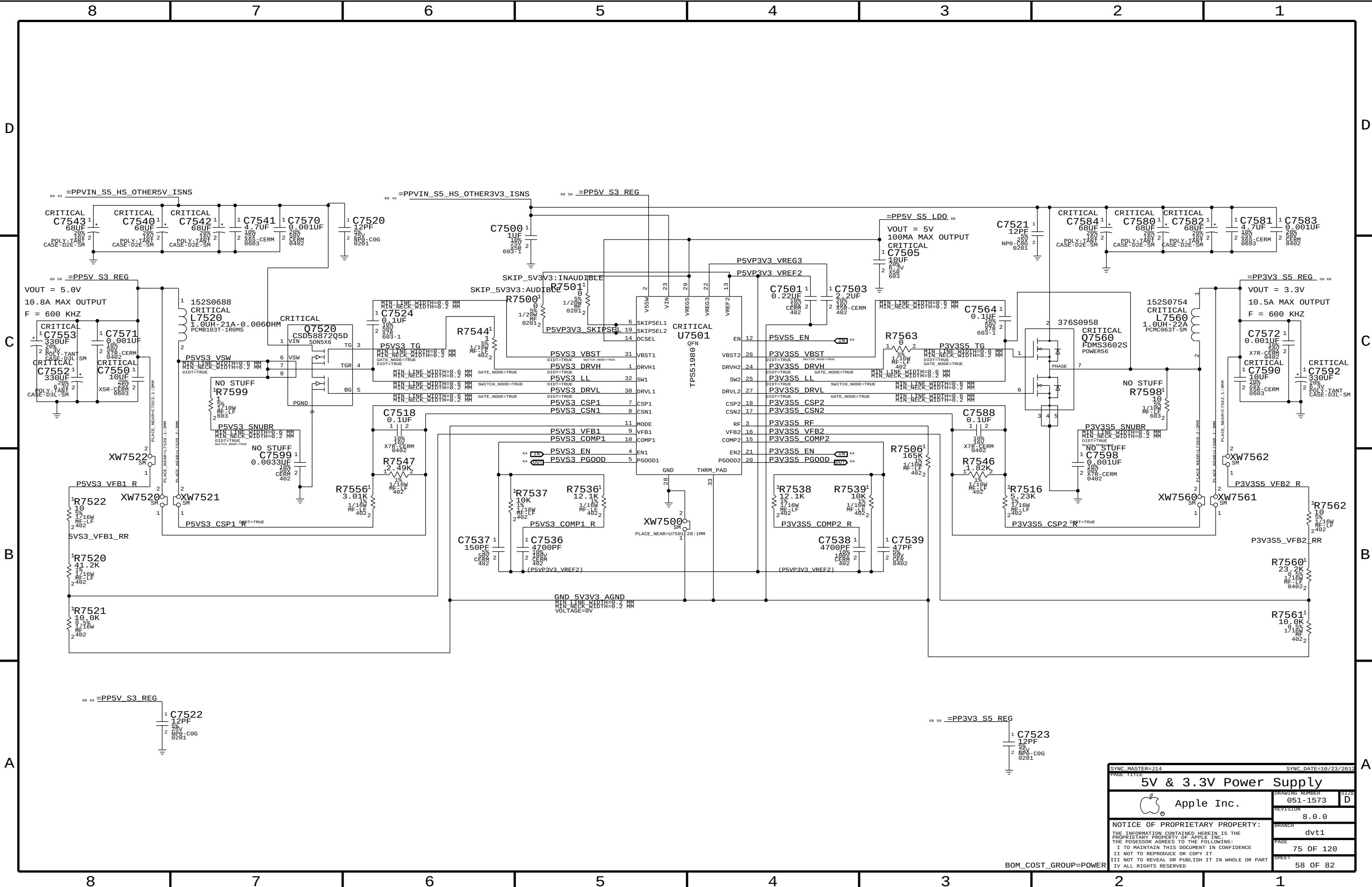
CPU VR12.6 VCC Power Stage		DRAWING NUMBER		SIZE
Apple Inc.		051-1573		D
REVISION		8.0.0		
BRANCH		dvt1		
PAGE		73 OF 120		
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1.2V S3 Regulator

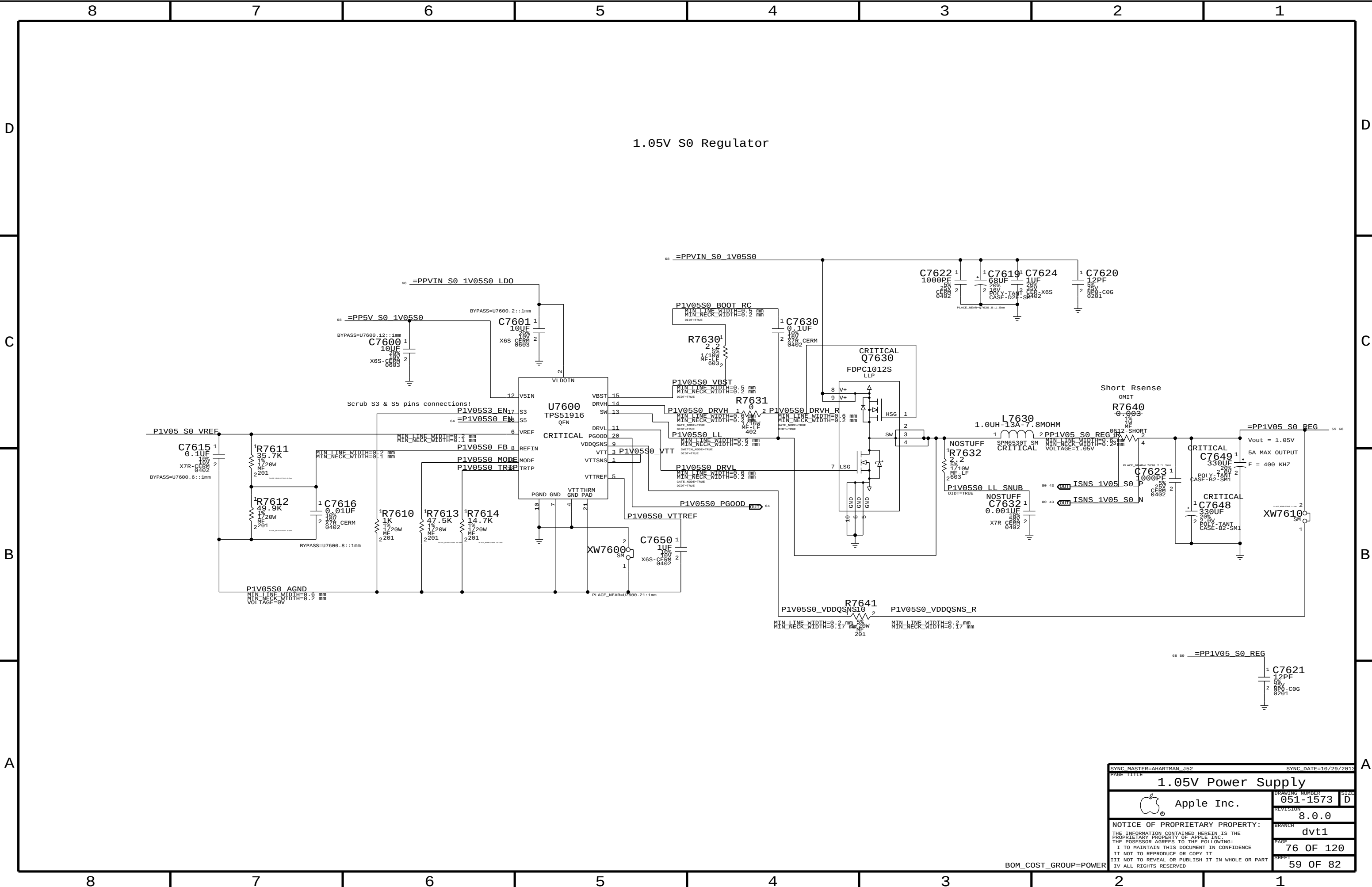


SYNC MASTER=J41 MLB		SYNC DATE=05/21/2013	
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LPDDR3 Supply			
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		SIZE	D
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		BRANCH	dvt1
		PAGE	74 OF 120
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BOM_COST_GROUP=POWER



SYNC MASTER=J14		SYNC DATE=10/23/2012	
PAGE TITLE			
5V & 3.3V Power Supply			
 Apple Inc.	DRAWING NUMBER	051-1573	
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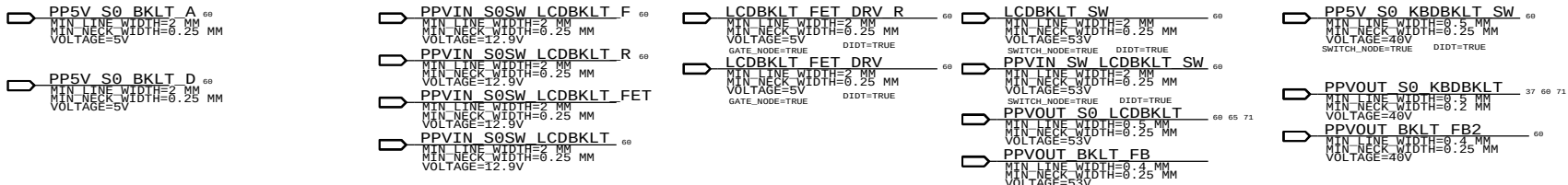


SYNC MASTER=AHARTMAN_352		SYNC DATE=10/29/2013	
PAGE TITLE		1.05V Power Supply	
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		PAGE	76 OF 120
		SHEET	59 OF 82

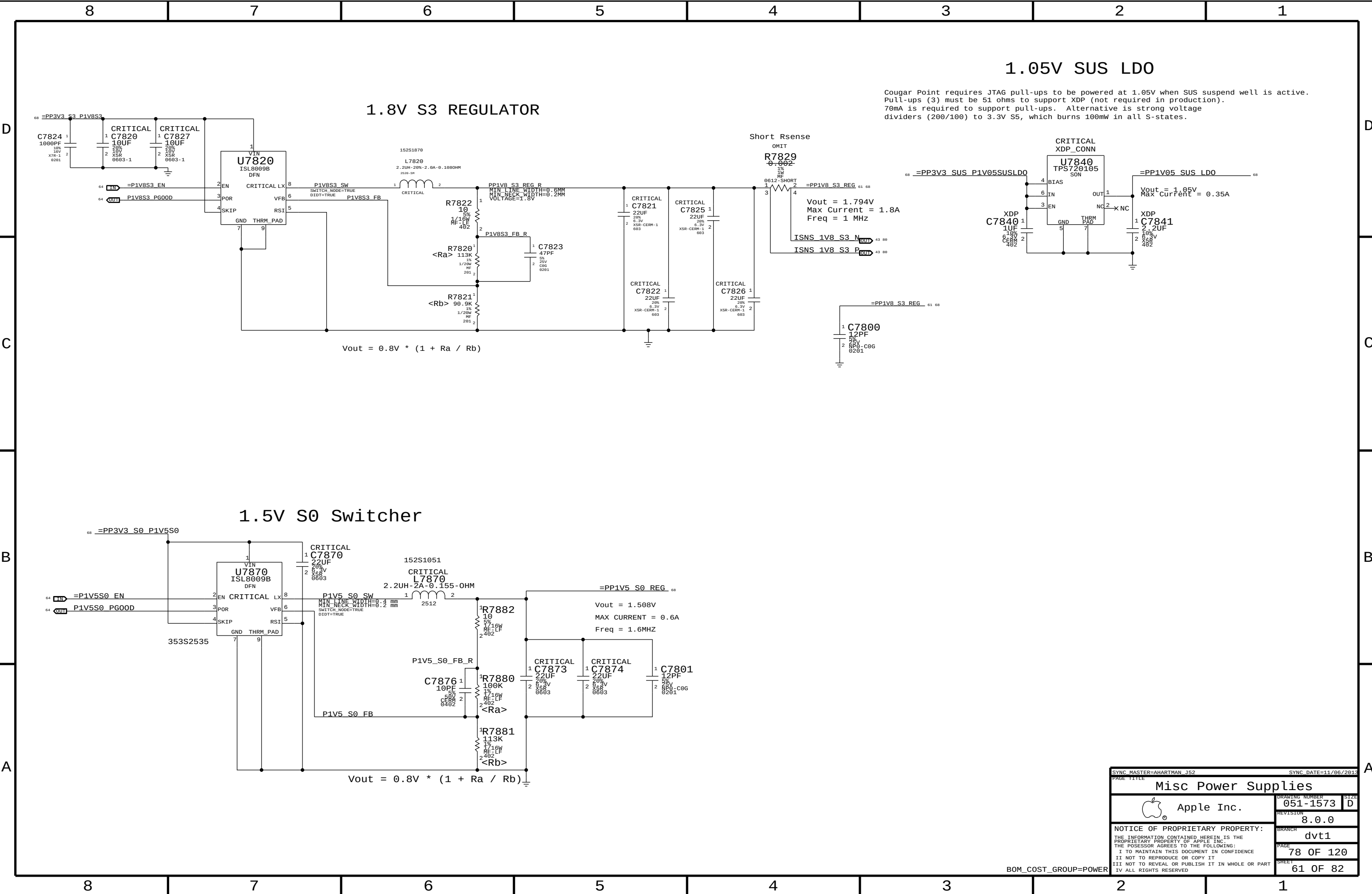
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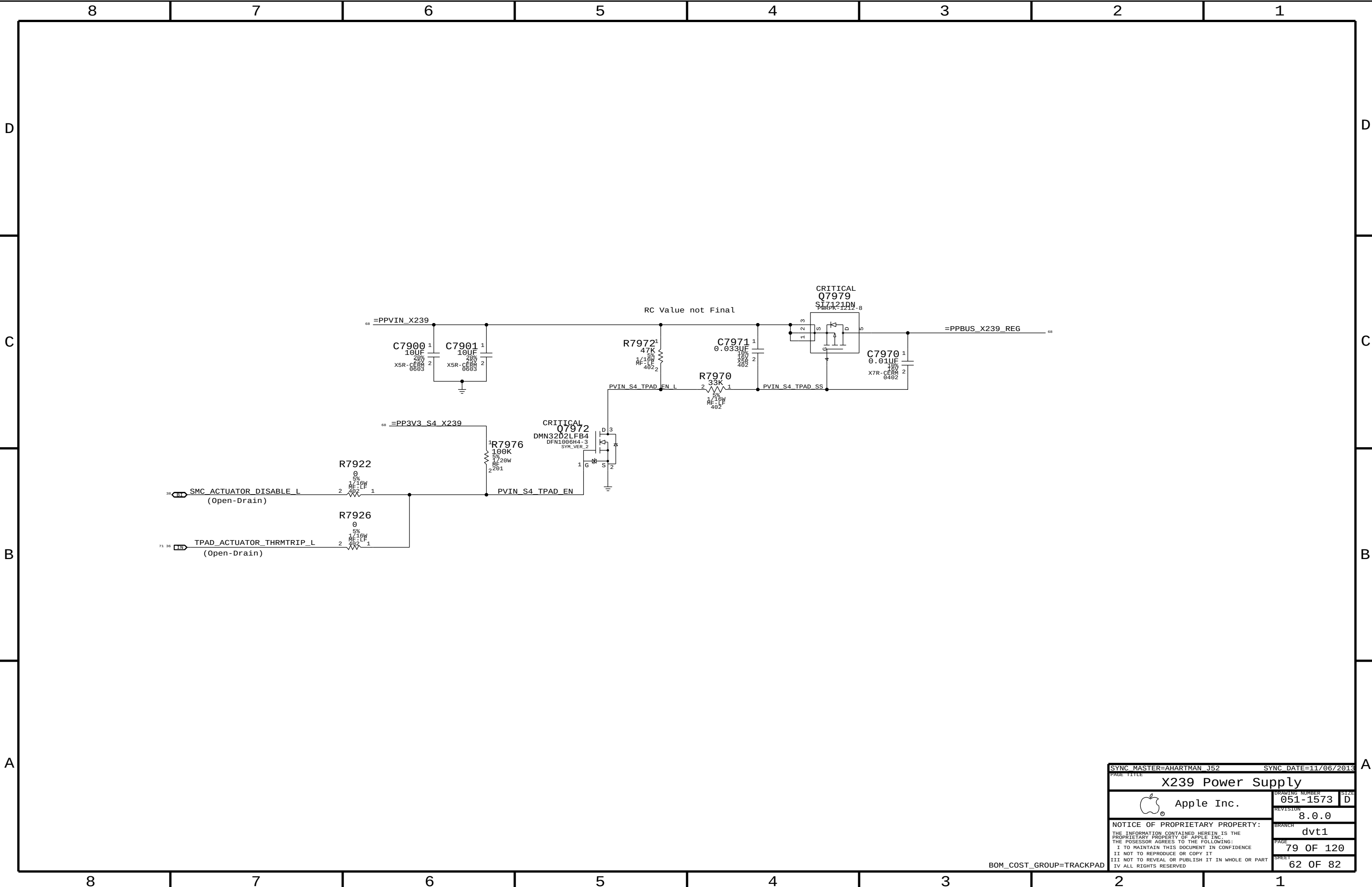
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
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SYNCH MASTER-SHART J44		SYNCH DATE=11/28/2012	
PAGE TITLE			
LCD & KBD Backlight Driver			
 Apple Inc.		DRAWING NUMBER 051-1573	SIZE D
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SYNC MASTER=AHARTMAN_352		SYNC DATE=11/06/2013	
PAGE TITLE		Misc Power Supplies	
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SYNC_DATE=11/06/2013

PAGE TITLE

X239 Power Supply

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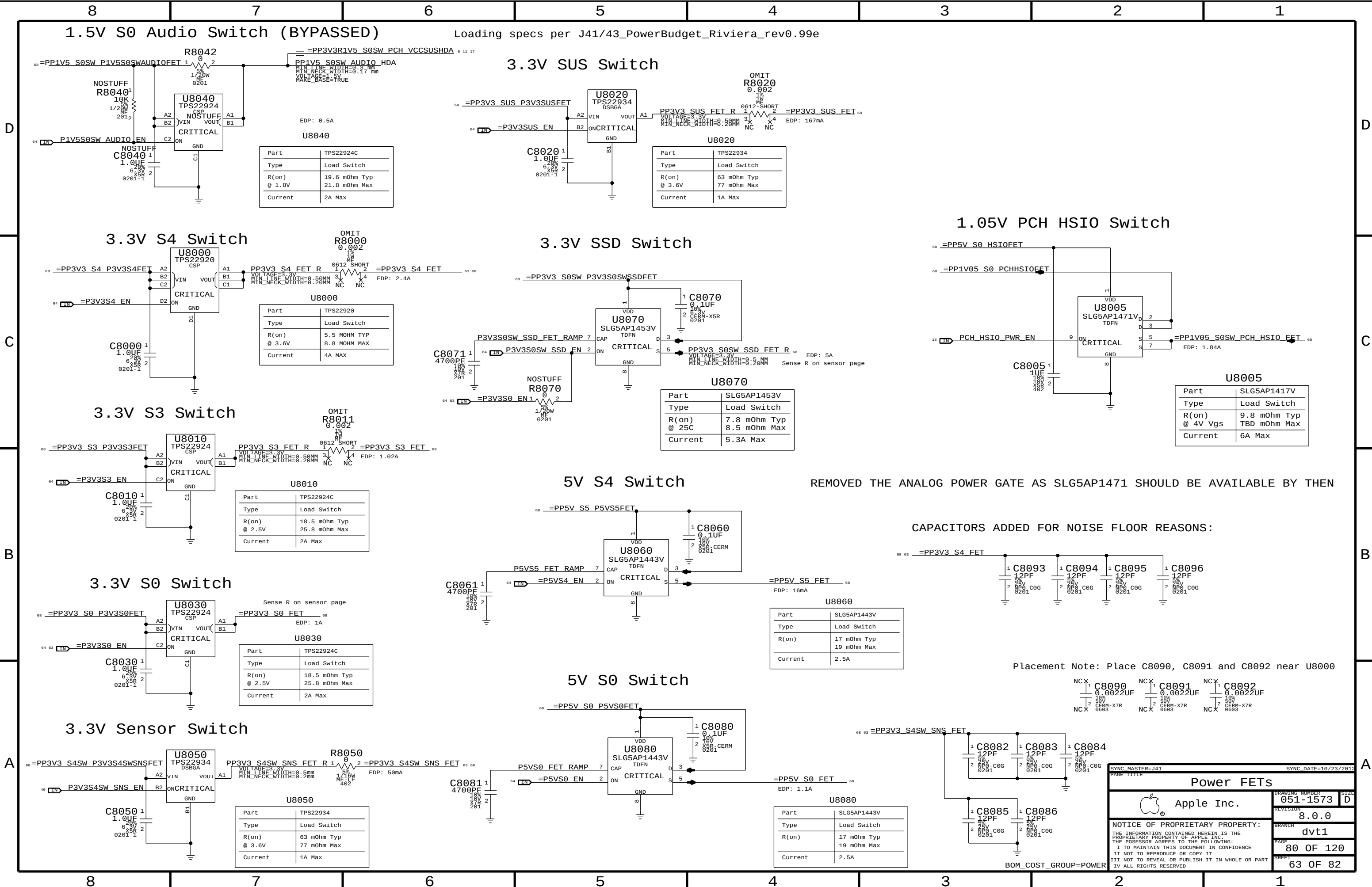
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BOM_COST_GROUP=TRACKPAD



1.5V S0 Audio Switch (BYPASSED)

Loading specs per J41/43_PowerBudget_Riviera_rev0.99e

3.3V SUS Switch

1.05V PCH HSI0 Switch

3.3V S4 Switch

3.3V SSD Switch

3.3V S3 Switch

5V S4 Switch

REMOVED THE ANALOG POWER GATE AS SLG5AP1471 SHOULD BE AVAILABLE BY THEN

3.3V S0 Switch

CAPACITORS ADDED FOR NOISE FLOOR REASONS:

Placement Note: Place C8090, C8091 and C8092 near U8000

3.3V Sensor Switch

5V S0 Switch

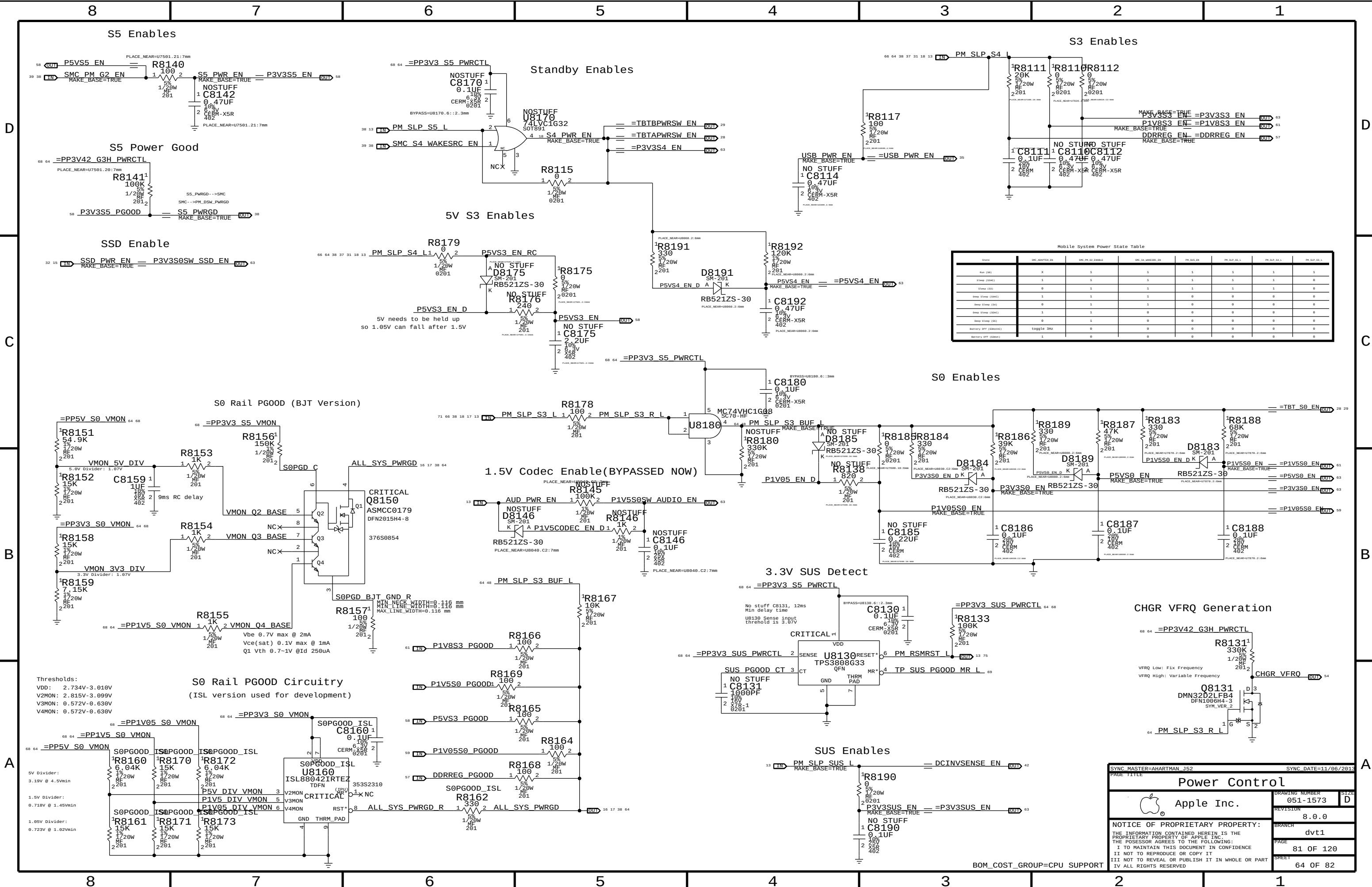
SYNC MASTER=J41
PAGE TITLE

SYNC DATE=10/23/2012

Power FETs		
 Apple Inc.	DRAWING NUMBER 051-1573	SIZE D
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BOM_COST_GROUP=POWER



Mobile System Power State Table							
State	SMC_ADAPTER_EN	SMC_PM_S0_ENABLE	SMC_SH_WAKEUP_EN	PM_SLP_S0	PM_SLP_S0_L	PM_SLP_S0_U	PM_SLP_S0_L1
Run (S0)	X	1	1	1	1	1	1
Standby (S0)	1	1	1	1	1	1	0
Standby (S1)	0	1	1	1	1	1	0
Deep Sleep (S0)	1	1	1	0	0	0	0
Deep Sleep (S1)	0	1	1	0	0	0	0
Deep Sleep (S2)	1	0	0	0	0	0	0
Deep Sleep (S3)	0	1	0	0	0	0	0
Battery Off (S0)	toggle SHZ	0	0	0	0	0	0
Battery Off (S0)	1	0	0	0	0	0	0

Power Control

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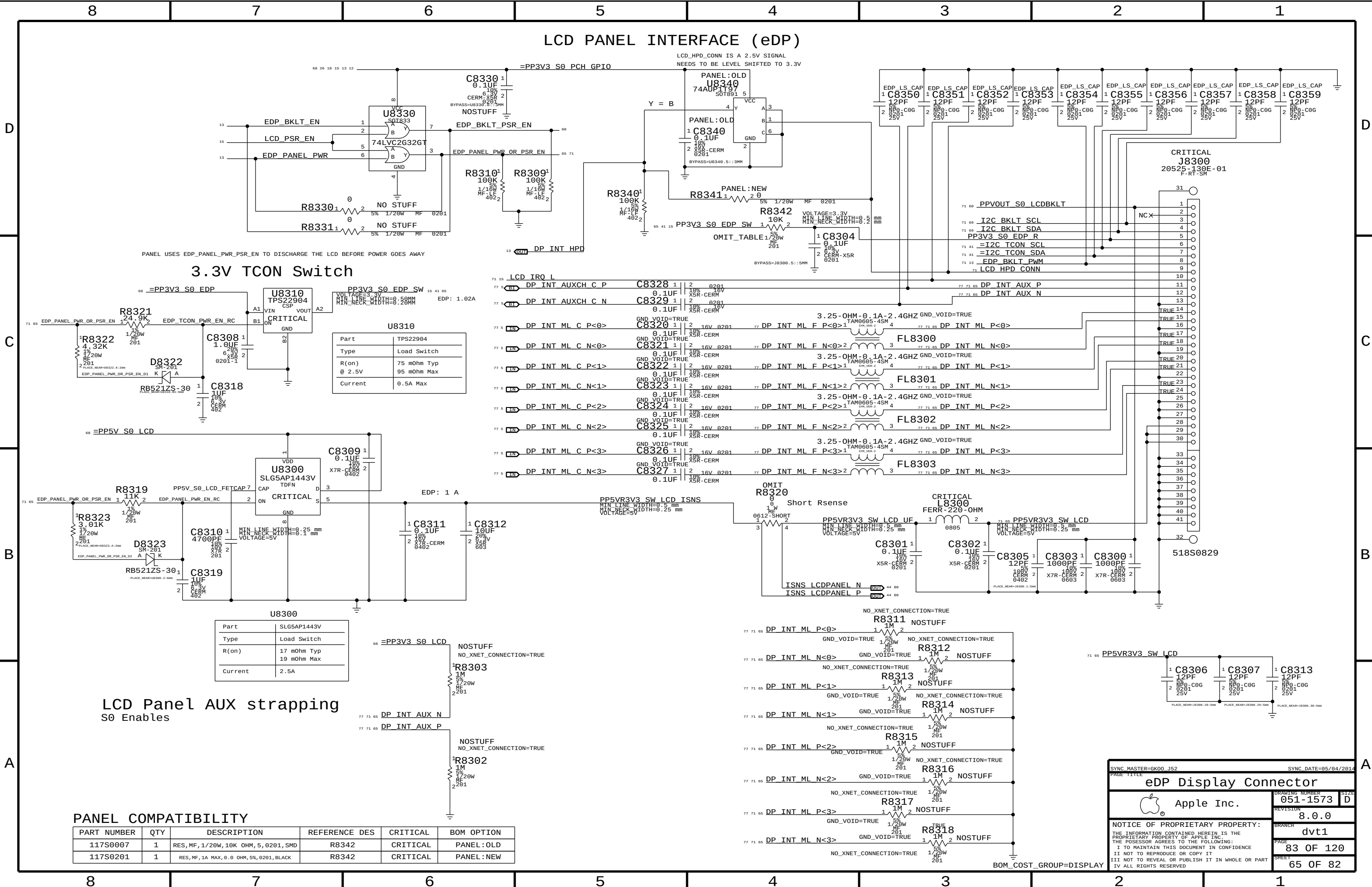
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SYNC MASTER=AHARTMAN_152

SYNC DATE=11/06/2013

PAGE TITLE

BOM_COST_GROUP=CPU_SUPPORT



PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S0007	1	RES,MF,1/20W,10K OHM,5,0201,SMD	R8342	CRITICAL	PANEL:OLD
117S0201	1	RES,MF,1A MAX,0.0 OHM,5%,0201,BLACK	R8342	CRITICAL	PANEL:NEW

SYNC MASTER=GK00 J52

SYNC DATE=05/04/2014

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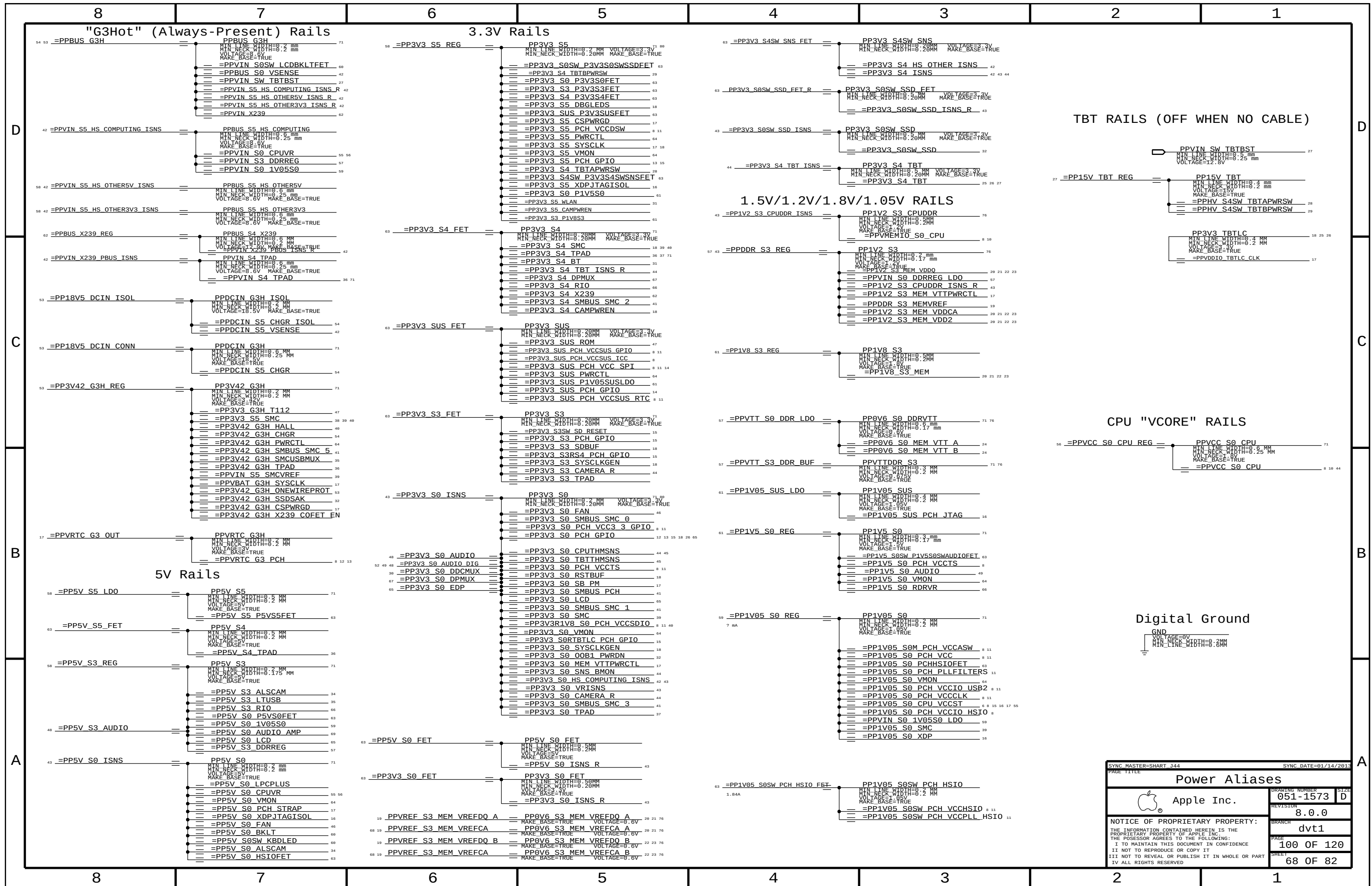
DRAWING NUMBER
051-1573

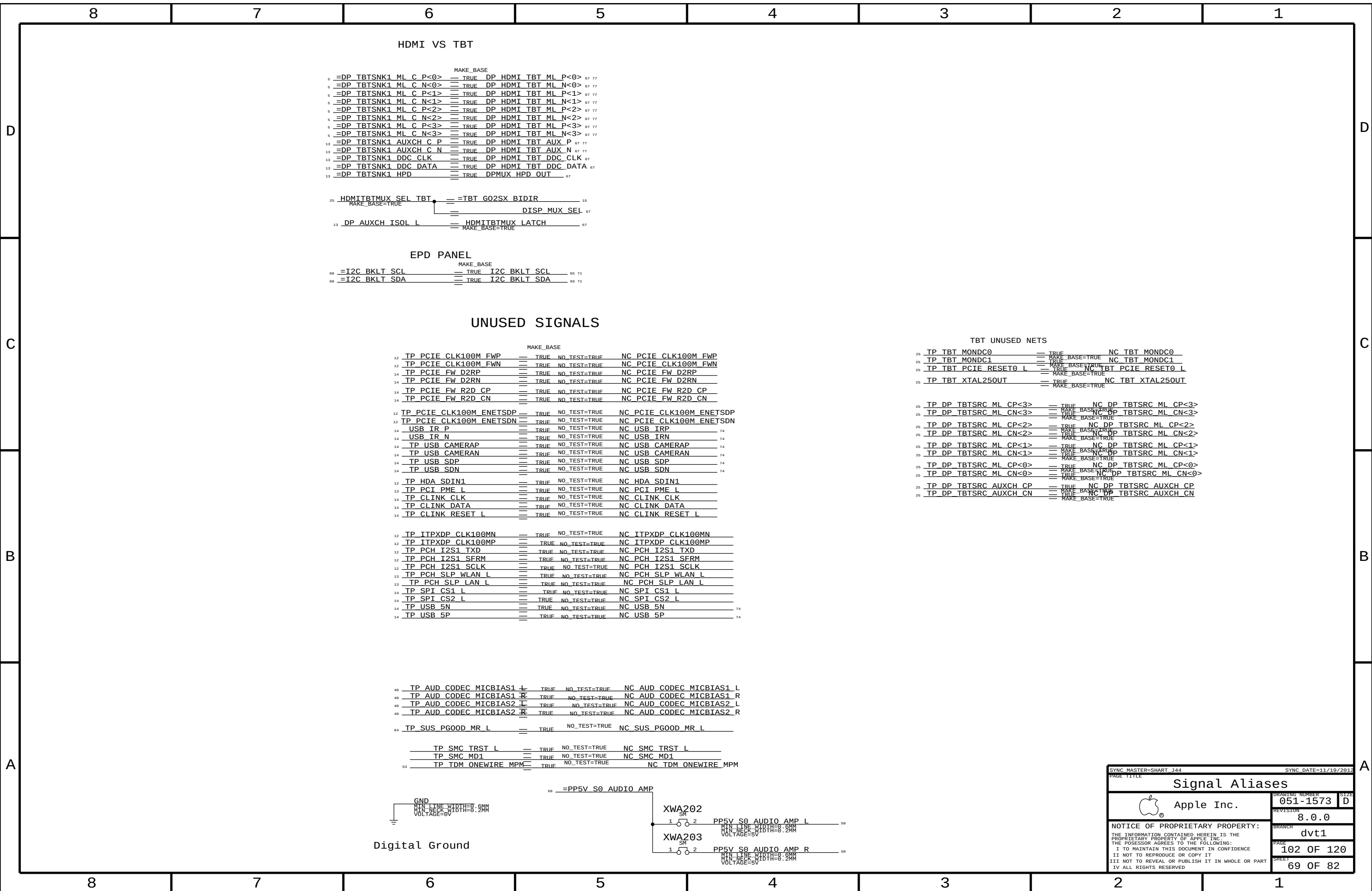
REVISION
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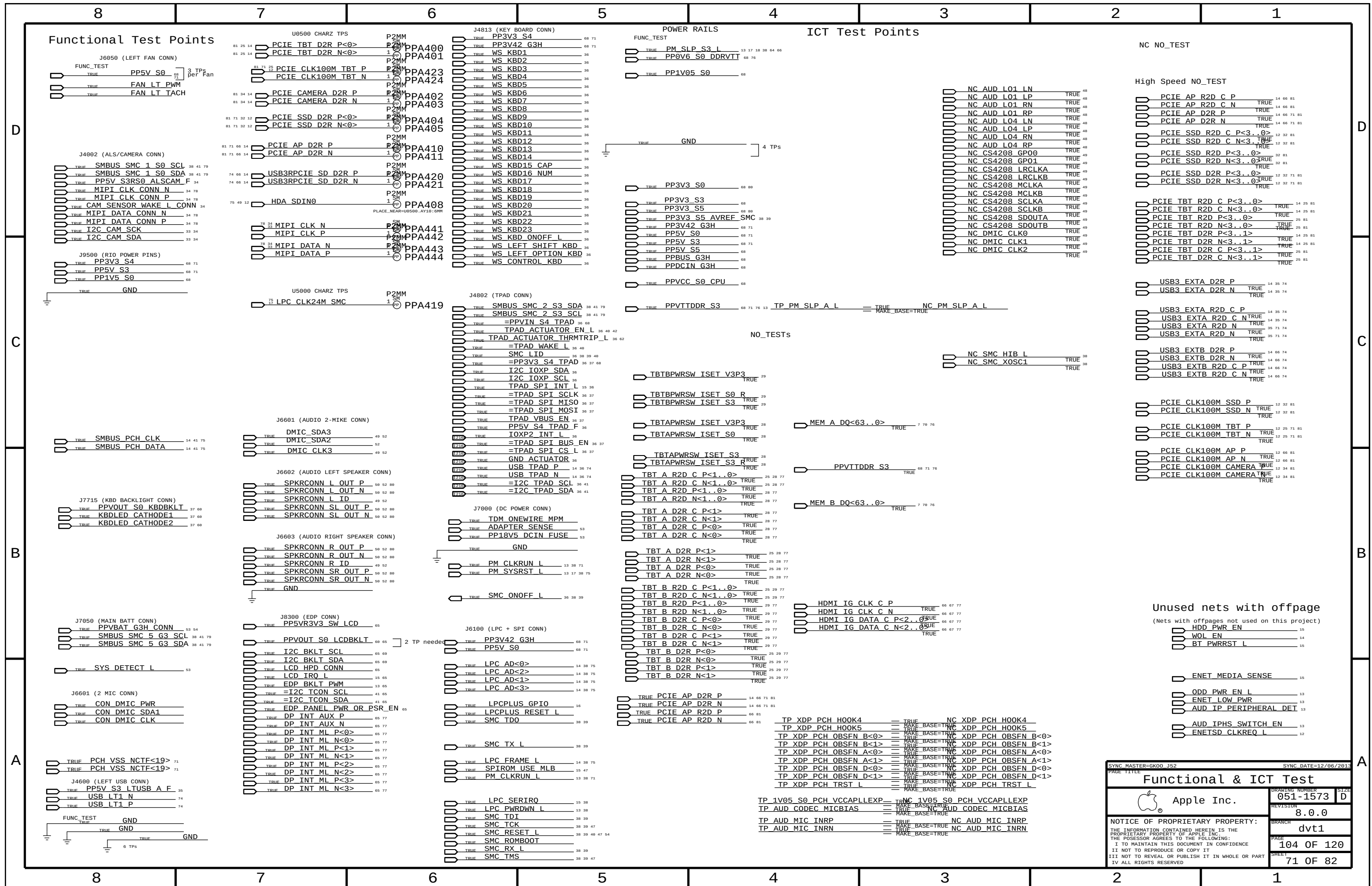
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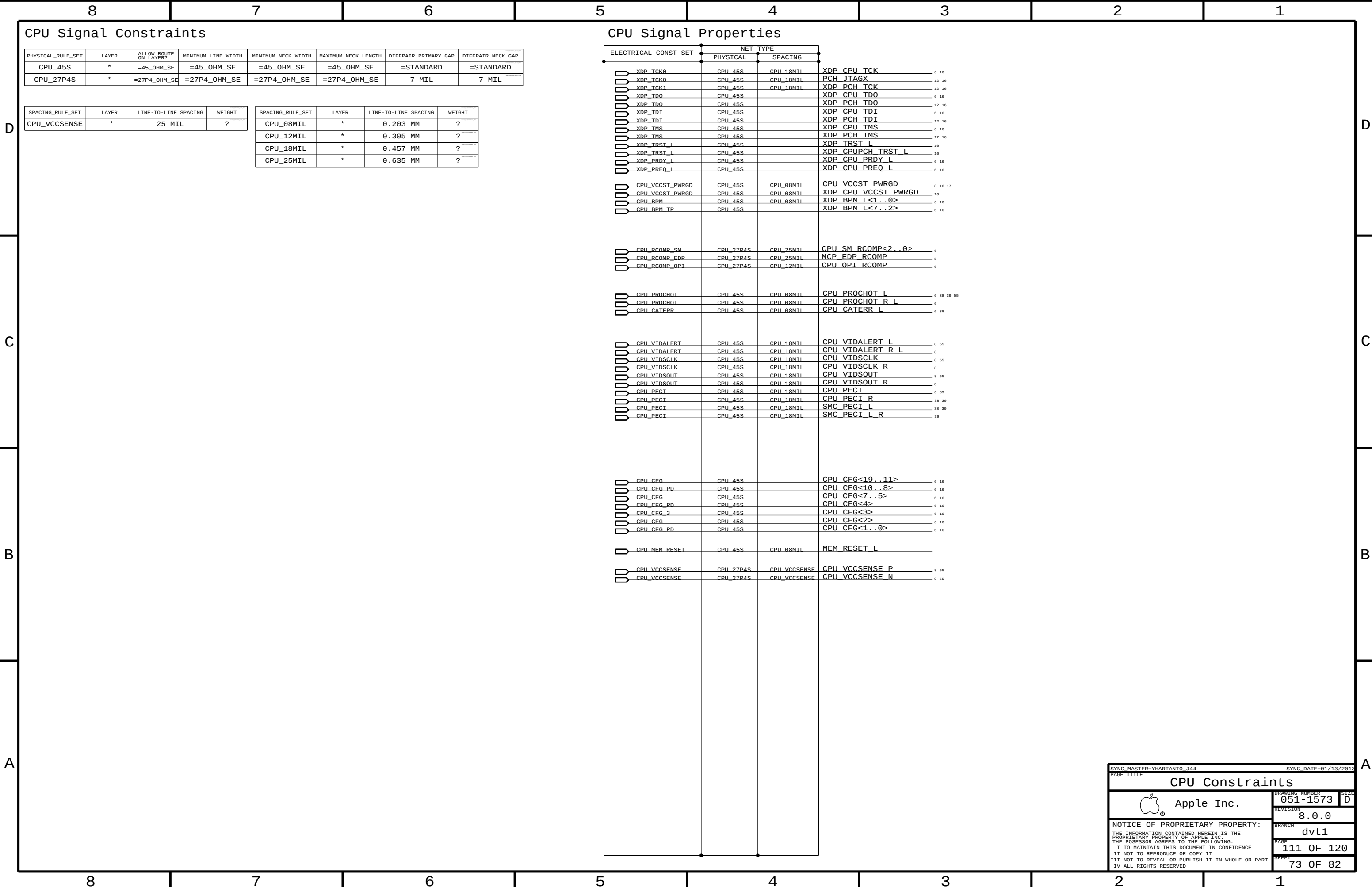
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USB 2 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_USB_RBIA5	*	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD	=STANDARD
USB_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB	*	=4X_DIELECTRIC	?	USB	TOP,BOTTOM	=6X_DIELECTRIC	?
USB_RBIA5	*	=6X_DIELECTRIC	?	USB_RBIA5	TOP,BOTTOM	=10X_DIELECTRIC	?

USB 3 Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
USB3_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_2SAME	*	=3X_DIELECTRIC	?	USB3_2SAME	TOP,BOTTOM	=4X_DIELECTRIC	?
USB3_TXRX	*	=6X_DIELECTRIC	?	USB3_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
USB3_20THER	*	=4X_DIELECTRIC	?	USB3_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
USB3_*	*	*	USB3_20THER
USB3_*	=SAME	*	USB3_2SAME
USB3_TX	*_RX	*	USB3_TXRX
USB3_RX	*_TX	*	USB3_TXRX

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_25M	*	=5x_DIELECTRIC	?

SATA Interface Constraints (Not Used)

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SATA_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF
SATA_45SE	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA_2SAME	*	=3X_DIELECTRIC	?	SATA_2SAME	TOP,BOTTOM	=4x_DIELECTRIC	?
SATA_TXRX	*	=6X_DIELECTRIC	?	SATA_TXRX	TOP,BOTTOM	=10X_DIELECTRIC	?
SATA_20THER	*	=4X_DIELECTRIC	?	SATA_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
SATA_*	*	*	SATA_20THER
SATA_*	=SAME	*	SATA_2SAME
SATA_TX	*_RX	*	SATA_TXRX
SATA_RX	*_TX	*	SATA_TXRX

USB Constraints

ELECTRICAL CONST SET		NET TYPE		
	PHYSICAL	SPACING		
	USB_BT	USB_85D	USB	USB_BT_P 14 31
	USB_BT	USB_85D	USB	USB_BT_N 14 31
	USB_BT	USB_85D	USB	USB_BT_CONN_P 31 66
	USB_BT	USB_85D	USB	USB_BT_CONN_N 31 66
	USB_EXT_A	USB_85D	USB	USB_EXT_A_P 14 35
	USB_EXT_A	USB_85D	USB	USB_EXT_A_N 14 35
		DEFAULT_T	DEFAULT_T	SMC_DEBUGPRT_RX_L 35 38
			DEFAULT_T	SMC_DEBUGPRT_TX_L 35 38
	USB_EXT_A	USB_85D	USB	USB2_EXT_A_MUXED_P 35
	USB_EXT_A	USB_85D	USB	USB2_EXT_A_MUXED_N 35
	USB_EXT_A	USB_85D	USB	USB2_EXT_A_MUXED_F_P 35
	USB_EXT_A	USB_85D	USB	USB2_EXT_A_MUXED_F_N 35
	USB_EXT_A	USB_85D	USB	USB_LT1_P 71
	USB_EXT_A	USB_85D	USB	USB_LT1_N 71
	USB_EXT_B	USB_85D	USB	USB_EXT_B_P 14 66
	USB_EXT_B	USB_85D	USB	USB_EXT_B_N 14 66
	USB_TPAD	USB_85D	USB	USB_TPAD_P 14 36 71
	USB_TPAD	USB_85D	USB	USB_TPAD_N 14 36 71
	USB3_EXT_A_D2R	USB_85D	USB3_RX	USB3_EXT_A_D2R_P 14 35 71
	USB3_EXT_A_D2R	USB_85D	USB3_RX	USB3_EXT_A_D2R_N 14 35 71
	USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A_R2D_P 35
	USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A_R2D_N 35 71
	USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A_R2D_C_P 14 35 71
	USB3_EXT_A_R2D	USB_85D	USB3_TX	USB3_EXT_A_R2D_C_N 14 35 71
	USB3_EXT_B_D2R	USB_85D	USB3_RX	USB3_EXT_B_D2R_P 14 66 71
	USB3_EXT_B_D2R	USB_85D	USB3_RX	USB3_EXT_B_D2R_N 14 66 71
	USB3_EXT_B_R2D	USB_85D	USB3_TX	USB3_EXT_B_R2D_C_P 14 66 71
	USB3_EXT_B_R2D	USB_85D	USB3_TX	USB3_EXT_B_R2D_C_N 14 66 71
	USB3_SD_D2R	USB3_85D	USB3_RX	USB3RPCIE_SD_D2R_P 14 66 71
	USB3_SD_D2R	USB3_85D	USB3_RX	USB3RPCIE_SD_D2R_N 14 66 71
	USB3_SD_R2D	USB3_85D	USB3_TX	USB3RPCIE_SD_R2D_C_P 14 66
	USB3_SD_R2D	USB3_85D	USB3_TX	USB3RPCIE_SD_R2D_C_N 14 66
	USB_NC	USB_85D	USB	NC_USB_IRP 69
	USB_NC	USB_85D	USB	NC_USB_IRN 69
	USB_NC	USB_85D	USB	NC_USB_5P 69
	USB_NC	USB_85D	USB	NC_USB_5N 69
	USB_NC	USB_85D	USB	NC_USB_SDP 69
	USB_NC	USB_85D	USB	NC_USB_SDN 69
	USB_NC	USB_85D	USB	NC_USB_CAMERAP 69
	USB_NC	USB_85D	USB	NC_USB_CAMERAN 69
	PCH_USB_RBIAS	PCH_USB_RBIAS	USB_RBIAS	PCH_USB_RBIAS 14
	SATA_85D	SATA_85D	SATA_RX	DUMMY_SATA_D2R_P
	SATA_85D	SATA_85D	SATA_RX	DUMMY_SATA_D2R_N
	SATA_85D	SATA_85D	SATA_TX	DUMMY_SATA_R2D_P
	SATA_85D	SATA_85D	SATA_TX	DUMMY_SATA_R2D_N
	SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X1 17
	SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2 17
	SYSCLK_CLK25M	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2_R 17
	SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_CAMERA 17 34
	SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKP 33 34
	SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP_R 34
	SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALP 34
	SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_XTALN 34
	SYSCLK_CLK25M_CAM	CLK_25M_45S	CLK_25M	CLK25M_CAM_CLKN 33 34
	SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT 17 25
	SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R 25

Notes:
This is here to keep the SATA rules.

SYNC MASTER=YHARTANTO_344		SYNC DATE=01/07/2013	
PAGE TITLE		USB Constraints	
 Apple Inc.		DRAWING NUMBER	051-1573
		REVISION	8.0.0
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LPC Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	6 MIL	?
CLK_LPC	*	8 MIL	?

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	8 MIL	?

PCH Single Net Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
PCH_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	7 MIL	7 MIL

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_12MIL	*	0.305 MM	?
PCH_15MIL	*	0.381 MM	?
PCH_18MIL	*	0.457 MM	?
PCH_20MIL	*	0.508 MM	?

PCH Net Properties

ELECTRICAL CONST SET		NET TYPE		
	PHYSICAL	SPACING		
LPC_AD	LPC_45S	LPC	LPC_AD<3..0>	14 38 73
LPC_AD	LPC_45S	LPC	LPC_FRAME_L	14 38 73
LPC_CLK24M_SMC	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC_R	12 17
LPC_CLK24M_SMC	CLK_LPC_45S	CLK_LPC	LPC_CLK24M_SMC	17 73
SMBUS_PCH	SMB_45S	SMB	SMBUS_PCH_CLK	14 41 73
SMBUS_PCH	SMB_45S	SMB	SMBUS_PCH_DATA	14 41 73
SML_PCH_0	SMB_45S	SMB	SML_PCH_0_CLK	14 41
SML_PCH_0	SMB_45S	SMB	SML_PCH_0_DATA	14 41
	SMB_45S	SMB	SML_PCH_1_CLK	14 41
	SMB_45S	SMB	SML_PCH_1_DATA	14 41
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK	12 49
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK_R	12
HDA_SYNC	HDA_45S	HDA	HDA_SYNC	12 49
HDA_SYNC	HDA_45S	HDA	HDA_SYNC_R	12
HDA_RST	HDA_45S	HDA	HDA_RST_R_L	12
HDA_RST	HDA_45S	HDA	HDA_RST_L	12 49
HDA_SDTN	HDA_45S	HDA	HDA_SDTN0	12 49 73
HDA_SDTN	HDA_45S	HDA	CS4208_HDA_SDOU0_R	49
HDA_SDOU0T	HDA_45S	HDA	HDA_SDOU0T	12 49
HDA_SDOU0T	HDA_45S	HDA	HDA_SDOU0T_R	12 17
SPT_MLB	SPT_45S	SPT	SPI_ALT_CLK	47
SPT_MLB	SPT_45S	SPT	SPI_CLK	47
SPT_MLB	SPT_45S	SPT	SPT_CLK_R	14 47
SPT_MLB	SPT_45S	SPT	SPT_MLB_CLK	47
SPT_MLB	SPT_45S	SPT	SPT_SMC_CLK	38 47
SPT_MLB	SPT_45S	SPT	SPI_ALT_CS_L	47
SPT_MLB	SPT_45S	SPT	SPI_CS0_L	47
SPT_MLB	SPT_45S	SPT	SPI_CS0_R_L	14 47
SPT_MLB	SPT_45S	SPT	SPT_MLB_CS_L	47
SPT_MLB	SPT_45S	SPT	SPT_SMC_CS_L	38 47
SPT_MLB	SPT_45S	SPT	SPI_ALT_I01_MISO	47
SPT_MLB	SPT_45S	SPT	SPI_MISO	14 47
SPT_MLB	SPT_45S	SPT	SPI_MISO_R	47
SPT_MLB	SPT_45S	SPT	SPT_MLB_I01_MISO	47
SPT_MLB	SPT_45S	SPT	SPI_SMC_MISO	38 47
SPT_MLB	SPT_45S	SPT	SPI_ALT_I00_MOSI	47
SPT_MLB	SPT_45S	SPT	SPI_MOSI	47
SPT_MLB	SPT_45S	SPT	SPI_MOSI_R	14 47
SPT_MLB	SPT_45S	SPT	SPT_MLB_I00_MOSI	47
SPT_MLB	SPT_45S	SPT	SPI_SMC_MOSI	38 47
SPT_MLB_I02	SPT_45S	SPT	SPI_I0<2>	14 47
SPT_MLB_I02	SPT_45S	SPT	SPI_I02_R	47
SPT_MLB_I02	SPT_45S	SPT	SPT_MLB_I02_WP_L	47
SPT_MLB_I02	SPT_45S	SPT	SPI_ALT_I02_WP_L	47
SPT_MLB_I03	SPT_45S	SPT	SPI_I0<3>	14 47
SPT_MLB_I03	SPT_45S	SPT	SPI_I03_R	47
SPT_MLB_I03	SPT_45S	SPT	SPT_MLB_I03_HOLD_L	47
SPT_MLB_I03	SPT_45S	SPT	SPI_ALT_I03_HOLD_L	47
SPT_TPAD	SPT_45S	SPT	TPAD_SPI_CLK	15 37
SPT_TPAD_CS	SPT_45S	SPT	TPAD_SPI_CS_L	15 37
SPT_TPAD	SPT_45S	SPT	TPAD_SPI_MISO	15 37
SPT_TPAD	SPT_45S	SPT	TPAD_SPI_MOSI	15 37
PCH_RTCX	PCH_45S	PCH_15MTI	PCH_CLK32K_RTCX1	12 17
PCH_SRTCST	PCH_45S	PCH_15MTI	PCH_SRTCST_L	12
PCH_RTCRST	PCH_45S	PCH_15MTI	PCH_RESET_L	12
PCH_THRMTRIP	PCH_45S	PCH_18MTI	PM_THRMTRIP_L	15 39
PCH_THRMTRIP	PCH_45S	PCH_18MTI	PM_THRMTRIP_R_L	39
	PCH_45S	PCH_15MTI	PCH_INTRUDER_L	12
	PCH_45S	PCH_15MTI	PCH_INTVRMEN	12
	PCH_45S	PCH_15MTI	PCH_DSWVRMEN	13
	PCH_45S	PCH_15MTI	PM_RSMRST_L	13 64
	PCH_45S	PCH_15MTI	PM_SYSRST_L	13 17 38 73
	PCH_45S	PCH_15MTI	XDP_DBRESET_L	16 17
	PCH_45S	PCH_15MTI	PM_PCH_SYS_PWROK	13 16 17 38
	PCH_45S	PCH_15MTI	XDP_SYS_PWROK	16
	PCH_45S	PCH_15MTI	SYS_PWROK_R	17
	PCH_45S	PCH_15MTI	PM_PCH_PWROK	13 17
	PCH_45S	PCH_15MTI	PM_S0_PG00D	17
	PCH_45S	PCH_15MTI	SMC_DELAYED_PWRGD	17 26 27 38
	PCH_45S	PCH_15MTI	PM_DSW_PWRGD	13 38
	PCH_45S	PCH_15MTI	PM_PWRBTN_L	13 16 38
	PCH_45S	PCH_15MTI	XDP_CPU_PWRBTN_L	16
	PCH_45S	PCH_15MTI	PCIE_WAKE_L	13 31 33
	PCH_45S	PCH_15MTI	AP_PCIE_WAKE_L	31 66
	PCH_45S	PCH_15MTI	CAM_PCIE_WAKE_L	33
	PCH_45S	PCH_15MTI	TBT_CIO_PLUG_EVENT_L	16 25
PCH_CLK24M_XTAL	PCH_45S	PCH_20MTI	PCH_CLK24M_XTALIN	12 17
PCH_CLK24M_XTAL	PCH_45S	PCH_20MTI	PCH_CLK24M_XTALOUT	12 17
PCH_CLK24M_XTAL	PCH_45S	PCH_20MTI	PCH_CLK24M_XTALOUT_R	17
PCH_RCOMP_PCIE	PCH_27P4S	PCH_12MTI	PCH_PCIE_RCOMP	14
PCH_RCOMP_OPT	PCH_27P4S	PCH_12MTI	PCH_OPI_COMP	15
PCH_RCOMP_SATA	PCH_27P4S	PCH_12MTI	PCH_SATA_RCOMP	16

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PAGE TITLE			
PCH Constraints			
 Apple Inc.	DRAWING NUMBER	SIZE	
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Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_40S	*	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE	=40_OHM_SE
MEM_50S	*	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE
MEM_70D	*	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF	=70_OHM_DIFF
MEM_73D	*	=73_OHM_DIFF	=73_OHM_DIFF	0.066 MM	=73_OHM_DIFF	=73_OHM_DIFF	=73_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	*	=2x_DIELECTRIC	?
MEM_DQS2OWNDATA	*	=3x_DIELECTRIC	?
MEM_CMD2CMD	*	=3x_DIELECTRIC	?
MEM_CMD2CTL	*	=3x_DIELECTRIC	?
MEM_CTL2CTL	*	=3x_DIELECTRIC	?
MEM_CLK2CLK	*	=6x_DIELECTRIC	?
MEM_DATA20THERMEM	*	=8x_DIELECTRIC	?
MEM_20THERMEM	*	=4x_DIELECTRIC	?
MEM_2PWR	*	=2x_DIELECTRIC	?
MEM_2GND	*	=2x_DIELECTRIC	?
MEM_20THER	*	=6x_DIELECTRIC	?
MEM_CMD2CMD_BM	*	=3x_DIELECTRIC	?
MEM_CMD2CTL_BM	*	=3x_DIELECTRIC	?
MEM_CTL2CTL_BM	*	=3x_DIELECTRIC	?
MEM_12MIL	*	0.305 MM	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
MEM_DATA2SELF	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_DQS2OWNDATA	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CMD	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CMD2CTL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CTL2CTL	TOP,BOTTOM	=5x_DIELECTRIC	?
MEM_CLK2CLK	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_20THERMEM	TOP,BOTTOM	=8x_DIELECTRIC	?
MEM_2PWR	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_2GND	TOP,BOTTOM	=4x_DIELECTRIC	?
MEM_20THER	TOP,BOTTOM	=10x_DIELECTRIC	?
MEM_CMD2CMD_BM	TOP,BOTTOM	=3x_DIELECTRIC	?
MEM_CMD2CTL_BM	TOP,BOTTOM	=3x_DIELECTRIC	?
MEM_CTL2CTL_BM	TOP,BOTTOM	=3x_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	*	*	MEM_20THER
MEM_*_DQS_*	*	*	MEM_20THER
MEM_CMD	*	*	MEM_20THER
MEM_CTL	*	*	MEM_20THER
MEM_CLK	*	*	MEM_20THER
MEM_*	MEM_*	*	MEM_20THERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	=SAME	*	MEM_DATA2SELF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_*_DQBYTE_*	MEM_*	*	MEM_DATA20THERMEM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_CMD	MEM_CMD	*	MEM_CMD2CMD
MEM_CMD	MEM_CTL	*	MEM_CMD2CTL
MEM_CTL	MEM_CTL	*	MEM_CTL2CTL
MEM_CLK	MEM_CLK	*	MEM_CLK2CLK
MEM_CMD	MEM_CMD	BGA_MEM	MEM_CMD2CMD_BM
MEM_CMD	MEM_CTL	BGA_MEM	MEM_CMD2CTL_BM
MEM_CTL	MEM_CTL	BGA_MEM	MEM_CTL2CTL_BM

Broadwell ULT Memory Down LPDDR3 1x4 Length Matching

LPDDR3 Signal Group	Unit	Min Length	Max Length
CTL/CKEmax - CTL/CKEmin	mils	0	50
CTL/CKE to CLK	mils	CLK - 100	0
(CMDmax - CMDmin)	mils	0	50
CMD to CLK	mils	CLK - 250	CLK + 250
DQmax - DQmin per byte	mils	125	0
DQmax to DQs per byte	mils	DQS - 200	DQS + 50
DQS to DQS#	mils	-2.5	2.5
DQS to CLK (Rule 1)	mils	CLK - 750	CLK + 1250
CLK to CLK#	mils	-2.5	2.5

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_PWR	MEM_*	*	MEM_2PWR
MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	MEM_*	*	MEM_2GND

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
MEM_70D	BGA_MEM	MEM_73D
MEM_40S	BGA_MEM	MEM_50S

Memory Net Properties

ELECTRICAL CONST SET	PHYSICAL	NET TYPE	SPACING
MEM_A_CLK0	MEM_70D	MEM_CLK	MEM_A_CLK P<0>
MEM_A_CLK0	MEM_70D	MEM_CLK	MEM_A_CLK N<0>
MEM_A_CLK1	MEM_70D	MEM_CLK	MEM_A_CLK P<1>
MEM_A_CLK1	MEM_70D	MEM_CLK	MEM_A_CLK N<1>
MEM_A_CTL	MEM_40S	MEM_CTL	MEM_A CS_L<1..0>
MEM_A_CTL	MEM_40S	MEM_CTL	MEM_A ODT<0>
MEM_A_CKE0	MEM_40S	MEM_CMD	MEM_A CKE<1..0>
MEM_A_CKE1	MEM_40S	MEM_CMD	MEM_A CKE<3..2>
MEM_A_CMD0	MEM_40S	MEM_CMD	MEM_A CAA<9..0>
MEM_A_CMD1	MEM_40S	MEM_CMD	MEM_A CAB<9..0>
MEM_A_DQBYTE0	MEM_40S	MEM_A_DQBYTE_0	MEM_A DQ<7..0>
MEM_A_DQBYTE1	MEM_40S	MEM_A_DQBYTE_1	MEM_A DQ<15..8>
MEM_A_DQBYTE2	MEM_40S	MEM_A_DQBYTE_2	MEM_A DQ<23..16>
MEM_A_DQBYTE3	MEM_40S	MEM_A_DQBYTE_3	MEM_A DQ<31..24>
MEM_A_DQBYTE4	MEM_40S	MEM_A_DQBYTE_4	MEM_A DQ<39..32>
MEM_A_DQBYTE5	MEM_40S	MEM_A_DQBYTE_5	MEM_A DQ<47..40>
MEM_A_DQBYTE6	MEM_40S	MEM_A_DQBYTE_6	MEM_A DQ<55..48>
MEM_A_DQBYTE7	MEM_40S	MEM_A_DQBYTE_7	MEM_A DQ<63..56>
MEM_A_DQS0	MEM_70D	MEM_A_DQS_0	MEM_A DQS P<0>
MEM_A_DQS0	MEM_70D	MEM_A_DQS_0	MEM_A DQS N<0>
MEM_A_DQS1	MEM_70D	MEM_A_DQS_1	MEM_A DQS P<1>
MEM_A_DQS1	MEM_70D	MEM_A_DQS_1	MEM_A DQS N<1>
MEM_A_DQS2	MEM_70D	MEM_A_DQS_2	MEM_A DQS P<2>
MEM_A_DQS2	MEM_70D	MEM_A_DQS_2	MEM_A DQS N<2>
MEM_A_DQS3	MEM_70D	MEM_A_DQS_3	MEM_A DQS P<3>
MEM_A_DQS3	MEM_70D	MEM_A_DQS_3	MEM_A DQS N<3>
MEM_A_DQS4	MEM_70D	MEM_A_DQS_4	MEM_A DQS P<4>
MEM_A_DQS4	MEM_70D	MEM_A_DQS_4	MEM_A DQS N<4>
MEM_A_DQS5	MEM_70D	MEM_A_DQS_5	MEM_A DQS P<5>
MEM_A_DQS5	MEM_70D	MEM_A_DQS_5	MEM_A DQS N<5>
MEM_A_DQS6	MEM_70D	MEM_A_DQS_6	MEM_A DQS P<6>
MEM_A_DQS6	MEM_70D	MEM_A_DQS_6	MEM_A DQS N<6>
MEM_A_DQS7	MEM_70D	MEM_A_DQS_7	MEM_A DQS P<7>
MEM_A_DQS7	MEM_70D	MEM_A_DQS_7	MEM_A DQS N<7>
MEM_B_CLK0	MEM_70D	MEM_CLK	MEM_B_CLK P<0>
MEM_B_CLK0	MEM_70D	MEM_CLK	MEM_B_CLK N<0>
MEM_B_CLK1	MEM_70D	MEM_CLK	MEM_B_CLK P<1>
MEM_B_CLK1	MEM_70D	MEM_CLK	MEM_B_CLK N<1>
MEM_B_CTL	MEM_40S	MEM_CTL	MEM_B CS_L<1..0>
MEM_B_CTL	MEM_40S	MEM_CTL	MEM_B ODT<0>
MEM_B_CKE0	MEM_40S	MEM_CMD	MEM_B CKE<1..0>
MEM_B_CKE1	MEM_40S	MEM_CMD	MEM_B CKE<3..2>
MEM_B_CMD0	MEM_40S	MEM_CMD	MEM_B CAA<9..0>
MEM_B_CMD1	MEM_40S	MEM_CMD	MEM_B CAB<9..0>
MEM_B_DQBYTE0	MEM_40S	MEM_B_DQBYTE_0	MEM_B DQ<7..0>
MEM_B_DQBYTE1	MEM_40S	MEM_B_DQBYTE_1	MEM_B DQ<15..8>
MEM_B_DQBYTE2	MEM_40S	MEM_B_DQBYTE_2	MEM_B DQ<23..16>
MEM_B_DQBYTE3	MEM_40S	MEM_B_DQBYTE_3	MEM_B DQ<31..24>
MEM_B_DQBYTE4	MEM_40S	MEM_B_DQBYTE_4	MEM_B DQ<39..32>
MEM_B_DQBYTE5	MEM_40S	MEM_B_DQBYTE_5	MEM_B DQ<47..40>
MEM_B_DQBYTE6	MEM_40S	MEM_B_DQBYTE_6	MEM_B DQ<55..48>
MEM_B_DQBYTE7	MEM_40S	MEM_B_DQBYTE_7	MEM_B DQ<63..56>
MEM_B_DQS0	MEM_70D	MEM_B_DQS_0	MEM_B DQS P<0>
MEM_B_DQS0	MEM_70D	MEM_B_DQS_0	MEM_B DQS N<0>
MEM_B_DQS1	MEM_70D	MEM_B_DQS_1	MEM_B DQS P<1>
MEM_B_DQS1	MEM_70D	MEM_B_DQS_1	MEM_B DQS N<1>
MEM_B_DQS2	MEM_70D	MEM_B_DQS_2	MEM_B DQS P<2>
MEM_B_DQS2	MEM_70D	MEM_B_DQS_2	MEM_B DQS N<2>
MEM_B_DQS3	MEM_70D	MEM_B_DQS_3	MEM_B DQS P<3>
MEM_B_DQS3	MEM_70D	MEM_B_DQS_3	MEM_B DQS N<3>
MEM_B_DQS4	MEM_70D	MEM_B_DQS_4	MEM_B DQS P<4>
MEM_B_DQS4	MEM_70D	MEM_B_DQS_4	MEM_B DQS N<4>
MEM_B_DQS5	MEM_70D	MEM_B_DQS_5	MEM_B DQS P<5>
MEM_B_DQS5	MEM_70D	MEM_B_DQS_5	MEM_B DQS N<5>
MEM_B_DQS6	MEM_70D	MEM_B_DQS_6	MEM_B DQS P<6>
MEM_B_DQS6	MEM_70D	MEM_B_DQS_6	MEM_B DQS N<6>
MEM_B_DQS7	MEM_70D	MEM_B_DQS_7	MEM_B DQS P<7>
MEM_B_DQS7	MEM_70D	MEM_B_DQS_7	MEM_B DQS N<7>
		MEM_PWR	PP1V2 S3
		MEM_PWR	PP1V2 S3 CPUDDR
		MEM_PWR	PP0V6 S0 DDRVTT
		MEM_PWR	PPVTTDDR S3
		MEM_12MIL	CPU DIMMA VREFDQ
		MEM_12MIL	CPU DIMMB VREFDQ
		MEM_12MIL	CPU DIMM VREFCA
		MEM_12MIL	PP0V6 S3 MEM VREFDQ_A
		MEM_12MIL	PP0V6 S3 MEM VREFDQ_B
		MEM_12MIL	PP0V6 S3 MEM VREFCA_A
		MEM_12MIL	PP0V6 S3 MEM VREFCA_B

SYNC MASTER=YHARTANTO_344

SYNC DATE=01/02/2013

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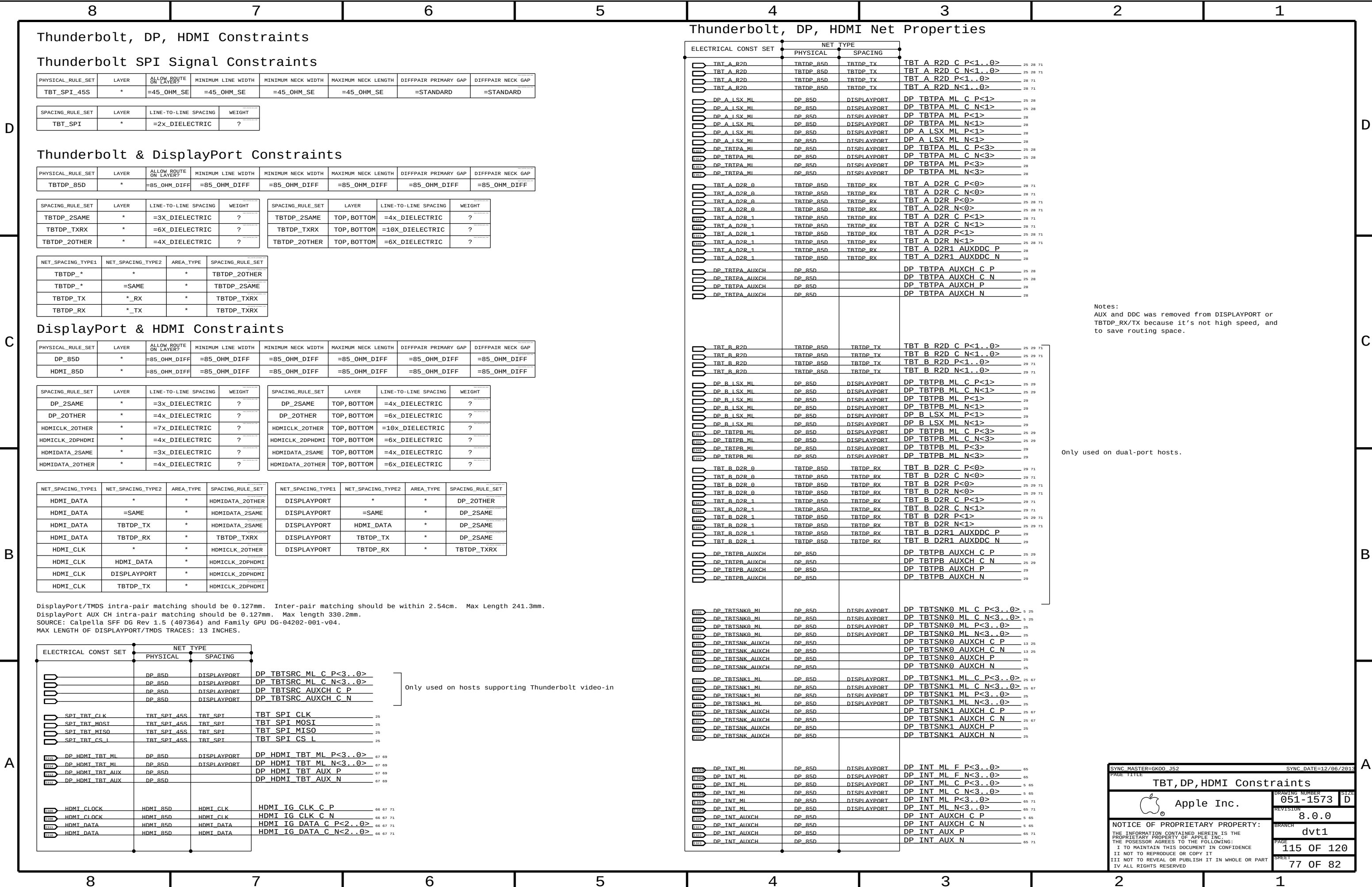
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MIPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
MIPI_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
MIPI_20THER	*	=4X_DIELECTRIC	?	MIPI_20THER	TOP,BOTTOM	=6X_DIELECTRIC	?
MIPI_2CLK	*	=6X_DIELECTRIC	?	MIPI_2CLK	TOP,BOTTOM	=8X_DIELECTRIC	?
MIPICLK_20THER	*	=7X_DIELECTRIC	?	MIPICLK_20THER	TOP,BOTTOM	=10X_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MIPI_DATA	*	*	MIPI_20THER
MIPI_DATA	CLK_MIPI	*	MIPI_2CLK
CLK_MIPI	*	*	MIPICLK_20THER

Memory Bus Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
S2_MEM_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
S2_MEM_85D	*	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF	=85_OHM_DIFF

Spacing Rule Sets

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
S2_DATA2SELF	*	=2X_DIELECTRIC	?	S2_DATA2SELF	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_DQS20WNDATA	*	=2X_DIELECTRIC	?	S2_DQS20WNDATA	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CMD	*	=2X_DIELECTRIC	?	S2_CMD2CMD	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CMD2CTRL	*	=2X_DIELECTRIC	?	S2_CMD2CTRL	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_CTRL2CTRL	*	=2X_DIELECTRIC	?	S2_CTRL2CTRL	TOP,BOTTOM	=4X_DIELECTRIC	?
S2_20THERMEM	*	=4X_DIELECTRIC	?	S2_20THERMEM	TOP,BOTTOM	=6X_DIELECTRIC	?
S2MEM_2PWR	*	=2X_DIELECTRIC	?	S2MEM_2PWR	TOP,BOTTOM	=4X_DIELECTRIC	?
S2MEM_2GND	*	=2X_DIELECTRIC	?	S2MEM_2GND	TOP,BOTTOM	=4X_DIELECTRIC	?
S2MEM_20THER	*	=6X_DIELECTRIC	?	S2MEM_20THER	TOP,BOTTOM	=10X_DIELECTRIC	?

Memory Bus Spacing Group Assignments

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET	NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_DATA*	*	*	S2MEM_20THER	S2_MEM_DQS1	S2_MEM_DATA1	*	S2_DQS20WNDATA
S2_MEM_DQS*	*	*	S2MEM_20THER	S2_MEM_DQS0	S2_MEM_DATA0	*	S2_DQS20WNDATA
S2_MEM_CMD	*	*	S2MEM_20THER				
S2_MEM_CTRL	*	*	S2MEM_20THER				
S2_MEM_CLK	*	*	S2MEM_20THER				
S2_MEM_DATA*	=SAME	*	S2_DATA2SELF				
S2_MEM_CMD	S2_MEM_CMD	*	S2_CMD2CMD				
S2_MEM_CMD	S2_MEM_CTRL	*	S2_CMD2CTRL				
S2_MEM_CTRL	S2_MEM_CTRL	*	S2_CTRL2CTRL				
S2_MEM_*	S2_MEM_*	*	S2_20THERMEM				

Memory to Power Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
S2_MEM_PWR	S2_MEM_*	*	S2MEM_2PWR
S2_MEM_PWR	*	*	DEFAULT

Memory to GND Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	S2_MEM_*	*	S2MEM_2GND

Camera Net Properties

ELECTRICAL CONST SET		NET TYPE			
		PHYSICAL	SPACING		
	S2 MEM_CLK	S2 MEM_85D	S2 MEM_CLK	MEM CAM CLK P	33 34
	S2 MEM_CLK	S2 MEM_85D	S2 MEM_CLK	MEM CAM CLK N	33 34
	S2 MEM_CKE	S2 MEM_45S	S2 MEM_CTRL	MEM CAM CKE	33 34
	S2 MEM_CS	S2 MEM_45S	S2 MEM_CTRL	MEM CAM CS L	33 34
	S2 MEM_CS	S2 MEM_45S	S2 MEM_CTRL	MEM CAM ODT	34
	S2 MEM_CMD	S2 MEM_45S	S2 MEM_CTRL	MEM CAM CAS L	33 34
	S2 MEM_CMD	S2 MEM_45S	S2 MEM_CTRL	MEM CAM RAS L	33 34
	S2 MEM_CMD	S2 MEM_45S	S2 MEM_CMD	MEM CAM WE L	33 34
	S2 MEM_CMD	S2 MEM_45S	S2 MEM_CMD	MEM CAM BA<0>	33 34
	S2 MEM_CMD	S2 MEM_45S	S2 MEM_CMD	MEM CAM BA<1>	33 34
	S2 MEM_CMD	S2 MEM_45S	S2 MEM_CMD	MEM CAM BA<2>	33 34
	S2 MEM_DQS0	S2 MEM_85D	S2 MEM_DQS0	MEM CAM DQS P<0>	33 34
	S2 MEM_DQS0	S2 MEM_85D	S2 MEM_DQS0	MEM CAM DQS N<0>	33 34
	S2 MEM_DQS1	S2 MEM_85D	S2 MEM_DQS1	MEM CAM DQS P<1>	33 34
	S2 MEM_DQS1	S2 MEM_85D	S2 MEM_DQS1	MEM CAM DQS N<1>	33 34
	S2 MEM_DATA_0	S2 MEM_45S	S2 MEM_DATA0	MEM CAM DM<0>	33 34
	S2 MEM_DATA_1	S2 MEM_45S	S2 MEM_DATA1	MEM CAM DM<1>	33 34
	S2 MEM_A	S2 MEM_45S	S2 MEM_CMD	MEM CAM A<14..0>	33 34
	S2 MEM_DATA_0	S2 MEM_45S	S2 MEM_DATA0	MEM CAM DQ<7..0>	33 34
	S2 MEM_DATA_1	S2 MEM_45S	S2 MEM_DATA1	MEM CAM DQ<15..8>	33 34
	MIPT_DATA_S2	MIPT_85D	MIPT_DATA	MIPI DATA P	33 34 71
	MIPT_DATA_S2	MIPT_85D	MIPT_DATA	MIPI DATA N	33 34 71
	MIPT_DATA_S2	MIPT_85D	MIPT_DATA	MIPI DATA CONN P	34 71
	MIPT_DATA_S2	MIPT_85D	MIPT_DATA	MIPI DATA CONN N	34 71
	MIPT_CLK_S2	MIPT_85D	CLK_MIPT	MIPI CLK P	33 34 71
	MIPT_CLK_S2	MIPT_85D	CLK_MIPT	MIPI CLK N	33 34 71
	MIPT_CLK_S2	MIPT_85D	CLK_MIPT	MIPI CLK CONN P	34 71
	MIPT_CLK_S2	MIPT_85D	CLK_MIPT	MIPI CLK CONN N	34 71
			S2 MFM_PWR	PP1V35_CAM	33 34
			S2 MFM_PWR	PP0V675_CAM_VREF	33 34
			S2 MFM_PWR	PP0V675_MEM_CAM_VREFCA	34
			S2 MFM_PWR	PP0V675_MEM_CAM_VREFDQ	34

	8	7	6	5	4	3	2	1	
D									D
C									C
B									B
A									A
	8	7	6	5	4	3	2	1	

SMC SMBus & Charger Net Properties

ELECTRICAL CONST SET	NET TYPE			
	PHYSICAL	SPACING		
SMBUS_SMC_2	SMB_45S	SMR	SMBUS_SMC_2_S3_SCL	30 41 71
SMBUS_SMC_2	SMB_45S	SMR	SMBUS_SMC_2_S3_SDA	30 41 71
SMBUS_SMC_1	SMB_45S	SMR	SMBUS_SMC_1_S0_SCL	30 41 71
SMBUS_SMC_1	SMB_45S	SMR	SMBUS_SMC_1_S0_SDA	30 41 71
SMBUS_SMC_0	SMB_45S	SMR	SMBUS_SMC_0_S0_SCL	30 41
SMBUS_SMC_0	SMB_45S	SMR	SMBUS_SMC_0_S0_SDA	30 41
SMBUS_SMC_5	SMB_45S	SMR	SMBUS_SMC_5_G3_SCL	30 41 71
SMBUS_SMC_5	SMB_45S	SMR	SMBUS_SMC_5_G3_SDA	30 41 71
SMBUS_SMC_3	SMB_45S	SMR	SMBUS_SMC_3_SCL	30 41
SMBUS_SMC_3	SMB_45S	SMR	SMBUS_SMC_3_SDA	30 41

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PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SENSE_45S	*	=1T01_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.1 MM	0.1 MM
THERM_45S	*	=1T01_DIFFPAIR	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	0.1 MM	0.1 MM
DIG_AUDIO	*	=1T01_DIFFPAIR	=1T01_DIFFPAIR	=1T01_DIFFPAIR	=1T01_DIFFPAIR	0.1 MM	0.1 MM
ANL_AUDIO	*	=1T01_DIFFPAIR	0.1 MM	0.1 MM	10 MM	0.1 MM	0.1 MM
ANL_AUDIO_WIDE	*	=1T01_DIFFPAIR	0.3 MM	0.3 MM	10 MM	0.1 MM	0.1 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SENSE	*	=2X_DIELECTRIC	?
THERM	*	=2X_DIELECTRIC	?
AUDIO	*	=2X_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND	*	=STANDARD	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
GND_P2MM	*	0.20 MM	1000
PWR_P2MM	*	0.20 MM	1000

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_VCCSENSE	GND	*	GND_P2MM

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	GND	*	GND_P2MM
GND	PCIE_*	*	GND_P2MM
GND	SATA_*	*	GND_P2MM
USB	GND	*	GND_P2MM
CLK_PCIE	SB_POWER	*	PWR_P2MM
SB_POWER	SATA_*	*	PWR_P2MM
USB	SB_POWER	*	PWR_P2MM

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
MEM_45S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.070 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_40S OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_72D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
MEM_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	100 MIL OVERRIDE	OVERRIDE	OVERRIDE
PCIE_85D OVERRIDE	* OVERRIDE	OVERRIDE	OVERRIDE	0.090 MM OVERRIDE	10 MM OVERRIDE	OVERRIDE	OVERRIDE
USB_85D	TOP			0.100 MM	500 MIL		
CPU_27P4S	BOTTOM			0.230 MM	100 MIL		
USB3_85D	TOP			0.100 MM	500 MIL		
USB3_85D	ISL10			0.075 MM			0.090 MM
DP_85D	ISL9			0.075 MM			0.090 MM
PCIE_85D	ISL10			0.075 MM			0.090 MM

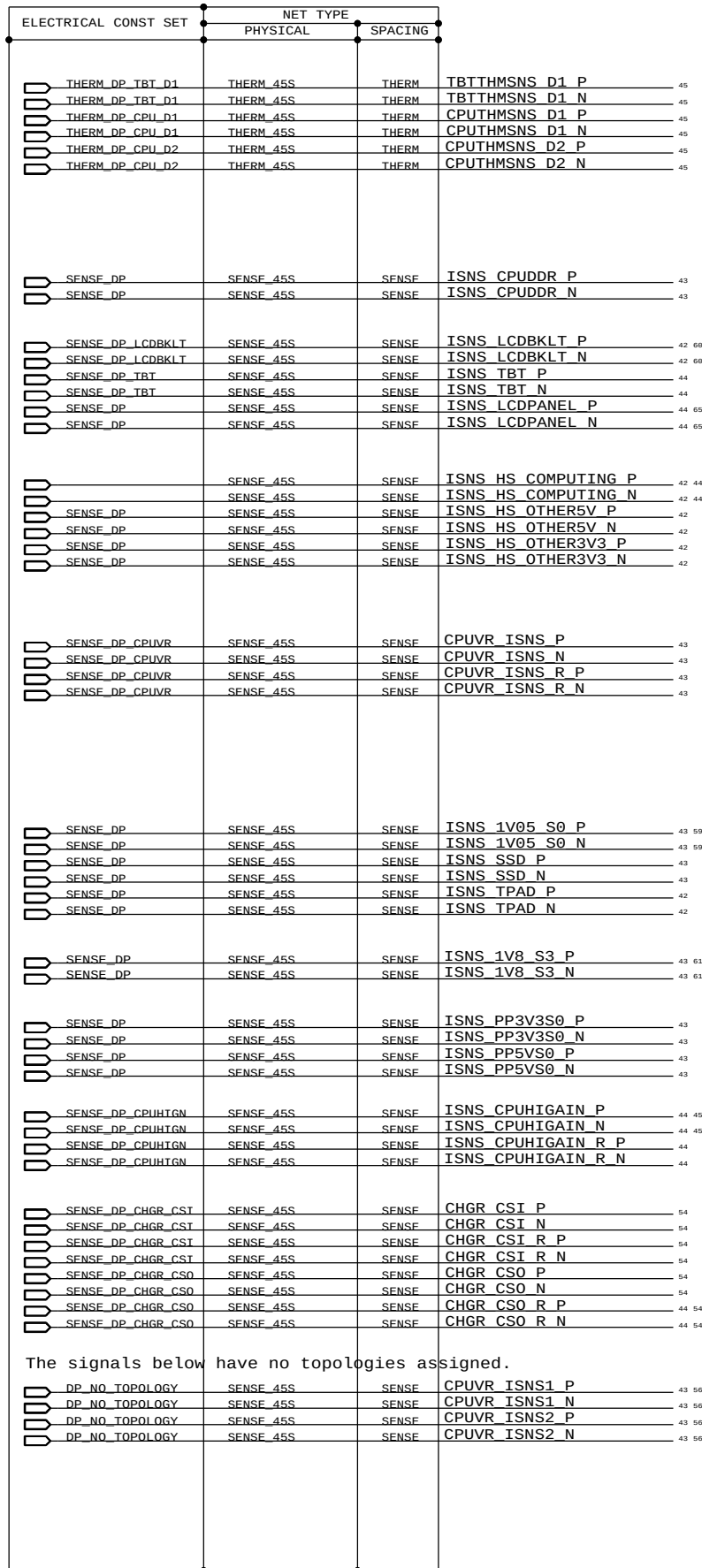
DP, SATA, HDMI, PCIE CONSTRAINT RELAXATIONS

Alternate diffpair width/gap through BGA fanout areas (95-ohm diff)

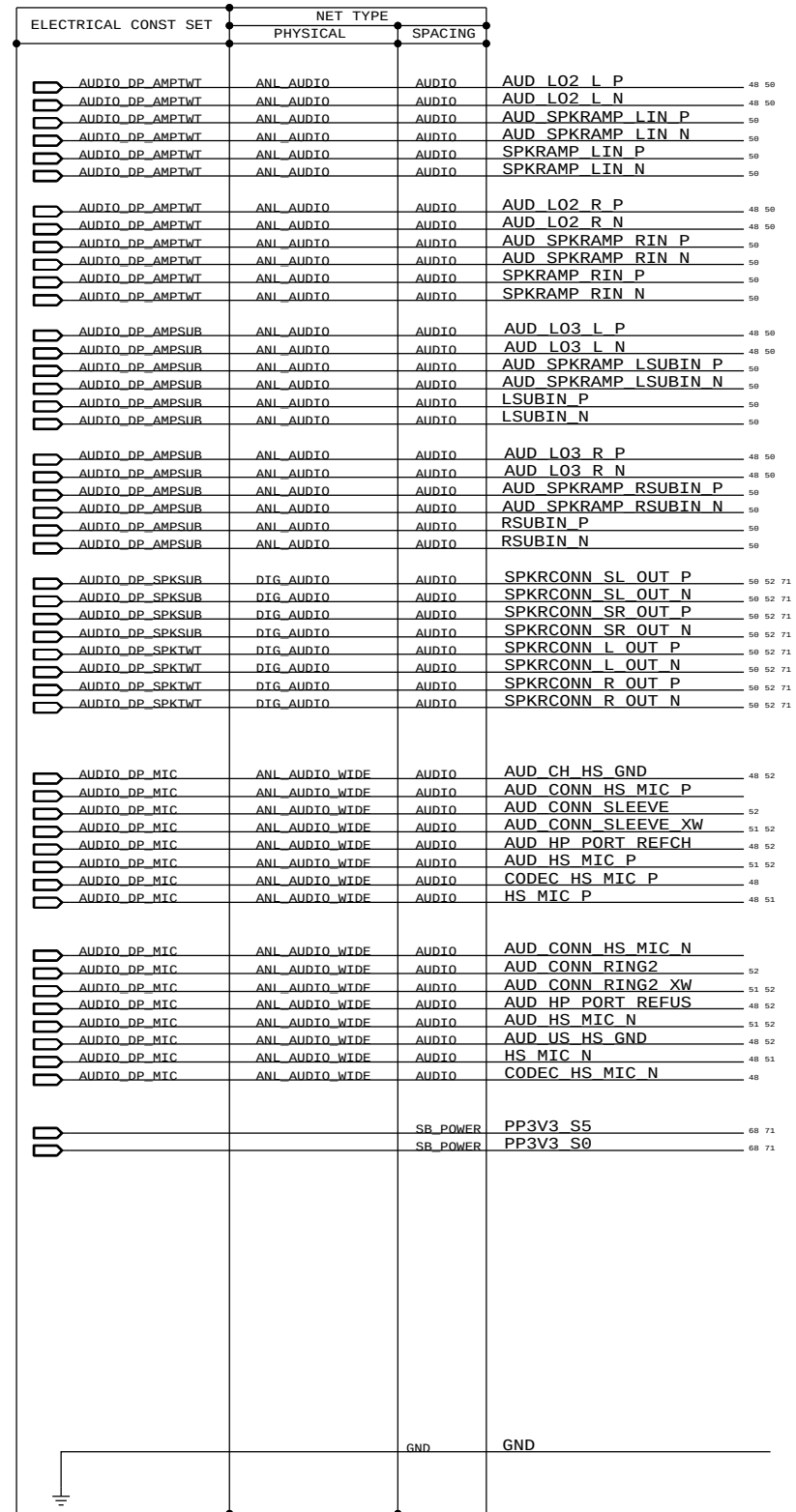
NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
DP_85D	BGA	P65_BGA
PCIE_85D	BGA	P65_BGA
CLK_PCIE_85D	BGA	P65_BGA
HDMI_85D	BGA	P65_BGA

NET_PHYSICAL_TYPE	AREA_TYPE	PHYSICAL_RULE_SET
SENSE_45S	*	SENSE_45S
THERM_45S	*	THERM_45S
DIG_AUDIO	*	DIG_AUDIO
ANL_AUDIO	*	ANL_AUDIO

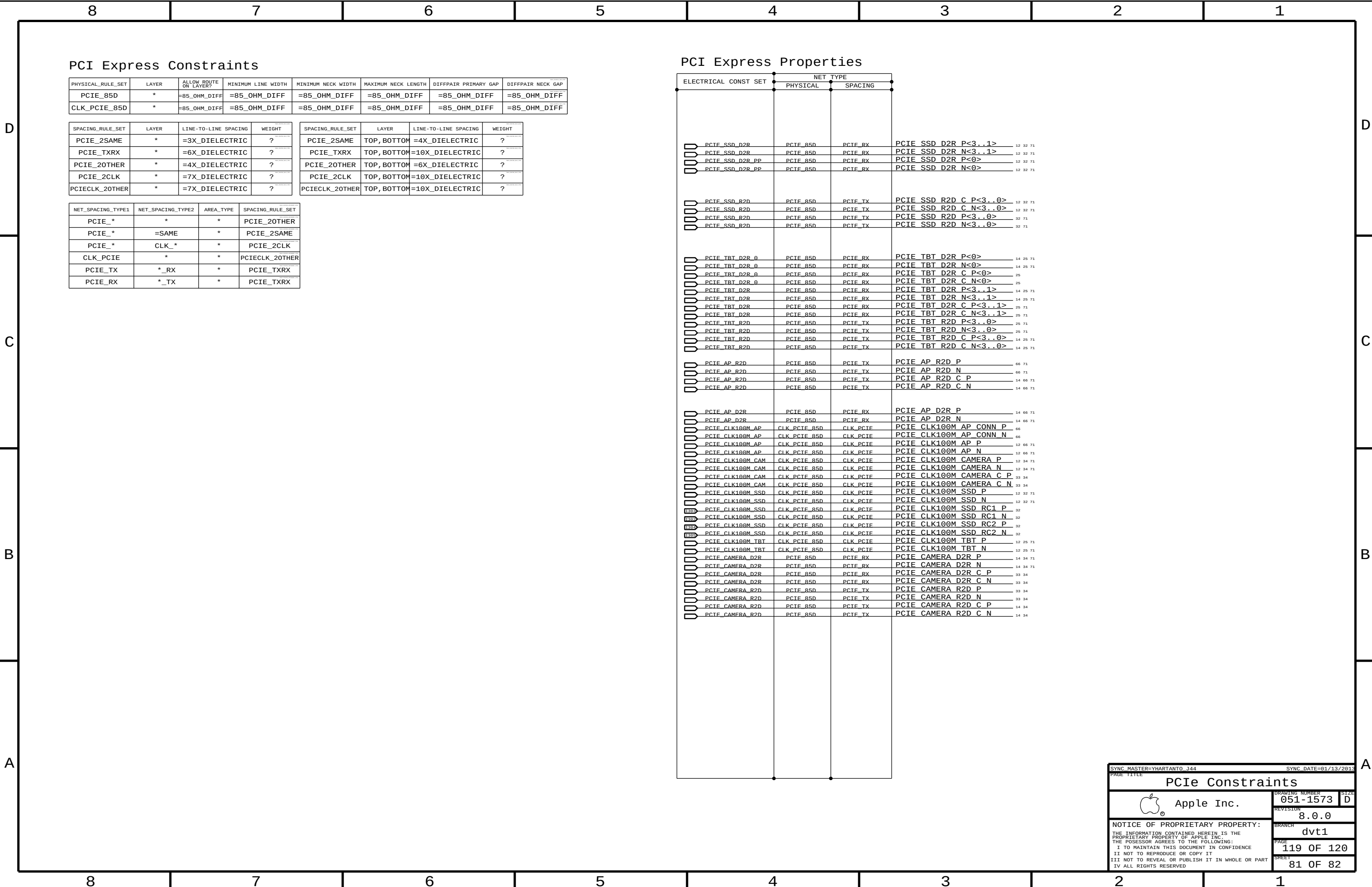
X304 Specific Net Properties



X304 Specific Net Properties



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Change List: <RDAR://COMPONENT/XXXXXX> X304 HW EE SCHEMATIC PROTO 0							
Kismet: AFP://KISMET.APPLE.COM/KISMET-PROJECTS/X304							
Useful Wiki Links: Schematic Conventions - https://hmts.ecs.apple.com/wiki/index.php/User:Wferry/SchConventions Schematic Design Wiki - https://hmts.ecs.apple.com/wiki/index.php/Schematic_Design							
MobileMac HW Radar: <rdar://component/497591> MobileMac HW Task <rdar://component/497587> MobileMac HW Schematic <rdar://component/497585> MobileMac HW New Bugs <rdar://component/497588> MobileMac HW Layout <rdar://component/497590> MobileMac HW Investigation <rdar://component/497589> MobileMac HW Architecture							
Other Info: Page Allocations - <rdar://problem/11791318> 2012 Schematic Page Allocations							
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